Multiphoton Interference and Representation Theory in the Many-Photon Limit

Mathematical Appendices

PhD Candidacy Exam

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Bogotá, Colombia
2026

Note on organization

These appendices present the mathematical derivations and numerical calculations supporting the doctoral candidacy proposal. Their organization follows the logical dependencies of the analysis. Appendices A and B establish the exact probabilistic formulation, the Schur–Weyl decomposition, and the geometry of compatible unitary frames. Appendices C–E develop the construction of irreducible representations, the projection onto weight spaces, and the description of their multiplicities. Appendices F and G examine the kernels of projected states and the physical sectorial densities and detection probabilities, maintaining the distinction between projected pure states and physical mixed states. Finally, Appendices H and I derive the stationary conditions, the Hessian of the toroidal integral, and the sectorial classical map, together with conditional results on the normalization of projected states and exponential suppression outside the classical image. The main candidacy document presents the physical motivation, established results, and research objectives, while the detailed derivations, assumptions, and supporting calculations are collected here.

Computational reproducibility. The numerical calculations and figures presented in these appendices were obtained using the companion Wolfram Language code `Multiphoton_Interference_Calculations.wl`. The code and the Mathematical Appendices are available in the GitHub repository, with the version associated with the candidacy examination identified by the release `candidacy-2026`. The implementation includes routines for constructing compatible frames, evaluating projected-state Gram matrices, physical sectorial densities and detection probabilities, and computing reduced Hessians and toroidal maps. The calculations were performed using Wolfram Mathematica 14.5 at machine precision. Numerical ranks were estimated from the singular values s_k using the relative criterion $s_k > 10^{-8} s_{\max}$, where $s_{\max} = \max_k s_k$. These numerical calculations complement the analytical results but do not replace the hypotheses required for the asymptotic statements.

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A Exact probabilistic formulation and Schur–Weyl reduction

This appendix derives an exact decomposition of multiphotonic scattering probabilities into irreducible sectors. For the correlated-mode preparations considered in the main text, the physical spatial input state is obtained by conditioning an auxiliary tensor-power state on the fixed occupation $\boldsymbol{n}$. Schur–Weyl duality then separates sector weights from the density operators on their internal weight spaces. This finite-N reduction relies on the action of diagonal matrices on weight spaces [1, 2] and the Gram-matrix description of partial indistinguishability [3, 4].

A.1 Effective single-particle matrix and initial occupation

Consider N photons distributed among M input modes, with occupation

$$\boldsymbol{n} = (n_1, \ldots, n_M), \qquad \sum_{i=1}^{M} n_i = N, \tag{A.1}$$

and normalized occupation

$$\overline{\boldsymbol{n}} = \frac{\boldsymbol{n}}{N}, \qquad \sum_{i=1}^{M} \overline{n}_i = 1. \tag{A.2}$$

Within the correlated-mode family, photons entering the same port i share a normalized internal state $|\chi_i\rangle$. Their internal overlaps define the Gram matrix

$$\Gamma_{ij} = \langle \chi_j | \chi_i \rangle , \qquad \Gamma \succeq 0, \qquad \Gamma_{ii} = 1. \tag{A.3}$$

Generalized multiphoton interference [5] and the discrimination of photon distinguishability [6] provide complementary settings in which internal-state overlaps are relevant. Here we work with the correlated-mode preparation specified above. The corresponding effective single-particle matrix is

$$\rho_{\overline{\boldsymbol{n}},\Gamma} = [\overline{\boldsymbol{n}}]^{1/2} \Gamma [\overline{\boldsymbol{n}}]^{1/2}, \qquad [\overline{\boldsymbol{n}}] = \mathrm{diag}(\overline{n}_1, \ldots, \overline{n}_M). \tag{A.4}$$

We write $\rho \equiv \rho_{\overline{\boldsymbol{n}},\Gamma}$ when no ambiguity arises.

Since $\Gamma_{ii} = 1$, the diagonal elements of ρ satisfy

$$\rho_{ii} = \overline{n}_i, \tag{A.5}$$

and, consequently,

$$\mathrm{Tr}\, \rho = \sum_{i=1}^{M} \overline{n}_i = 1. \tag{A.6}$$

The matrix ρ is positive semidefinite, has unit trace, and has a diagonal fixed by the input occupation. It is the spatial marginal of the auxiliary single-particle state used in the toroidal construction of [7], not the spatial one-particle reduced density matrix of the physical fixed-occupation input, which is $[\overline{\boldsymbol{n}}]$. Its spectrum also governs the concentration of irreducible-sector labels under the conditions discussed below [8].

A.2 State conditioned on the initial occupation

Consider the tensor-power operator $\rho^{\otimes N}$ on $(\mathbb{C}^M)^{\otimes N}$. In the product basis $|i_1, \ldots, i_N\rangle$, the type of a sequence records the number of occurrences of each mode index. The projector onto sequences of type $\boldsymbol{n}$ is

$$\Pi_{\boldsymbol{n}} = \sum_{\mathrm{type}(i_1,\ldots,i_N)=\boldsymbol{n}} |i_1, \ldots, i_N\rangle \langle i_1, \ldots, i_N| . \tag{A.7}$$

Projecting this auxiliary tensor-power operator onto the fixed initial occupation gives the unnormalized operator

$$\widetilde{R}_{\boldsymbol{n},\Gamma} = \Pi_{\boldsymbol{n}} \rho^{\otimes N} \Pi_{\boldsymbol{n}}, \tag{A.8}$$

and the normalized conditioned state is

$$R_{\boldsymbol{n},\Gamma} = \frac{\Pi_{\boldsymbol{n}} \rho^{\otimes N} \Pi_{\boldsymbol{n}}}{\mathrm{Tr}[\Pi_{\boldsymbol{n}} \rho^{\otimes N} \Pi_{\boldsymbol{n}}]}. \tag{A.9}$$

This construction reproduces the spatial density operator obtained by tracing over the unresolved internal degrees of freedom of the physical correlated-mode input state. Indeed, for two sequences of the same type $\boldsymbol{n}$, the matrix elements of the conditioned state are

$$\langle i_1, \ldots, i_N | \, R_{\boldsymbol{n},\Gamma} \, | j_1, \ldots, j_N \rangle = \frac{1}{C_{\boldsymbol{n}}} \prod_{r=1}^{N} \Gamma_{i_r j_r}, \tag{A.10}$$

with all other matrix elements vanishing. Here $C_{\boldsymbol{n}}$ is the number of sequences of type $\boldsymbol{n}$, defined below. Equation (A.10) follows from the symmetrized correlated-mode input state. Thus, conditioning the auxiliary state is an exact representation of the physical preparation within this family, not an additional physical postselection.

A.2.1 Multinomial normalization

The normalization follows from the projector property and cyclicity of the trace:

$$\mathrm{Tr}[\Pi_{\boldsymbol{n}} \rho^{\otimes N} \Pi_{\boldsymbol{n}}] = \mathrm{Tr}[\Pi_{\boldsymbol{n}} \rho^{\otimes N}]. \tag{A.11}$$

Every sequence of type $\boldsymbol{n}$ contains n_i occurrences of mode i. Because $\rho_{ii} = \overline{n}_i$, each contributes the same diagonal matrix element,

$$\langle i_1, \ldots, i_N | \, \rho^{\otimes N} \, | i_1, \ldots, i_N \rangle = \prod_{\substack{i=1 \\ n_i > 0}}^{M} (\overline{n}_i)^{n_i}. \tag{A.12}$$

There are $C_{\boldsymbol{n}}$ such sequences, with

$$C_{\boldsymbol{n}} = \frac{N!}{n_1! \cdots n_M!}, \tag{A.13}$$

and therefore

$$\mathcal{Z}_{\boldsymbol{n}} := \mathrm{Tr}[\Pi_{\boldsymbol{n}} \rho^{\otimes N}] = C_{\boldsymbol{n}} \prod_{\substack{i=1 \\ n_i > 0}}^{M} (\overline{n}_i)^{n_i}. \tag{A.14}$$

The normalized state becomes

$$R_{\boldsymbol{n},\Gamma} = \frac{\Pi_{\boldsymbol{n}} \rho^{\otimes N} \Pi_{\boldsymbol{n}}}{C_{\boldsymbol{n}} \prod_{\substack{i=1 \\ n_i > 0}}^{M} (\overline{n}_i)^{n_i}}. \tag{A.15}$$

A.3 Interferometric evolution and the Born rule

The Hong–Ou–Mandel effect illustrates the role of indistinguishability in two-photon interference [9]. For fully indistinguishable photons, transition amplitudes in linear-optical networks admit expressions in terms of matrix permanents [10]; their computational complexity is central to boson sampling [11]. The density-operator formulation below also allows for partial distinguishability. Let $U \in U(M)$ describe the lossless interferometer. Its action on the spatial N-particle state is $U^{\otimes N}$, giving the

output density operator $U^{\otimes N} R_{\boldsymbol{n},\Gamma} U^{\dagger\otimes N}$. For an output occupation $\boldsymbol{n}'$, the Born rule yields

$$P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma) = \mathrm{Tr}\left[\Pi_{\boldsymbol{n}'} U^{\otimes N} R_{\boldsymbol{n},\Gamma} U^{\dagger\otimes N}\right]. \tag{A.16}$$

Using (A.15),

$$P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma) = \frac{1}{C_{\boldsymbol{n}} \prod_{\substack{i=1 \\ n_i>0}}^{M} (\overline{n}_i)^{n_i}} \mathrm{Tr}\left[\Pi_{\boldsymbol{n}'} U^{\otimes N} \Pi_{\boldsymbol{n}} \rho^{\otimes N} \Pi_{\boldsymbol{n}} U^{\dagger\otimes N}\right]. \tag{A.17}$$

This expression is exact but still acts on the full tensor-product space. Schur–Weyl duality reduces it to irreducible blocks.

A.4 Schur–Weyl decomposition

Schur–Weyl duality gives [1, 2]

$$(\mathbb{C}^M)^{\otimes N} = \bigoplus_{\substack{\Lambda \vdash N \\ \ell(\Lambda)\le M}} \mathcal{H}_\Lambda \otimes \mathcal{M}_\Lambda, \tag{A.18}$$

Here $\mathcal{H}_\Lambda$ carries the irreducible representation of $U(M)$ labeled by Λ, and $\mathcal{M}_\Lambda$ carries the associated irreducible representation of the symmetric group S_N. We distinguish

$$d_{[\Lambda]} := \dim \mathcal{M}_\Lambda, \qquad m_\Lambda(\boldsymbol{n}) := \dim \mathcal{H}_{\Lambda,\boldsymbol{n}}. \tag{A.19}$$

The quantity $d_{[\Lambda]}$ is the Schur–Weyl multiplicity of the sector, whereas $m_\Lambda(\boldsymbol{n})$ is the multiplicity of the weight $\boldsymbol{n}$ within $\mathcal{H}_\Lambda$. These two multiplicities should not be identified.

For an $M \times M$ matrix A, let $D^\Lambda(A)$ denote its polynomial action on $\mathcal{H}_\Lambda$. In particular,

$$D^\Lambda(AB) = D^\Lambda(A) D^\Lambda(B) \tag{A.20}$$

holds also for nonunitary matrices, including ρ, Γ, and $[\overline{\boldsymbol{n}}]^{1/2}$.

The relevant operators decompose as

$$\rho^{\otimes N} = \bigoplus_\Lambda D^\Lambda(\rho) \otimes I_{[\Lambda]}, \tag{A.21}$$

$$U^{\otimes N} = \bigoplus_\Lambda D^\Lambda(U) \otimes I_{[\Lambda]}, \tag{A.22}$$

$$\Pi_{\boldsymbol{n}} = \bigoplus_\Lambda P_{\Lambda,\boldsymbol{n}} \otimes I_{[\Lambda]}, \tag{A.23}$$

$$\Pi_{\boldsymbol{n}'} = \bigoplus_\Lambda P_{\Lambda,\boldsymbol{n}'} \otimes I_{[\Lambda]}, \tag{A.24}$$

where $P_{\Lambda,\boldsymbol{n}}$ projects onto the weight space $\mathcal{H}_{\Lambda,\boldsymbol{n}}$.

Substituting the block decompositions into (A.17) and tracing over $\mathcal{M}_\Lambda$ gives the factor $\mathrm{Tr}\, I_{[\Lambda]} = d_{[\Lambda]}$ and yields

$$P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma) = \frac{1}{C_{\boldsymbol{n}} \prod_{\substack{i=1 \\ n_i>0}}^{M} (\overline{n}_i)^{n_i}} \sum_\Lambda d_{[\Lambda]}\, \mathrm{Tr}_{\mathcal{H}_\Lambda}\left[P_{\Lambda,\boldsymbol{n}'} D^\Lambda(U) P_{\Lambda,\boldsymbol{n}} D^\Lambda(\rho) P_{\Lambda,\boldsymbol{n}} D^\Lambda(U^\dagger)\right]. \tag{A.25}$$

A.5 Exact cancellation of the diagonal factors

Let

$$S = [\overline{\boldsymbol{n}}]^{1/2}, \qquad \rho = S\Gamma S. \tag{A.26}$$

By multiplicativity,

$$D^\Lambda(\rho) = D^\Lambda(S) D^\Lambda(\Gamma) D^\Lambda(S). \tag{A.27}$$

On a weight vector $|\Lambda, \boldsymbol{n}, a\rangle \in \mathcal{H}_{\Lambda,\boldsymbol{n}}$, a diagonal matrix $S = \mathrm{diag}(s_1, \ldots, s_M)$ acts as

$$D^\Lambda(S)\,|\Lambda, \boldsymbol{n}, a\rangle = \left(\prod_{i=1}^{M} s_i^{n_i} \right) |\Lambda, \boldsymbol{n}, a\rangle\,. \tag{A.28}$$

The scalar depends on the weight $\boldsymbol{n}$ but not on the multiplicity index a. For $s_i = \sqrt{\overline{n}_i}$, set

$$c_{\boldsymbol{n}} := \prod_{\substack{i=1 \\ n_i>0}}^{M} (\overline{n}_i)^{n_i/2}. \tag{A.29}$$

Thus,

$$P_{\Lambda,\boldsymbol{n}} D^\Lambda(S) = c_{\boldsymbol{n}} P_{\Lambda,\boldsymbol{n}}, \qquad D^\Lambda(S) P_{\Lambda,\boldsymbol{n}} = c_{\boldsymbol{n}} P_{\Lambda,\boldsymbol{n}}. \tag{A.30}$$

Equation (A.27) then gives

$$P_{\Lambda,\boldsymbol{n}} D^\Lambda(\rho) P_{\Lambda,\boldsymbol{n}} = \left(\prod_{\substack{i=1 \\ n_i>0}}^{M} (\overline{n}_i)^{n_i} \right) P_{\Lambda,\boldsymbol{n}} D^\Lambda(\Gamma) P_{\Lambda,\boldsymbol{n}}. \tag{A.31}$$

The scalar factor cancels the occupation-dependent denominator in (A.25), leaving

$$P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma) = \frac{1}{C_{\boldsymbol{n}}} \sum_{\Lambda} d_{[\Lambda]}\, \mathrm{Tr}_{\mathcal{H}_\Lambda} \left[P_{\Lambda,\boldsymbol{n}'} D^\Lambda(U) P_{\Lambda,\boldsymbol{n}} D^\Lambda(\Gamma) P_{\Lambda,\boldsymbol{n}} D^\Lambda(U^\dagger) \right]. \tag{A.32}$$

The reduction is exact. The diagonal factors of $[\overline{\boldsymbol{n}}]^{1/2}$ act as a scalar on each fixed weight space and cancel after normalization. Partial indistinguishability thus enters through the restriction of $D^\Lambda(\Gamma)$ to the initial weight space.

A.6 Probability of an irreducible sector

The statistical weight of sector Λ is the trace of its block in the conditioned input state. Before applying the interferometer, this weight is

$$P(\Lambda \mid \boldsymbol{n}, \Gamma) = \frac{d_{[\Lambda]}}{C_{\boldsymbol{n}} \displaystyle\prod_{\substack{i=1 \\ n_i>0}}^{M} (\overline{n}_i)^{n_i}}\, \mathrm{Tr}_{\mathcal{H}_\Lambda} \left[P_{\Lambda,\boldsymbol{n}} D^\Lambda(\rho) P_{\Lambda,\boldsymbol{n}} \right]. \tag{A.33}$$

Using the cancellation identity (A.31),

$$P(\Lambda \mid \boldsymbol{n}, \Gamma) = \frac{d_{[\Lambda]}}{C_{\boldsymbol{n}}}\, \mathrm{Tr}_{\mathcal{H}_\Lambda} \left[P_{\Lambda,\boldsymbol{n}} D^\Lambda(\Gamma) P_{\Lambda,\boldsymbol{n}} \right]. \tag{A.34}$$

The sector weights are normalized:

$$\sum_{\Lambda} P(\Lambda \mid \boldsymbol{n}, \Gamma) = 1. \tag{A.35}$$

The factor $d_{[\Lambda]}$ comes from the S_N multiplicity space $\mathcal{M}_\Lambda$. It differs from the weight multiplicity $m_\Lambda(\boldsymbol{n})$, which determines the dimension of the space supporting the conditioned sector density. The sector weights depend on the input occupation and Γ, but not on U or the output occupation $\boldsymbol{n}'$.

A.7 Concentration of sector labels after occupation conditioning

The concentration of sector labels for tensor-power states can also be transferred to the fixed-occupation preparation. Let B be a set of partitions of N and let Π_B project onto the corresponding Schur–Weyl sectors. Since Π_B commutes with the occupation projector $\Pi_{\boldsymbol{n}}$, positivity gives

$$\begin{aligned} P(\Lambda \in B \mid \boldsymbol{n}, \Gamma) &= \frac{\operatorname{Tr}[\Pi_B \Pi_{\boldsymbol{n}} \rho^{\otimes N} \Pi_{\boldsymbol{n}}]}{\mathcal{Z}_{\boldsymbol{n}}} \\ &\leq \frac{\operatorname{Tr}[\Pi_B \rho^{\otimes N}]}{\mathcal{Z}_{\boldsymbol{n}}}. \end{aligned} \tag{A.36}$$

The multinomial estimate for the type $\overline{\boldsymbol{n}} = \boldsymbol{n}/N$ implies $\mathcal{Z}_{\boldsymbol{n}} \geq (N+1)^{-M}$ [7]. Therefore,

$$P(\Lambda \in B \mid \boldsymbol{n}, \Gamma) \leq (N+1)^M \operatorname{Tr}[\Pi_B \rho^{\otimes N}]. \tag{A.37}$$

For fixed M, the exponential concentration bounds of Keyl and Werner for $\rho^{\otimes N}$ [8] remain exponential after conditioning, up to a polynomial factor. Along sequences for which $\operatorname{spec}\rho$ converges to $\boldsymbol{\lambda}$, this controls the concentration of Λ/N around $\boldsymbol{\lambda}$. It does not imply concentration of populations within a fixed weight space, whether in the physical density $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ or in the projected-state distribution $|c_j(V)|^2$.

A.8 Physical density on a weight-multiplicity space

For a fixed sector Λ and initial weight $\boldsymbol{n}$, the unnormalized block on $\mathcal{H}_{\Lambda,\boldsymbol{n}}$ is

$$X_{\Lambda,\boldsymbol{n}}(\rho) = P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(\rho) P_{\Lambda,\boldsymbol{n}}. \tag{A.38}$$

Whenever

$$\operatorname{Tr} X_{\Lambda,\boldsymbol{n}}(\rho) > 0, \tag{A.39}$$

or, equivalently, $P(\Lambda \mid \boldsymbol{n}, \Gamma) > 0$, its normalization defines

$$\sigma_{\Lambda,\boldsymbol{n}}(\rho) = \frac{P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(\rho) P_{\Lambda,\boldsymbol{n}}}{\operatorname{Tr}[P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(\rho) P_{\Lambda,\boldsymbol{n}}]}. \tag{A.40}$$

The scalar in (A.31) cancels between the numerator and denominator, so that

$$\sigma_{\Lambda,\boldsymbol{n}}(\rho) = \sigma_{\Lambda,\boldsymbol{n}}(\Gamma) = \frac{P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(\Gamma) P_{\Lambda,\boldsymbol{n}}}{\operatorname{Tr}[P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(\Gamma) P_{\Lambda,\boldsymbol{n}}]}. \tag{A.41}$$

Thus,

$$\sigma_{\Lambda,\boldsymbol{n}}(\Gamma) \succeq 0, \qquad \operatorname{Tr} \sigma_{\Lambda,\boldsymbol{n}}(\Gamma) = 1. \tag{A.42}$$

This defines the map

$$\Gamma \longmapsto \sigma_{\Lambda,\boldsymbol{n}}(\Gamma) \in \mathcal{D}(\mathcal{H}_{\Lambda,\boldsymbol{n}}), \tag{A.43}$$

defined for sectors with nonzero probability.

Equation (A.41) is the central sectorial reduction: conditional on Λ and $\boldsymbol{n}$, the spatial state relevant to detection is described by a normalized density operator on $\mathcal{H}_{\Lambda,\boldsymbol{n}}$. If Γ is positive definite, its rank equals $m_\Lambda(\boldsymbol{n})$; hence it is mixed whenever $m_\Lambda(\boldsymbol{n}) > 1$. A pure projected frame state must not, in general, be identified with this physical sector density.

Upon further fixing

$$\operatorname{spec}\rho = \boldsymbol{\lambda}, \qquad \operatorname{diag}\rho = \overline{\boldsymbol{n}}, \tag{A.44}$$

the admissible sector densities are obtained by varying Γ over Gram matrices compatible with these data. The spectrum and diagonal of ρ do not, in general, determine a unique $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$; by contrast, if all $\overline{n}_i > 0$, the full matrix ρ fixes Γ through $\Gamma = [\overline{\boldsymbol{n}}]^{-1/2} \rho [\overline{\boldsymbol{n}}]^{-1/2}$.

A.9 Sector decomposition of the probability

For $P(\Lambda \mid \boldsymbol{n}, \Gamma) > 0$, the probability of detecting $\boldsymbol{n}'$ conditional on the sector is

$$P(\boldsymbol{n}' \mid \boldsymbol{n}, \Lambda, U, \Gamma) = \mathrm{Tr}_{\mathcal{H}_\Lambda}\left[P_{\Lambda,\boldsymbol{n}'} D^\Lambda(U) \sigma_{\Lambda,\boldsymbol{n}}(\Gamma) D^\Lambda(U^\dagger)\right]. \tag{A.45}$$

Because the output-weight projectors resolve the identity on $\mathcal{H}_\Lambda$,

$$\sum_{\boldsymbol{n}'} P(\boldsymbol{n}' \mid \boldsymbol{n}, \Lambda, U, \Gamma) = 1. \tag{A.46}$$

Combining (A.34), (A.41), and (A.45), the total probability (A.32) takes the form

$$P(\boldsymbol{n}' \mid U, \boldsymbol{n}, \Gamma) = \sum_\Lambda P(\Lambda \mid \boldsymbol{n}, \Gamma) P(\boldsymbol{n}' \mid \boldsymbol{n}, \Lambda, U, \Gamma). \tag{A.47}$$

The sum runs over sectors with nonzero statistical weight; sectors of zero weight do not contribute. The first factor specifies the weight of each sector, while $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ and the Born rule determine its conditional detection distribution.

If

$$\{|\Lambda, \boldsymbol{n}, a\rangle\}_{a=1}^{m_\Lambda(\boldsymbol{n})} \tag{A.48}$$

is an orthonormal basis of $\mathcal{H}_{\Lambda,\boldsymbol{n}}$, the conditioned density can be written as

$$\sigma_{\Lambda,\boldsymbol{n}}(\Gamma) = \sum_{a,b=1}^{m_\Lambda(\boldsymbol{n})} \sigma_{ab} \, |\Lambda, \boldsymbol{n}, a\rangle \langle \Lambda, \boldsymbol{n}, b| \,. \tag{A.49}$$

In a specified weight basis, the diagonal coefficients σ_{aa} are populations and the off-diagonal coefficients σ_{ab}, $a \neq b$, are coherences. The interferometric action $D^\Lambda(U)$ mixes these weight-space directions before projection onto the detected occupation. Thus, (A.47) separates sector weights from conditional transition probabilities exactly, but does not determine their large-N asymptotics or their relation to the induced classical measure.

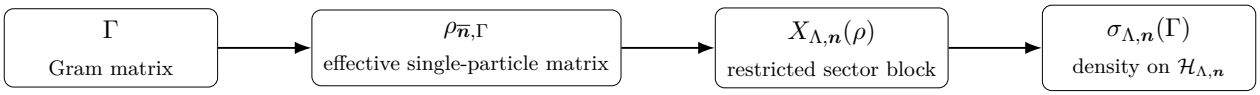


Figure 1: Reduction from the Gram matrix Γ to the physical density operator $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ on the weight space $\mathcal{H}_{\Lambda,\boldsymbol{n}}$, for fixed irreducible sector Λ and initial occupation $\boldsymbol{n}$.

B Spectrum–diagonal geometry and unistochastic region of $SU(3)$

This appendix characterizes unitary frames compatible with a prescribed normalized spectrum $\boldsymbol{\lambda}$ and diagonal $\overline{\boldsymbol{n}}$ in three modes. The condition $Q\boldsymbol{\lambda} = \overline{\boldsymbol{n}}$ defines a section of the bistochastic polytope, whose unistochastic subset contains the matrices $Q_{ij} = |V_{ij}|^2$. For matrices of order three, column orthogonality reduces to the closure of a triangle, tested by Heron's criterion. In the nondegenerate interior, this yields two conjugate unitary branches up to diagonal rephasings. The associated matrices $\rho(V)$ and $\Gamma(V)$ connect this construction to the compatible frames discussed in the main text. The bistochastic polytope was characterized by Birkhoff [12], while the geometry of unistochastic matrices in low dimensions is discussed in [13, 14, 15].

B.1 Spectrum–diagonal section of the bistochastic polytope

Let

$$Q = (q_{ij})_{i,j=1}^{3} \tag{B.1}$$

be a bistochastic matrix. Its three row and three column sums impose five independent normalization conditions on nine entries. Thus,

$$\dim \mathcal{B}_3 = 4. \tag{B.2}$$

We choose as independent coordinates

$$q_{11} = x, \qquad q_{12} = u, \qquad q_{21} = y, \qquad q_{22} = v. \tag{B.3}$$

The remaining entries are determined by bistochasticity,

$$Q(x,u,y,v) = \begin{pmatrix} x & u & 1-x-u \\ y & v & 1-y-v \\ 1-x-y & 1-u-v & x+u+y+v-1 \end{pmatrix}. \tag{B.4}$$

We now fix the normalized spectrum and diagonal,

$$\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \lambda_3)^T, \qquad \overline{\boldsymbol{n}} = (\overline{n}_1, \overline{n}_2, \overline{n}_3)^T, \tag{B.5}$$

with

$$\sum_{i=1}^{3} \lambda_i = \sum_{i=1}^{3} \overline{n}_i = 1. \tag{B.6}$$

We take $\boldsymbol{\lambda}$ to be a nonnegative spectrum, ordered in nonincreasing order. Spectrum–diagonal compatibility requires

$$Q\boldsymbol{\lambda} = \overline{\boldsymbol{n}}. \tag{B.7}$$

For $\lambda_2 \neq \lambda_3$, the first two components of (B.7) determine

$$u(x) = \frac{\overline{n}_1 - \lambda_3 - (\lambda_1 - \lambda_3)x}{\lambda_2 - \lambda_3}, \tag{B.8}$$

and

$$v(y) = \frac{\overline{n}_2 - \lambda_3 - (\lambda_1 - \lambda_3)y}{\lambda_2 - \lambda_3}. \tag{B.9}$$

The third equation is not independent, since the normalization of $\boldsymbol{\lambda}$ and $\overline{\boldsymbol{n}}$, together with the bistochasticity of Q, fixes it automatically.

These two independent constraints reduce the affine dimension from four to two, before imposing

nonnegativity. Substituting (B.8) and (B.9), we obtain

$$Q(x,y)=\begin{pmatrix} x & u(x) & 1-x-u(x) \\ y & v(y) & 1-y-v(y) \\ 1-x-y & 1-u(x)-v(y) & x+y+u(x)+v(y)-1 \end{pmatrix}. \tag{B.10}$$

The denominators in (B.8)–(B.9) vanish when $\lambda_2=\lambda_3$. If the spectrum is not fully degenerate, the affine section can instead be parametrized using a different pair of free entries of Q. For the fully degenerate spectrum $\boldsymbol{\lambda}=(1,1,1)/3$, every bistochastic matrix satisfies $Q\boldsymbol{\lambda}=(1,1,1)/3$. In that case, compatible frames exist only for $\overline{\boldsymbol{n}}=(1,1,1)/3$, and the constraint does not reduce the bistochastic polytope.

B.2 Bistochastic region and existence condition

The bistochastic region of the section is obtained by imposing the nonnegativity of all entries of (B.10),

$$\begin{aligned} x\geq 0, \quad & u(x)\geq 0, \quad & 1-x-u(x)\geq 0, \\ y\geq 0, \quad & v(y)\geq 0, \quad & 1-y-v(y)\geq 0, \\ 1-x-y\geq 0, \quad & 1-u(x)-v(y)\geq 0, \quad & x+y+u(x)+v(y)-1\geq 0. \end{aligned} \tag{B.11}$$

Since all these inequalities are affine in (x,y), the intersection

$$\mathcal{B}_3\cap\{Q: Q\boldsymbol{\lambda}=\overline{\boldsymbol{n}}\} \tag{B.12}$$

is a convex polygon in the (x,y) plane, possibly of lower dimension for special spectrum–diagonal data. The existence of the section can be formulated without resorting to the previous parametrization. The Schur–Horn theorem [16, 17] states that

$$\overline{\boldsymbol{n}}\prec\boldsymbol{\lambda} \tag{B.13}$$

is equivalent to the existence of a Hermitian matrix with spectrum $\boldsymbol{\lambda}$ and diagonal $\overline{\boldsymbol{n}}$ and, therefore, of a unitary matrix V such that

$$\operatorname{diag}\left(V[\boldsymbol{\lambda}]V^{\dagger}\right)=\overline{\boldsymbol{n}}. \tag{B.14}$$

Defining

$$Q_{ij}(V)=|V_{ij}|^2, \tag{B.15}$$

this condition can be written precisely as

$$Q(V)\boldsymbol{\lambda}=\overline{\boldsymbol{n}}. \tag{B.16}$$

Schur–Horn guarantees the existence of a compatible unitary frame, but not that every bistochastic matrix in this section is unistochastic. The latter condition is addressed below.

B.3 Unistochastic region and Heron's criterion

Let Q be a point in the bistochastic section. To reconstruct a unitary matrix V with

$$Q_{ij}=|V_{ij}|^2, \tag{B.17}$$

we fix the row and column phases so that the first column is real and nonnegative,

$$V_{i1}=\sqrt{Q_{i1}}, \tag{B.18}$$

and write the second column as

$$V_{i2}=\sqrt{Q_{i2}}\,\mathrm{e}^{\mathrm{i}\phi_i}. \tag{B.19}$$

Orthogonality between the two columns requires

$$\sum_{i=1}^{3} \sqrt{Q_{i1}Q_{i2}}\, \mathrm{e}^{\mathrm{i}\phi_i} = 0. \tag{B.20}$$

We define

$$L_i = \sqrt{Q_{i1}Q_{i2}}, \qquad i = 1, 2, 3. \tag{B.21}$$

Equation (B.20) admits a solution if and only if three complex vectors of moduli L_1, L_2, L_3 can close a triangle. Therefore,

$$L_1 \leq L_2 + L_3, \qquad L_2 \leq L_1 + L_3, \qquad L_3 \leq L_1 + L_2. \tag{B.22}$$

For bistochastic matrices of order three, these inequalities constitute a necessary and sufficient condition for unistochasticity [13, 14].

The same criterion can be written in terms of a single algebraic quantity. If $\mathcal{A}$ is the area of the triangle with sides L_1, L_2, L_3, Heron's formula gives

$$16\mathcal{A}^2 = 2\left(L_1^2L_2^2 + L_2^2L_3^2 + L_3^2L_1^2\right) - \left(L_1^4 + L_2^4 + L_3^4\right). \tag{B.23}$$

Defining

$$\xi_i = L_i^2 = Q_{i1}Q_{i2}, \tag{B.24}$$

we introduce

$$H_{\mathrm{Heron}}(Q) := 2\left(\xi_1\xi_2 + \xi_2\xi_3 + \xi_3\xi_1\right) - \left(\xi_1^2 + \xi_2^2 + \xi_3^2\right) = 16\mathcal{A}^2. \tag{B.25}$$

Within the bistochastic section,

$$\begin{cases} H_{\mathrm{Heron}}(Q) > 0, & Q \text{ lies in the interior of the unistochastic region,} \\ H_{\mathrm{Heron}}(Q) = 0, & Q \text{ lies on its boundary,} \\ H_{\mathrm{Heron}}(Q) < 0, & Q \text{ is not unistochastic.} \end{cases} \tag{B.26}$$

For $H_{\mathrm{Heron}}(Q) > 0$, the triangle is nondegenerate and the closure equation has two conjugate solutions after fixing the row and column phases. When $H_{\mathrm{Heron}}(Q) = 0$, the triangle is degenerate and the branches coalesce up to these phase freedoms.

B.4 Geometry for the three occupations

To compare the dependence of the unistochastic region on the initial occupation, we fix in all three cases

$$\boldsymbol{\lambda} = \overline{\Lambda} = \frac{1}{15}(11, 3, 1), \qquad N = 45, \qquad \Lambda = (33, 9, 3), \tag{B.27}$$

and consider

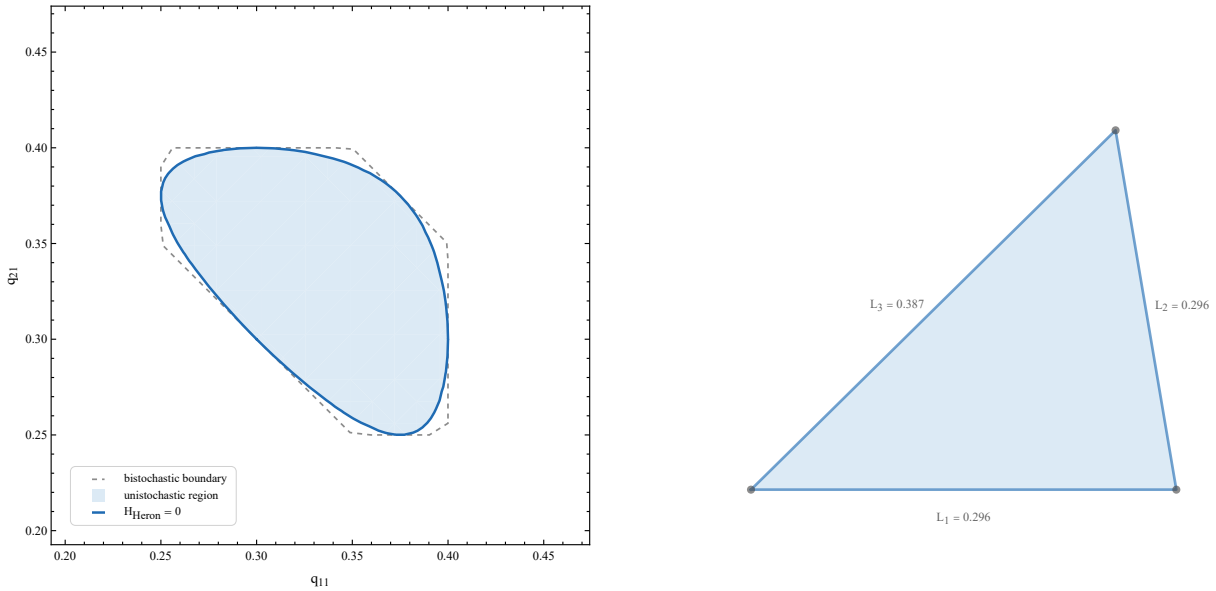
$$\overline{\boldsymbol{n}}^{(1)} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right), \qquad \overline{\boldsymbol{n}}^{(2)} = \frac{1}{5}(3, 1, 1), \qquad \overline{\boldsymbol{n}}^{(3)} = \frac{1}{9}(3, 2, 4), \tag{B.28}$$

corresponding to the occupations

$$\boldsymbol{n}^{(1)} = (15, 15, 15), \qquad \boldsymbol{n}^{(2)} = (27, 9, 9), \qquad \boldsymbol{n}^{(3)} = (15, 10, 20). \tag{B.29}$$

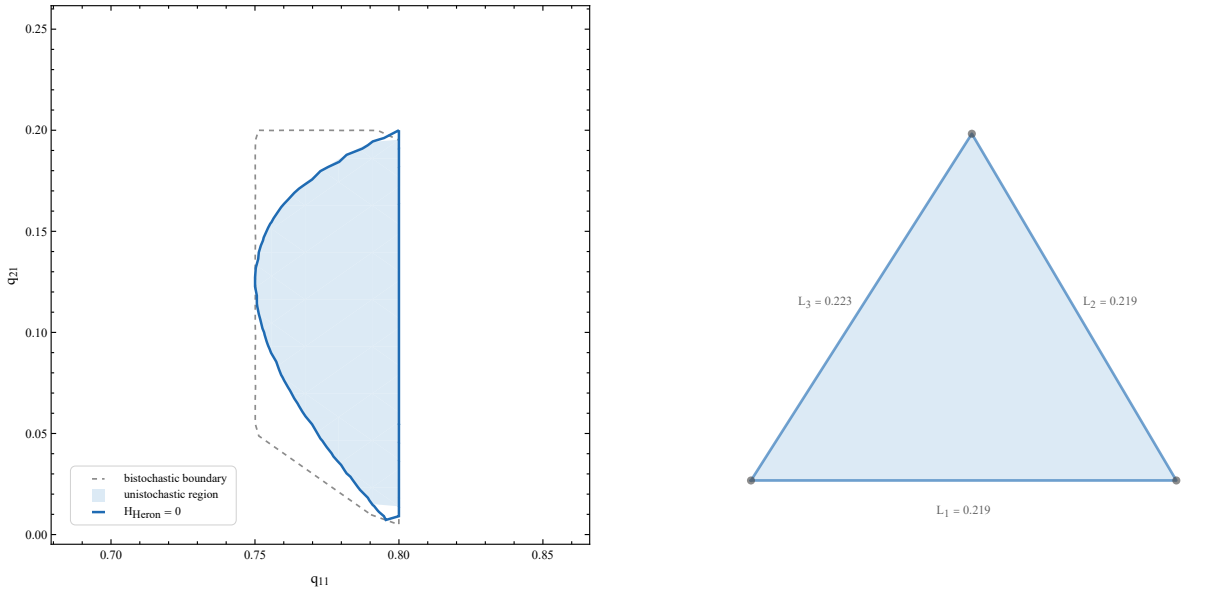
We use the labels balanced, partially unbalanced, and asymmetric for these three configurations, without imposing a quantitative ordering of asymmetry. For each occupation, we select the compatible frame labeled V_1 in the numerical data. The frame banks are obtained by sampling the bistochastic section, retaining points satisfying $H_{\mathrm{Heron}}(Q) \geq 0$, and reconstructing the conjugate branches when distinct. The numerical data contain up to twelve frames per configuration; the sampling does not exhaust the compatible frames.

Figs. 2–4 show the bistochastic section and the region selected by $H_{\text{Heron}} \geq 0$, together with the triangle associated with the selected frame.



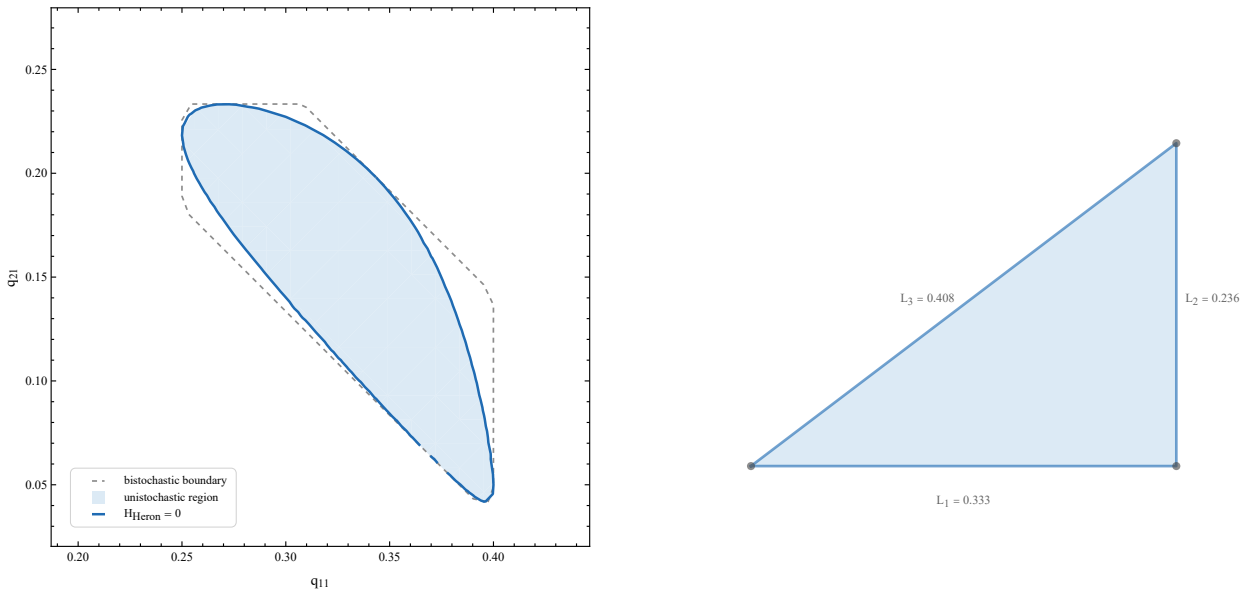
(a) (b)

Figure 2: Bistochastic and unistochastic regions for the balanced configuration $\overline{\boldsymbol{n}} = (1/3, 1/3, 1/3)$, with $\boldsymbol{\lambda} = (0.7333, 0.2000, 0.0667)$, $N = 45$, and $\Lambda = (33, 9, 3)$. (a) Affine section $Q\boldsymbol{\lambda} = \overline{\boldsymbol{n}}$ in the (q_{11}, q_{21}) plane. The dashed curve is the bistochastic boundary and the light-blue region is the unistochastic subset, $H_{\text{Heron}} \geq 0$. (b) Triangle associated with the selected frame V_1. The condition $H_{\text{Heron}} > 0$ places the frame in the interior of the unistochastic region.



(a) (b)

Figure 3: Bistochastic and unistochastic regions for the partially unbalanced configuration $\overline{\boldsymbol{n}} = (0.6000, 0.2000, 0.2000)$, with $\boldsymbol{\lambda} = (0.7333, 0.2000, 0.0667)$, $N = 45$, and $\Lambda = (33, 9, 3)$. (a) Affine section $Q\boldsymbol{\lambda} = \overline{\boldsymbol{n}}$ in the (q_{11}, q_{21}) plane. The dashed curve is the bistochastic boundary and the light-blue region is the unistochastic subset, $H_{\text{Heron}} \geq 0$. (b) Triangle associated with the selected frame V_1. The condition $H_{\text{Heron}} > 0$ places the frame in the interior of the unistochastic region.



(a) (b)

Figure 4: Bistochastic and unistochastic regions for the asymmetric configuration $\overline{\boldsymbol{n}} = (0.3333, 0.2222, 0.4444)$, with $\boldsymbol{\lambda} = (0.7333, 0.2000, 0.0667)$, $N = 45$, and $\Lambda = (33, 9, 3)$. (a) Affine section $Q\boldsymbol{\lambda} = \overline{\boldsymbol{n}}$ in the (q_{11}, q_{21}) plane. The dashed curve is the bistochastic boundary and the light-blue region is the unistochastic subset, $H_{\text{Heron}} \geq 0$. (b) Triangle associated with the selected frame V_1. The condition $H_{\text{Heron}} > 0$ places the frame in the interior of the unistochastic region.

The three figures illustrate the dependence of the compatible unistochastic region on the diagonal $\overline{\boldsymbol{n}}$ at fixed spectrum $\boldsymbol{\lambda}$. The condition $Q\boldsymbol{\lambda} = \overline{\boldsymbol{n}}$ determines in each case a different affine section of the bistochastic polytope and, within it, $H_{\text{Heron}}(Q) \geq 0$ selects the points that admit a unitary reconstruction.

For the balanced occupation, Fig. 2(a), the unistochastic region occupies a broad and approximately central part of the section. Moving to the partially unbalanced configuration, Fig. 3(a), the section shifts toward larger values of q_{11} and the admissible region acquires a narrower shape. The asymmetric configuration, Fig. 4(a), produces a region with a different extent and orientation. These differences do not arise from a change in the spectrum, which is common to the three cases, but from the different diagonal constraint imposed by $\overline{\boldsymbol{n}}$.

The frames selected in Figs. 2(b), 3(b) and 4(b) satisfy $H_{\text{Heron}} > 0$. The corresponding triangles are therefore nondegenerate, and the three points belong to the interior of their respective unistochastic regions. The boundary $H_{\text{Heron}} = 0$ marks the limit at which the triangle becomes degenerate and the two phase solutions coincide.

B.5 Reconstruction of the unitary branches

Consider a point with $H_{\text{Heron}}(Q) > 0$, so that the associated triangle is nondegenerate. Using row phases to make the first column real and nonnegative, we write

$$v_1 = \begin{pmatrix} \sqrt{Q_{11}} \\ \sqrt{Q_{21}} \\ \sqrt{Q_{31}} \end{pmatrix}, \tag{B.30}$$

and use the global phase of the second column to set $\phi_1 = 0$. Orthogonality between v_1 and v_2 then requires

$$L_1 + L_2 e^{i\phi_2} + L_3 e^{i\phi_3} = 0. \tag{B.31}$$

Since $L_1, L_2, L_3 > 0$ in this case,

$$\cos\phi_2 = \frac{L_3^2 - L_1^2 - L_2^2}{2L_1L_2}, \tag{B.32}$$

from which the two solutions are obtained,

$$\phi_2^{(\pm)} = \pm \arccos\left[\frac{L_3^2 - L_1^2 - L_2^2}{2L_1L_2}\right]. \tag{B.33}$$

For each of them,

$$\phi_3^{(\pm)} = \arg\left[-\left(L_1 + L_2 e^{i\phi_2^{(\pm)}}\right)\right]. \tag{B.34}$$

The two solutions are related by complex conjugation under the adopted phase convention. In the degenerate cases in which one of the L_i vanishes, the phases are obtained directly from (B.31).

The second column is then given by

$$v_2 = \begin{pmatrix} \sqrt{Q_{12}} \\ \sqrt{Q_{22}} e^{i\phi_2} \\ \sqrt{Q_{32}} e^{i\phi_3} \end{pmatrix}, \tag{B.35}$$

with

$$v_1^\dagger v_1 = v_2^\dagger v_2 = 1, \qquad v_1^\dagger v_2 = 0. \tag{B.36}$$

Finally, we choose a unit vector v_3, unique up to a phase, spanning the orthogonal complement of $\mathrm{span}\{v_1, v_2\}$ and form

$$V = (v_1, v_2, v_3) \in U(3). \tag{B.37}$$

Bistochasticity guarantees that

$$|V_{i3}|^2 = 1 - |V_{i1}|^2 - |V_{i2}|^2 = Q_{i3}, \tag{B.38}$$

and therefore

$$|V|^{\circ 2} = Q. \tag{B.39}$$

If a representative in $SU(3)$ is required, we remove the determinant through the global phase

$$V_{SU(3)} = e^{-i\arg(\det V)/3} V. \tag{B.40}$$

This transformation does not modify $Q(V)$ or the matrices $\rho(V)$ and $\Gamma(V)$.

After fixing the row and column phases, a point with $H_{\mathrm{Heron}}(Q) > 0$ has two representatives related by complex conjugation,

$$V_+, \qquad V_-, \tag{B.41}$$

which satisfy

$$|V_+|^{\circ 2} = |V_-|^{\circ 2} = Q. \tag{B.42}$$

General frames with the same Q also include left and right diagonal rephasings of these representatives. Right rephasings leave $V[\boldsymbol{\lambda}]V^\dagger$ unchanged, whereas left rephasings act on this matrix by diagonal conjugation. Thus, Q fixes the moduli but not all the phase information. On the boundary $H_{\mathrm{Heron}}(Q) = 0$, the conjugate branches coalesce up to these phase freedoms.

B.6 Reconstruction of ρ and the Gram matrix

Given an admissible unitary frame, we define

$$\rho(V) = V[\boldsymbol{\lambda}]V^\dagger. \tag{B.43}$$

Its diagonal satisfies

$$\begin{aligned}\rho_{ii}(V) &= \sum_{j=1}^{3} V_{ij}\lambda_j V_{ij}^* \\ &= \sum_{j=1}^{3} |V_{ij}|^2 \lambda_j = (Q\boldsymbol{\lambda})_i = \overline{n}_i,\end{aligned} \tag{B.44}$$

and therefore

$$\operatorname{diag}\rho(V) = \overline{\boldsymbol{n}}. \tag{B.45}$$

If all components of $\overline{\boldsymbol{n}}$ are strictly positive, the corresponding Gram matrix is

$$\Gamma(V) = [\overline{\boldsymbol{n}}]^{-1/2}\rho(V)[\overline{\boldsymbol{n}}]^{-1/2}. \tag{B.46}$$

Since $\rho(V) \succeq 0$,

$$\Gamma(V) \succeq 0, \tag{B.47}$$

and, using (B.45),

$$\Gamma_{ii}(V) = \frac{\rho_{ii}(V)}{\overline{n}_i} = 1. \tag{B.48}$$

Thus, $\Gamma(V)$ is a physically admissible Gram matrix, since it is positive semidefinite with unit diagonal. For a prescribed physical input Γ, equality $\rho(V) = \rho_{\overline{\boldsymbol{n}},\Gamma}$ requires $\boldsymbol{\lambda} = \operatorname{spec}\rho_{\overline{\boldsymbol{n}},\Gamma}$ and a frame V diagonalizing the physical effective matrix.

When one of the components $\overline{n}_i$ vanishes, $[\overline{\boldsymbol{n}}]^{-1/2}$ is not defined on the full space. The construction then applies to the occupied modes; Gram-matrix entries involving unoccupied modes are not fixed by $\rho(V)$.

The matrix Q does not generally determine the off-diagonal entries of $\rho(V)$ or $\Gamma(V)$: inequivalent conjugate branches may give different Gram matrices at fixed spectrum and diagonal. However, left diagonal rephasings conjugate $\Gamma(V)$ by a diagonal unitary and merely shift the relative phases of the associated classical map, leaving its image unchanged. Therefore, different matrices $\Gamma(V)$ do not always represent physically distinct preparations. This distinction matters when comparing projected pure states with physical sector densities in the following appendices.

C Construction through exterior powers and projection onto weights

In this appendix we realize the polynomial irrep of $U(M)$ associated with a partition Λ within a tensor product of exterior powers of the fundamental representation [1, 2]. The construction provides a highest-weight vector, its orbit under unitary frames, and a principal-minor formula for the corresponding highest-weight matrix element. We then project the orbit onto a weight $\boldsymbol{n}$ and express the weight projector as a Fourier integral over the maximal torus. The squared norm of the projection is a Fourier coefficient of the resulting generating function. For projected pure states, this exact representation provides the starting point for stationary analysis and the construction of the sectorial classical map.

C.1 Exterior construction and highest-weight vector

Let

$$\Lambda = (\Lambda_1, \dots, \Lambda_M), \qquad \Lambda_1 \geq \Lambda_2 \geq \cdots \geq \Lambda_M \geq 0, \qquad |\Lambda| = N. \tag{C.1}$$

The number of columns of height r in the Young diagram of Λ is

$$p_r = \Lambda_r - \Lambda_{r+1}, \qquad \Lambda_{M+1} = 0. \tag{C.2}$$

These multiplicities satisfy

$$\sum_{r=1}^{M} r p_r = |\Lambda| = N, \tag{C.3}$$

in agreement with the total number of boxes in the diagram.

Each column of height r is associated with the exterior representation $\wedge^r \mathbb{C}^M$. The tensor-product space

$$\mathcal{E}_\Lambda = \bigotimes_{r=1}^{M} \left(\wedge^r \mathbb{C}^M \right)^{\otimes p_r} \tag{C.4}$$

carries the action

$$\mathcal{R}_\Lambda(X) = \bigotimes_{r=1}^{M} \left(X^{\wedge r} \right)^{\otimes p_r}. \tag{C.5}$$

The space $\mathcal{E}_\Lambda$ is generally reducible. We identify the subrepresentation of highest weight Λ using the following vector.

Let $\{e_1, \dots, e_M\}$ be the canonical basis of $\mathbb{C}^M$ and define, in $\wedge^r \mathbb{C}^M$, the normalized exterior vector

$$|\Omega_r\rangle = |e_1 \wedge e_2 \wedge \cdots \wedge e_r\rangle. \tag{C.6}$$

The tensor-product vector

$$|\Omega_\Lambda\rangle = \bigotimes_{r=1}^{M} |\Omega_r\rangle^{\otimes p_r} \in \mathcal{E}_\Lambda \tag{C.7}$$

is a highest-weight vector: each exterior factor is annihilated by the positive-root raising operators, and the same holds for their tensor product. Its weight follows directly from the action of a diagonal matrix $D = \mathrm{diag}(z_1, \dots, z_M)$,

$$\begin{aligned} \mathcal{R}_\Lambda(D)\,|\Omega_\Lambda\rangle &= \prod_{r=1}^{M} (z_1 \cdots z_r)^{p_r}\,|\Omega_\Lambda\rangle \\ &= \prod_{a=1}^{M} z_a^{\sum_{r=a}^{M} p_r}\,|\Omega_\Lambda\rangle \\ &= \left(\prod_{a=1}^{M} z_a^{\Lambda_a} \right) |\Omega_\Lambda\rangle, \end{aligned} \tag{C.8}$$

where we have used $\sum_{r=a}^{M} p_r = \Lambda_a$. Thus, $|\Omega_\Lambda\rangle$ has weight Λ.

The irreducible representation of highest weight Λ can therefore be realized as the invariant subspace generated by the unitary orbit of $|\Omega_\Lambda\rangle$,

$$\mathcal{H}_\Lambda := \operatorname{span}\{\mathcal{R}_\Lambda(U)\,|\Omega_\Lambda\rangle : U \in U(M)\} \subseteq \mathcal{E}_\Lambda. \tag{C.9}$$

We denote by

$$D^\Lambda(U) = \mathcal{R}_\Lambda(U)|_{\mathcal{H}_\Lambda} \tag{C.10}$$

the corresponding action on $\mathcal{H}_\Lambda$.

By polynomiality, the action extends to any $X \in \mathrm{Mat}_M(\mathbb{C})$, including nonunitary and singular matrices, and its restriction to $\mathcal{H}_\Lambda$ defines $D^\Lambda(X)$. This extension is used for general matrix arguments in the following appendices.

The factor of height M is one-dimensional,

$$X^{\wedge M} = \det X. \tag{C.11}$$

For $U \in SU(M)$, this factor equals one; hence columns of height M do not affect the restriction to $SU(M)$, although they remain relevant to the $U(M)$ weight and the total degree N.

C.2 Highest-weight orbit and principal minors

Let

$$V = (v_1, \ldots, v_M) \in U(M), \tag{C.12}$$

where $v_a = Ve_a$ is the a-th column of the frame. The exterior action satisfies

$$V^{\wedge r}\,|e_1 \wedge \cdots \wedge e_r\rangle = |v_1 \wedge \cdots \wedge v_r\rangle\,, \tag{C.13}$$

and therefore

$$|\Omega_\Lambda(V)\rangle := D^\Lambda(V)\,|\Omega_\Lambda\rangle = \bigotimes_{r=1}^{M} |v_1 \wedge \cdots \wedge v_r\rangle^{\otimes p_r}\,. \tag{C.14}$$

Since $D^\Lambda(V)$ is unitary, $\|\Omega_\Lambda(V)\| = 1$. These vectors form the unitary orbit of the highest-weight state [18].

The exterior-power construction gives a determinant formula for the diagonal highest-weight matrix element. For decomposable exterior vectors,

$$\langle u_1 \wedge \cdots \wedge u_r | w_1 \wedge \cdots \wedge w_r\rangle = \det\left[\langle u_a | w_b\rangle\right]_{a,b=1}^{r}. \tag{C.15}$$

In particular,

$$\langle\Omega_r|\,X^{\wedge r}\,|\Omega_r\rangle = \det X_{[r],[r]}, \tag{C.16}$$

where $[r] = \{1, \ldots, r\}$ and $X_{[r],[r]}$ denotes the upper-left $r \times r$ block of X.

Taking the product over the exterior factors gives

$$\langle\Omega_\Lambda|\,D^\Lambda(X)\,|\Omega_\Lambda\rangle = \prod_{r=1}^{M} \left[\det X_{[r],[r]}\right]^{p_r}. \tag{C.17}$$

This identity will provide the generating function for the projected highest-weight orbit.

As a particular case, for two frames $V, W \in U(M)$,

$$\langle\Omega_\Lambda(V)|\Omega_\Lambda(W)\rangle = \prod_{r=1}^{M} \left[\det(V^\dagger W)_{[r],[r]}\right]^{p_r}. \tag{C.18}$$

This overlap will enter the construction of the kernels between projected states in Appendix F.

C.3 Exact projection onto a weight

The irreducible representation decomposes into weight spaces,

$$\mathcal{H}_\Lambda = \bigoplus_{\mathbf{m}} \mathcal{H}_{\Lambda,\mathbf{m}}. \tag{C.19}$$

We denote by

$$P_{\Lambda,\boldsymbol{n}} : \mathcal{H}_\Lambda \longrightarrow \mathcal{H}_{\Lambda,\boldsymbol{n}} \tag{C.20}$$

the orthogonal projector onto the weight $\boldsymbol{n}$. When $\boldsymbol{n}$ does not belong to the weight set of the representation, $P_{\Lambda,\boldsymbol{n}} = 0$.

For a weight $\boldsymbol{n}$ occurring in the representation, the unnormalized projection of the orbit state is

$$|\widetilde{\psi}_{\boldsymbol{n}}(V)\rangle = P_{\Lambda,\boldsymbol{n}} \, |\Omega_\Lambda(V)\rangle = P_{\Lambda,\boldsymbol{n}} D^\Lambda(V) \, |\Omega_\Lambda\rangle \, . \tag{C.21}$$

Its squared norm is

$$\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) := \| P_{\Lambda,\boldsymbol{n}} \, |\Omega_\Lambda(V)\rangle \|^2 = \langle \Omega_\Lambda(V)| \, P_{\Lambda,\boldsymbol{n}} \, |\Omega_\Lambda(V)\rangle \, . \tag{C.22}$$

Whenever $\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) > 0$, the normalized state is defined by

$$|\boldsymbol{n}, V; \Lambda\rangle = \frac{P_{\Lambda,\boldsymbol{n}} D^\Lambda(V) \, |\Omega_\Lambda\rangle}{\sqrt{\mathcal{N}_{\Lambda,\boldsymbol{n}}(V)}} \in \mathcal{H}_{\Lambda,\boldsymbol{n}}. \tag{C.23}$$

The map

$$V \longmapsto |\boldsymbol{n}, V; \Lambda\rangle \tag{C.24}$$

defines a pure state in $\mathcal{H}_{\Lambda,\boldsymbol{n}}$, up to an overall phase. This family need not contain every direction of the weight space. Appendix F examines the span of selected projected states using their kernels and Gram matrices.

The projected pure state is generally distinct from the physical sectorial density $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ of Appendix A, obtained by restricting and normalizing $D^\Lambda(\Gamma)$ on the weight space. Even when projected states span this space, such spanning does not imply a representation of the physical density as a positive mixture of them.

The construction applies to any $V \in U(M)$ with a nonzero projection. For fixed normalized shape $\overline{\Lambda} = \Lambda/N$ and occupation $\overline{\boldsymbol{n}}$, compatible frames satisfy $Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}}$ (Appendix B); this additional condition does not itself identify the frame with a prescribed physical Gram matrix Γ.

C.4 Toroidal representation of the projector

The projector onto a weight admits an exact Fourier representation over the maximal torus [7]. Define

$$t_{\boldsymbol{\phi}} = \operatorname{diag}\left(\mathrm{e}^{\mathrm{i}\phi_1}, \ldots, \mathrm{e}^{\mathrm{i}\phi_M}\right), \qquad \boldsymbol{\phi} = (\phi_1, \ldots, \phi_M). \tag{C.25}$$

Then

$$P_{\Lambda,\boldsymbol{n}} = \int_{[0,2\pi]^M} \frac{d^M\phi}{(2\pi)^M} \, \mathrm{e}^{-\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi}} D^\Lambda(t_{\boldsymbol{\phi}}). \tag{C.26}$$

To verify this identity, consider a weight vector $|\psi_{\mathbf{m}}\rangle \in \mathcal{H}_{\Lambda,\mathbf{m}}$, for which

$$\begin{aligned}\int_{[0,2\pi]^M} \frac{d^M\phi}{(2\pi)^M}\, \mathrm{e}^{-\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi}} D^\Lambda(t_{\boldsymbol{\phi}})\,|\psi_{\mathbf{m}}\rangle \\ = \left[\prod_{j=1}^M \int_0^{2\pi} \frac{d\phi_j}{2\pi}\, \mathrm{e}^{\mathrm{i}(m_j-n_j)\phi_j}\right] |\psi_{\mathbf{m}}\rangle \\ = \delta_{\mathbf{m},\boldsymbol{n}}\,|\psi_{\mathbf{m}}\rangle\,, \end{aligned} \tag{C.27}$$

which is precisely the action of $P_{\Lambda,\boldsymbol{n}}$.

Since D^Λ is homogeneous of degree $N = |\Lambda|$, every weight $\mathbf{m}$ satisfies

$$\sum_{j=1}^M m_j = N. \tag{C.28}$$

The target weight also satisfies $\sum_j n_j = N$. Thus, a common phase contributes equally to both factors in the projector integral. Taking the Mth mode as the reference phase, write

$$\phi_j = \varphi_j + \alpha, \qquad j = 1,\dots,M-1, \qquad \phi_M = \alpha. \tag{C.29}$$

Then

$$t_{\boldsymbol{\phi}} = \mathrm{e}^{\mathrm{i}\alpha}\widehat{t}_{\boldsymbol{\varphi}}, \qquad \widehat{t}_{\boldsymbol{\varphi}} = \mathrm{diag}\left(\mathrm{e}^{\mathrm{i}\varphi_1},\dots,\mathrm{e}^{\mathrm{i}\varphi_{M-1}},1\right). \tag{C.30}$$

The homogeneity of D^Λ implies

$$D^\Lambda(\mathrm{e}^{\mathrm{i}\alpha}X) = \mathrm{e}^{\mathrm{i}N\alpha} D^\Lambda(X), \tag{C.31}$$

while

$$\mathrm{e}^{-\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi}} = \mathrm{e}^{-\mathrm{i}N\alpha} \exp\left[-\mathrm{i}\sum_{j=1}^{M-1} n_j\varphi_j\right]. \tag{C.32}$$

The global-phase factors cancel exactly, and the projector reduces to an integral over the torus of relative phases,

$$P_{\Lambda,\boldsymbol{n}} = \int_{[0,2\pi]^{M-1}} \frac{d^{M-1}\varphi}{(2\pi)^{M-1}}\, \exp\left[-\mathrm{i}\sum_{j=1}^{M-1} n_j\varphi_j\right] D^\Lambda(\widehat{t}_{\boldsymbol{\varphi}}). \tag{C.33}$$

Only $M-1$ independent relative phases are therefore required; choosing the first mode as reference gives an equivalent representation.

C.5 Projected norm and generating function

Substituting (C.26) into (C.22), the squared norm of the projected vector is

$$\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = \int_{[0,2\pi]^M} \frac{d^M\phi}{(2\pi)^M}\, \mathrm{e}^{-\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi}}\, \langle\Omega_\Lambda(V)|\, D^\Lambda(t_{\boldsymbol{\phi}})\,|\Omega_\Lambda(V)\rangle\,. \tag{C.34}$$

Using $|\Omega_\Lambda(V)\rangle = D^\Lambda(V)\,|\Omega_\Lambda\rangle$ and the unitarity of V, we obtain

$$\langle\Omega_\Lambda(V)|\, D^\Lambda(t_{\boldsymbol{\phi}})\,|\Omega_\Lambda(V)\rangle = \langle\Omega_\Lambda|\, D^\Lambda(V^\dagger t_{\boldsymbol{\phi}} V)\,|\Omega_\Lambda\rangle\,. \tag{C.35}$$

Applying (C.17),

$$\langle\Omega_\Lambda(V)|\, D^\Lambda(t_{\boldsymbol{\phi}})\,|\Omega_\Lambda(V)\rangle = \prod_{r=1}^M \left[\det\left(V^\dagger t_{\boldsymbol{\phi}} V\right)_{[r],[r]}\right]^{p_r}. \tag{C.36}$$

It follows that

$$\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = \int_{[0,2\pi]^M} \frac{d^M\phi}{(2\pi)^M}\, \mathrm{e}^{-\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi}} \prod_{r=1}^{M} \left[\det\left(V^\dagger t_{\boldsymbol{\phi}} V\right)_{[r],[r]}\right]^{p_r}. \tag{C.37}$$

Upon eliminating the global phase through (C.33), this becomes

$$\begin{aligned}\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = & \int_{[0,2\pi]^{M-1}} \frac{d^{M-1}\varphi}{(2\pi)^{M-1}}\, \exp\left[-\mathrm{i}\sum_{j=1}^{M-1} n_j\varphi_j\right] \\ & \times \prod_{r=1}^{M} \left[\det\left(V^\dagger \widehat{t}_{\boldsymbol{\varphi}} V\right)_{[r],[r]}\right]^{p_r}.\end{aligned} \tag{C.38}$$

Thus, $\mathcal{N}_{\Lambda,\boldsymbol{n}}(V)$ is the Fourier coefficient at weight $\boldsymbol{n}$ of the generating function associated with the highest-weight orbit. This exact integral determines the squared norm, not the components of the state in a multiplicity basis. Its relative-phase form provides the starting point for the conditional stationary analysis in Appendix H; it does not by itself establish a general semiclassical approximation for transition probabilities.

D Equivalent formulations of the (p,q) irrep of $SU(3)$

In this appendix we specialize the exterior-power construction of Appendix C to $SU(3)$. A partition $\Lambda = (\Lambda_1, \Lambda_2, \Lambda_3)$ labeling a polynomial irrep of $U(3)$ determines an irrep (p,q) of $SU(3)$ after separating the determinant factor [1, 2]. Since $\wedge^2\mathbf{3} \simeq \overline{\mathbf{3}}$, we can describe its highest-weight orbit using the fundamental and antifundamental representations. We obtain an explicit expression for the projection onto a weight and derive equivalent formulas for its squared norm: a multinomial sum, a coefficient of a generating function, and a Fourier integral over the maximal torus. We also extend these formulas to overlaps between projected states. All results are exact and concern projected pure states; their relation to physical sector densities requires a separate analysis.

D.1 From a partition of $U(3)$ to an irrep (p,q) of $SU(3)$

Let

$$\Lambda = (\Lambda_1, \Lambda_2, \Lambda_3), \qquad \Lambda_1 \geq \Lambda_2 \geq \Lambda_3 \geq 0, \qquad |\Lambda| = N. \tag{D.1}$$

We define

$$p = \Lambda_1 - \Lambda_2, \qquad q = \Lambda_2 - \Lambda_3, \qquad r = \Lambda_3. \tag{D.2}$$

Then

$$\Lambda = (p+q+r, q+r, r) = (p+q, q, 0) + r\mathbf{1}. \tag{D.3}$$

The corresponding polynomial representation factorizes as

$$D^{\Lambda}(V) \simeq (\det V)^r D^{(p+q,q,0)}(V). \tag{D.4}$$

For $V \in SU(3)$, $\det V = 1$ and the noncentral part is characterized by the Dynkin labels (p,q). Therefore, partitions whose entries differ by a common integer shift determine equivalent irreps upon restriction to $SU(3)$ [1, 2]. Such shifts change the polynomial degree N and do not identify sectors of the same physical fixed-photon-number system.

Now let

$$\boldsymbol{n} = (n_1, n_2, n_3), \qquad |\boldsymbol{n}| = N, \tag{D.5}$$

be a weight of the polynomial representation. The corresponding weight of the reduced polynomial representation is

$$\mathbf{W} = \boldsymbol{n} - r\mathbf{1}. \tag{D.6}$$

Since

$$N = p + 2q + 3r, \tag{D.7}$$

the reduced weight satisfies

$$|\mathbf{W}| = p + 2q. \tag{D.8}$$

The central shift also cancels in the unnormalized spectrum–diagonal condition. Since a bistochastic matrix satisfies $Q\mathbf{1} = \mathbf{1}$,

$$Q\Lambda = \boldsymbol{n} \qquad \Longleftrightarrow \qquad Q(p+q, q, 0)^T = \mathbf{W}. \tag{D.9}$$

Thus, the representation label and its weight can be shifted together without changing this compatibility condition.

D.2 Fundamental–antifundamental construction

For $M = 3$, the column multiplicities introduced in Appendix C are

$$p_1 = p, \qquad p_2 = q, \qquad p_3 = r. \tag{D.10}$$

The highest-weight vector can be written as

$$|\Omega_\Lambda\rangle = |e_1\rangle^{\otimes p} \otimes |e_1 \wedge e_2\rangle^{\otimes q} \otimes |e_1 \wedge e_2 \wedge e_3\rangle^{\otimes r}\,. \tag{D.11}$$

For a frame

$$V = (v_1, v_2, v_3) \in SU(3), \tag{D.12}$$

the corresponding orbit is

$$|\Omega_\Lambda(V)\rangle = |v_1\rangle^{\otimes p} \otimes |v_1 \wedge v_2\rangle^{\otimes q} \otimes |v_1 \wedge v_2 \wedge v_3\rangle^{\otimes r}\,. \tag{D.13}$$

For $SU(3)$, the second exterior power is equivalent to the antifundamental representation [1, 2]. Let $\overline{\mathbb{C}^3}$ be a copy of $\mathbb{C}^3$ carrying the conjugate representation and define

$$\mathcal{J} : \wedge^2\mathbb{C}^3 \longrightarrow \overline{\mathbb{C}^3} \tag{D.14}$$

by

$$\mathcal{J}(e_i \wedge e_j) = \sum_{k=1}^{3} \epsilon_{kij}\, \overline{e}_k. \tag{D.15}$$

The cofactor identity

$$\sum_{i,j} \epsilon_{kij} V_{ia} V_{jb} = (\det V) \sum_c (V^{-1})_{ck} \epsilon_{cab} \tag{D.16}$$

implies, for $V \in SU(3)$,

$$\mathcal{J}(Vu \wedge Vv) = \overline{V}\, \mathcal{J}(u \wedge v). \tag{D.17}$$

Therefore,

$$\wedge^2\mathbf{3} \simeq \overline{\mathbf{3}}. \tag{D.18}$$

We now write

$$V = (\zeta, \beta, \eta). \tag{D.19}$$

Since $\mathcal{J}(e_1 \wedge e_2) = \overline{e}_3$, the equivariance of $\mathcal{J}$ gives

$$\mathcal{J}(\zeta \wedge \beta) = \overline{\eta}, \tag{D.20}$$

that is,

$$\zeta \wedge \beta \longleftrightarrow \overline{\eta}. \tag{D.21}$$

On the other hand,

$$v_1 \wedge v_2 \wedge v_3 = (\det V) e_1 \wedge e_2 \wedge e_3 = e_1 \wedge e_2 \wedge e_3, \tag{D.22}$$

so the rank-three exterior factor is independent of the frame and represents only the central shift $r\mathbf{1}$. The nontrivial content of the orbit can then be written as

$$|\Omega_{p,q}(V)\rangle = |\zeta\rangle^{\otimes p} \otimes |\overline{\eta}\rangle^{\otimes q}\,, \qquad \langle\eta|\zeta\rangle = 0. \tag{D.23}$$

This vector belongs to

$$\mathrm{Sym}^p(\mathbb{C}^3) \otimes \mathrm{Sym}^q(\overline{\mathbb{C}^3}). \tag{D.24}$$

The symmetric traceless subspace of this tensor product carries the irrep (p,q) [1, 2]. The orbit vector belongs to this subspace, as can be checked directly: if

$$T^{i_1 \cdots i_p}_{j_1 \cdots j_q} = \zeta^{i_1} \cdots \zeta^{i_p} \eta_{j_1} \cdots \eta_{j_q}, \qquad \eta_j = (\eta^j)^*, \tag{D.25}$$

contracting a fundamental index with an antifundamental index gives a factor

$$\sum_i \zeta^i \eta_i = \langle \eta | \zeta \rangle = 0. \tag{D.26}$$

All such contractions vanish, so the orbit vector (D.23) belongs to the irrep (p,q).

D.3 Weights and explicit projection

To recover the polynomial $U(3)$ weights associated with the partition $(p+q,q,0)$, we tensor the fundamental–antifundamental description with the determinant character:

$$(\det)^q \otimes \mathrm{Sym}^p(\mathbf{3}) \otimes \mathrm{Sym}^q(\overline{\mathbf{3}}), \tag{D.27}$$

restricted to the traceless subspace.

In this polynomial realization, the fundamental basis vector e_i has weight e_i, whereas the antifundamental factor together with its corresponding determinant character contributes the weight

$$\mathbf{1} - e_i. \tag{D.28}$$

This correspondence agrees with

$$\overline{e}_1 \leftrightarrow e_2 \wedge e_3, \qquad \overline{e}_2 \leftrightarrow e_3 \wedge e_1, \qquad \overline{e}_3 \leftrightarrow e_1 \wedge e_2. \tag{D.29}$$

We expand the orbit vector (D.23) in normalized occupation bases of the symmetric tensor factors. With $|\mathbf{a}| = p$ and $|\mathbf{b}| = q$,

$$|\Omega_{p,q}(V)\rangle = \sum_{\substack{|\mathbf{a}|=p \\ |\mathbf{b}|=q}} \sqrt{\frac{p!}{\prod_i a_i!}\frac{q!}{\prod_i b_i!}}\, \zeta^{\mathbf{a}} (\eta^*)^{\mathbf{b}} \, |\mathbf{a}\rangle \, |\overline{\mathbf{b}}\rangle \,. \tag{D.30}$$

Here

$$\zeta^{\mathbf{a}} = \prod_{i=1}^{3} \zeta_i^{a_i}, \qquad (\eta^*)^{\mathbf{b}} = \prod_{i=1}^{3} (\eta_i^*)^{b_i}. \tag{D.31}$$

The fundamental factor contributes the weight $\mathbf{a}$. The antifundamental contribution is

$$\sum_{i=1}^{3} b_i(\mathbf{1} - e_i) = q\mathbf{1} - \mathbf{b}. \tag{D.32}$$

Therefore, the total polynomial weight of $|\mathbf{a}\rangle \, |\overline{\mathbf{b}}\rangle$ is

$$\mathbf{W} = \mathbf{a} + q\mathbf{1} - \mathbf{b}, \tag{D.33}$$

or, equivalently,

$$\mathbf{a} - \mathbf{b} = \mathbf{W} - q\mathbf{1}. \tag{D.34}$$

Its sum satisfies $|\mathbf{W}| = p + 2q$, in agreement with the partition $(p+q,q,0)$.

Let $P_{\mathbf{W}}$ be the projector onto the polynomial weight $\mathbf{W}$ within the irrep (p,q). Applying this projector to (D.30), only the terms compatible with (D.34) remain,

$$P_{\mathbf{W}} \, |\Omega_{p,q}(V)\rangle = \sum_{\substack{|\mathbf{a}|=p,\ |\mathbf{b}|=q \\ \mathbf{a}-\mathbf{b}=\mathbf{W}-q\mathbf{1}}} \sqrt{\frac{p!}{\prod_i a_i!}\frac{q!}{\prod_i b_i!}}\, \zeta^{\mathbf{a}} (\eta^*)^{\mathbf{b}} \, |\mathbf{a}\rangle \, |\overline{\mathbf{b}}\rangle \,. \tag{D.35}$$

Although the expansion uses a basis of the full symmetric tensor product, the projected sum remains

in the irrep (p,q): the initial orbit vector belongs to that irrep, and its weight spaces are invariant under the diagonal torus.

The orthonormality of the tensor-product basis allows us to evaluate the norm directly,

$$\begin{aligned}\mathcal{Z}_{\mathbf{W}}(\zeta,\eta) &:= \|P_{\mathbf{W}}\,|\Omega_{p,q}(V)\rangle\|^2 \\ &= \sum_{\substack{|\mathbf{a}|=p,\ |\mathbf{b}|=q \\ \mathbf{a}-\mathbf{b}=\mathbf{W}-q\mathbf{1}}} \frac{p!}{\prod_i a_i!}\frac{q!}{\prod_i b_i!}\prod_{i=1}^{3}|\zeta_i|^{2a_i}|\eta_i|^{2b_i}.\end{aligned} \tag{D.36}$$

For an $SU(3)$ frame, the rank-three exterior factor has unit norm and shifts every polynomial weight by $r\mathbf{1}$. Therefore, with $\mathbf{W}=\boldsymbol{n}-r\mathbf{1}$,

$$\mathcal{Z}_{\mathbf{W}}(\zeta,\eta)=\mathcal{N}_{\Lambda,\boldsymbol{n}}(V). \tag{D.37}$$

This formula computes the norm of a projected pure state without constructing a basis of $\mathcal{H}_{\Lambda,\boldsymbol{n}}$. It should not be identified with the physical sector weight $P(\Lambda\mid\boldsymbol{n},\Gamma)$ or with the sector density $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ of Appendix A.

D.4 Generating function and toroidal representation

The multinomial sum can equivalently be obtained from a generating function. Let

$$Z=\mathrm{diag}(z_1,z_2,z_3). \tag{D.38}$$

For the fundamental factor,

$$g_1(z)=\zeta^\dagger Z\zeta=\sum_{i=1}^{3}|\zeta_i|^2 z_i. \tag{D.39}$$

For the rank-two exterior factor, we write $V_2=(\zeta,\beta)$. By Cauchy–Binet,

$$\begin{aligned}g_2(z)&=\det(V_2^\dagger Z V_2)\\ &=|\eta_1|^2z_2z_3+|\eta_2|^2z_1z_3+|\eta_3|^2z_1z_2\\ &=(z_1z_2z_3)\left(\frac{|\eta_1|^2}{z_1}+\frac{|\eta_2|^2}{z_2}+\frac{|\eta_3|^2}{z_3}\right).\end{aligned} \tag{D.40}$$

The generating function associated with the orbit vector is then

$$F_V(z)=g_1(z)^p g_2(z)^q, \tag{D.41}$$

or, explicitly,

$$F_V(z)=\left(\sum_i|\zeta_i|^2z_i\right)^p(z_1z_2z_3)^q\left(\sum_i|\eta_i|^2z_i^{-1}\right)^q. \tag{D.42}$$

Upon expanding both powers, the exponent of z_i is a_i+q-b_i. Therefore, the coefficient associated with the weight $\mathbf{W}$ selects exactly the condition (D.34), and we obtain

$$[z^{\mathbf{W}}]F_V(z)=\mathcal{Z}_{\mathbf{W}}(\zeta,\eta)=\mathcal{N}_{\Lambda,\boldsymbol{n}}(V). \tag{D.43}$$

The projected norm depends only on $|\zeta_i|^2$ and $|\eta_i|^2$, and hence on $Q(V)$. The projected vectors and their cross kernels, however, can retain relative-phase information not fixed by $Q(V)$.

The same coefficient can be extracted by Fourier integration. To specialize the toroidal projector of Appendix C to three modes, we define

$$t_{\boldsymbol{\phi}}=\mathrm{diag}(\mathrm{e}^{\mathrm{i}\phi_1},\mathrm{e}^{\mathrm{i}\phi_2},\mathrm{e}^{\mathrm{i}\phi_3}). \tag{D.44}$$

On the unitary diagonal torus, the polynomial $U(3)$ action is represented on the traceless subspace by

$$\widetilde{D}^{(p,q)}(t_{\boldsymbol{\phi}}) = (\det t_{\boldsymbol{\phi}})^q \left[t_{\boldsymbol{\phi}}^{\otimes p} \otimes \overline{t_{\boldsymbol{\phi}}}^{\otimes q}\right] \tag{D.45}$$

on the traceless subspace. The weight projector is then

$$P_{\mathbf{W}} = \int_{\mathbb{T}^3} \frac{\mathrm{d}^3\phi}{(2\pi)^3}\, \mathrm{e}^{-\mathrm{i}\mathbf{W}\cdot\boldsymbol{\phi}} \widetilde{D}^{(p,q)}(t_{\boldsymbol{\phi}}), \tag{D.46}$$

or, using $(\det t_{\boldsymbol{\phi}})^q = \mathrm{e}^{\mathrm{i}q\mathbf{1}\cdot\boldsymbol{\phi}}$,

$$P_{\mathbf{W}} = \int_{\mathbb{T}^3} \frac{\mathrm{d}^3\phi}{(2\pi)^3}\, \mathrm{e}^{-\mathrm{i}(\mathbf{W}-q\mathbf{1})\cdot\boldsymbol{\phi}} t_{\boldsymbol{\phi}}^{\otimes p} \otimes \overline{t_{\boldsymbol{\phi}}}^{\otimes q}. \tag{D.47}$$

In particular, the norm can be written directly as

$$\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = \int_{\mathbb{T}^3} \frac{\mathrm{d}^3\phi}{(2\pi)^3}\, \mathrm{e}^{-\mathrm{i}\mathbf{W}\cdot\boldsymbol{\phi}} F_V(\mathrm{e}^{\mathrm{i}\phi_1}, \mathrm{e}^{\mathrm{i}\phi_2}, \mathrm{e}^{\mathrm{i}\phi_3}). \tag{D.48}$$

The global phase is redundant. Under $\boldsymbol{\phi} \mapsto \boldsymbol{\phi} + \chi\mathbf{1}$, the tensor action produces the factor $\mathrm{e}^{\mathrm{i}(p-q)\chi}$, while the Fourier factor in (D.47) produces $\mathrm{e}^{-\mathrm{i}(p-q)\chi}$. The two cancel because

$$|\mathbf{W}| - 3q = p - q. \tag{D.49}$$

Only two relative phases are therefore independent. Here we take the first mode as reference, $\phi_1 = 0$, which is equivalent to the choice of the last mode in Appendix C. We obtain

$$\begin{aligned}\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = \int_{[0,2\pi]^2} &\frac{\mathrm{d}\phi_2\, \mathrm{d}\phi_3}{(2\pi)^2}\, \mathrm{e}^{-\mathrm{i}(W_2\phi_2 + W_3\phi_3)} \\ &\times F_V(1, \mathrm{e}^{\mathrm{i}\phi_2}, \mathrm{e}^{\mathrm{i}\phi_3}).\end{aligned} \tag{D.50}$$

This reduction is the $SU(3)$ version of the elimination of the global phase obtained in Appendix C. The same degree of freedom will appear later as the null mode of the Hessian associated with the projection integral.

The identities

$$\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = \mathcal{Z}_{\mathbf{W}}(\zeta,\eta) = [z^{\mathbf{W}}]F_V(z) \tag{D.51}$$

and (D.48) give equivalent exact expressions for the projected norm. The multinomial sum collects terms of the required weight; coefficient extraction and the toroidal integral recover the same quantity algebraically and by Fourier orthogonality, respectively. They do not by themselves provide a semiclassical approximation.

D.5 Extension to the cross kernel

Let

$$V = (\zeta, \beta, \eta), \qquad V' = (\zeta', \beta', \eta') \tag{D.52}$$

be two $SU(3)$ frames. The unnormalized projected kernel is defined by

$$\widetilde{K}(V, V') = \langle \Omega_{p,q}(V) |\, P_{\mathbf{W}}\, | \Omega_{p,q}(V') \rangle\,. \tag{D.53}$$

Using the projected expansion,

$$\widetilde{K}(V, V') = \sum_{\substack{|\mathbf{a}|=p,\ |\mathbf{b}|=q \\ \mathbf{a}-\mathbf{b}=\mathbf{W}-q\mathbf{1}}} \frac{p!}{\prod_i a_i!} \frac{q!}{\prod_i b_i!} \prod_{i=1}^{3} (\zeta_i^* \zeta_i')^{a_i} (\eta_i \eta_i'^*)^{b_i}. \tag{D.54}$$

We introduce the cross generating function

$$F_{V,V'}(z) = \left(\sum_i \zeta_i^* \zeta'_i z_i\right)^p (z_1 z_2 z_3)^q \left(\sum_i \eta_i \eta_i'^* z_i^{-1}\right)^q. \tag{D.55}$$

The weight condition is again incorporated into the coefficient extraction,

$$\widetilde{K}(V, V') = [z^{\mathbf{W}}] F_{V,V'}(z). \tag{D.56}$$

For $V' = V$ we recover

$$\widetilde{K}(V, V) = \mathcal{N}_{\Lambda,\boldsymbol{n}}(V). \tag{D.57}$$

The result agrees with the exterior-power expression for the unnormalized projected kernel,

$$[z^{\boldsymbol{n}}] \prod_{s=1}^{3} \det(V_s^\dagger Z V'_s)^{p_s}, \tag{D.58}$$

after separating the central factor $p_3 = r$ and using $\wedge^2 \mathbf{3} \simeq \overline{\mathbf{3}}$. Unlike the projected norm, the cross kernel retains phase information from the two frames that is generally not determined by their unistochastic matrices. The normalized kernel and the rank criterion for spanning the weight space are developed in Appendix F.

The exterior-power and fundamental–antifundamental constructions thus give equivalent descriptions of the projected highest-weight orbit. In three modes, its noncentral dependence on the frame can be expressed using the two columns ζ and η, with $\langle \eta | \zeta \rangle = 0$. This exact representation provides a starting point for computing projected kernels and coordinates in the weight spaces. It does not, by itself, determine the large-N behavior of general representation matrix elements or physical sectorial transition probabilities.

E Multiplicity in $SU(3)$: Gelfand–Tsetlin, isospin, and Wigner-D coordinates

We construct an orthonormal basis for a fixed weight space of an $SU(3)$ irrep. Gelfand–Tsetlin patterns give the exact weight multiplicity and label its independent directions by the isospin j associated with the chain $U(3) \supset U(2)$ [19]. Using the fundamental–antifundamental construction of the preceding appendix, we express the projected highest-weight orbit state in this basis through polynomial Wigner-D elements [20, 21]. The resulting expansion, $|\mathbf{W}, V\rangle = \sum_j c_j(V)\,|\mathbf{W}, j\rangle$, separates the allowed isospin channels from the frame-dependent amplitudes $c_j(V)$. These coordinates will also be used to describe projected kernels and the physical sector density in the same weight space, without identifying the projected pure state with that density.

E.1 Multiplicity and Gelfand–Tsetlin basis

Let $\mathcal{H}_\Lambda$ be the polynomial irrep of $U(3)$ associated with

$$\Lambda = (\Lambda_1, \Lambda_2, \Lambda_3), \qquad \Lambda_1 \geq \Lambda_2 \geq \Lambda_3 \geq 0. \tag{E.1}$$

Upon restricting the representation to the maximal torus,

$$\mathcal{H}_\Lambda = \bigoplus_{\boldsymbol{n}} \mathcal{H}_{\Lambda,\boldsymbol{n}}, \tag{E.2}$$

where $\mathcal{H}_{\Lambda,\boldsymbol{n}}$ is the weight space of $\boldsymbol{n}$. Its dimension

$$m_\Lambda(\boldsymbol{n}) = \dim \mathcal{H}_{\Lambda,\boldsymbol{n}} \tag{E.3}$$

is the weight multiplicity.

To determine it, we use the Gelfand–Tsetlin basis associated with the chain

$$U(3) \supset U(2) \supset U(1). \tag{E.4}$$

The vectors of this basis are labeled by patterns

$$\begin{array}{ccccc} \Lambda_1 & & \Lambda_2 & & \Lambda_3 \\ & \mu_1 & & \mu_2 & \\ & & \nu_1 & & \end{array} \tag{E.5}$$

satisfying the interlacing conditions

$$\Lambda_1 \geq \mu_1 \geq \Lambda_2, \qquad \Lambda_2 \geq \mu_2 \geq \Lambda_3, \qquad \mu_1 \geq \nu_1 \geq \mu_2. \tag{E.6}$$

The weight of the pattern is given by

$$n_1 = \nu_1, \qquad n_2 = \mu_1 + \mu_2 - \nu_1, \qquad n_3 = |\Lambda| - \mu_1 - \mu_2. \tag{E.7}$$

Therefore,

$$m_\Lambda(\boldsymbol{n}) = \#\{\text{GT patterns of weight } \boldsymbol{n}\}. \tag{E.8}$$

For a fixed weight,

$$\nu_1 = n_1, \qquad \mu_1 + \mu_2 = n_1 + n_2. \tag{E.9}$$

Defining

$$s_{12} = n_1 + n_2, \tag{E.10}$$

we have

$$\mu_2 = s_{12} - \mu_1, \tag{E.11}$$

and the interlacing conditions reduce to

$$L \leq \mu_1 \leq U, \tag{E.12}$$

with

$$L = \max\{\Lambda_2,\, s_{12} - \Lambda_2,\, n_1,\, n_2\}, \tag{E.13}$$

and

$$U = \min\{\Lambda_1,\, s_{12} - \Lambda_3\}. \tag{E.14}$$

The multiplicity is then given by

$$m_\Lambda(\boldsymbol{n}) = \max\{0, U - L + 1\}. \tag{E.15}$$

To relate this description to $SU(3)$, we write

$$p = \Lambda_1 - \Lambda_2, \qquad q = \Lambda_2 - \Lambda_3, \qquad r = \Lambda_3, \tag{E.16}$$

and remove the central shift through

$$\mathbf{W} = \boldsymbol{n} - r\mathbf{1}, \qquad |\mathbf{W}| = p + 2q. \tag{E.17}$$

The corresponding partition without the central component is

$$\Lambda^{(0)} = (p + q, q, 0). \tag{E.18}$$

After subtracting r from each entry of the GT pattern, we use the same symbols μ_1, μ_2, and ν_1 for the shifted labels. The reduced pattern is

$$\begin{array}{ccccc} p+q & & q & & 0 \\ & \mu_1 & & \mu_2 & \\ & & \nu_1 & & \end{array}, \tag{E.19}$$

with

$$p + q \geq \mu_1 \geq q \geq \mu_2 \geq 0, \qquad \mu_1 \geq \nu_1 \geq \mu_2. \tag{E.20}$$

The reduced weight satisfies

$$W_1 = \nu_1, \qquad W_2 = \mu_1 + \mu_2 - \nu_1, \qquad W_3 = p + 2q - \mu_1 - \mu_2. \tag{E.21}$$

For fixed $\mathbf{W}$, we introduce

$$j = \frac{\mu_1 - \mu_2}{2}. \tag{E.22}$$

Since $\mu_1 + \mu_2 = W_1 + W_2$, it follows that

$$\mu_1 = \frac{W_1 + W_2}{2} + j, \qquad \mu_2 = \frac{W_1 + W_2}{2} - j. \tag{E.23}$$

For a fixed weight, each admissible GT pattern therefore corresponds to one allowed value of j. We denote the associated vector in the $U(3) \downarrow U(2)$ branching basis by $|\mathbf{W}, j\rangle$ [19]. Here j labels directions within a fixed weight space, not different irreducible sectors.

The interlacing conditions determine the allowed interval of isospin labels. The lower bound is

$$j_{\min} = \frac{1}{2}\max\{|W_1 - W_2|, |p - W_3|\}, \tag{E.24}$$

while

$$j_{\max} = \frac{1}{2}\min\{p + W_3,\, p + 2q - W_3\}. \tag{E.25}$$

For an admissible weight, the values of j increase in integer steps,

$$j = j_{\min}, j_{\min}+1, \ldots, j_{\max}, \tag{E.26}$$

with integer or half-integer parity fixed by the weight. Consequently,

$$m_\Lambda(\boldsymbol{n}) = \#\left\{j : j_{\min} \le j \le j_{\max}\right\}, \tag{E.27}$$

and, when the interval is nonempty,

$$m_\Lambda(\boldsymbol{n}) = j_{\max} - j_{\min} + 1. \tag{E.28}$$

This expression agrees with the direct counting of GT patterns in (E.15).

E.2 Basis adapted to $SU(2)_{12}$

Appendix D gives the fundamental–antifundamental construction

$$|\Omega_{p,q}(V)\rangle = |\zeta\rangle^{\otimes p} \otimes |\overline{\eta}\rangle^{\otimes q}, \qquad V = (\zeta, v_2, \eta) \in SU(3), \qquad \langle\eta|\zeta\rangle = 0. \tag{E.29}$$

Its projection onto the reduced weight $\mathbf{W}$ can be written as

$$\begin{aligned} P_{\mathbf{W}}\,|\Omega_{p,q}(V)\rangle = \sum_{\substack{|\mathbf{a}|=p,\ |\mathbf{b}|=q \\ \mathbf{a}-\mathbf{b}=\mathbf{W}-q\mathbf{1}}} & \sqrt{\frac{p!q!}{a_1!a_2!a_3!b_1!b_2!b_3!}} \\ & \times \zeta_1^{a_1}\zeta_2^{a_2}\zeta_3^{a_3}(\eta_1^*)^{b_1}(\eta_2^*)^{b_2}(\eta_3^*)^{b_3}\,|\mathbf{a}\rangle\,|\overline{\mathbf{b}}\rangle\,. \end{aligned} \tag{E.30}$$

To organize the projected coefficients, we use the subgroup $SU(2)_{12}$ acting on the first two modes. In the fundamental representation,

$$e_1 \equiv |\uparrow\rangle\,, \qquad e_2 \equiv |\downarrow\rangle\,. \tag{E.31}$$

For an occupation $\mathbf{a}$, we define

$$j_1 = \frac{a_1+a_2}{2} = \frac{p-a_3}{2}, \qquad m_1 = \frac{a_1-a_2}{2}, \tag{E.32}$$

so that

$$a_1 = j_1 + m_1, \qquad a_2 = j_1 - m_1. \tag{E.33}$$

For the conjugate doublet we adopt the convention

$$\overline{e}_1 \longleftrightarrow -\,|\downarrow\rangle\,, \qquad \overline{e}_2 \longleftrightarrow |\uparrow\rangle\,, \tag{E.34}$$

and define

$$j_2 = \frac{b_1+b_2}{2} = \frac{q-b_3}{2}, \qquad m_2 = \frac{b_2-b_1}{2}. \tag{E.35}$$

Therefore,

$$b_1 = j_2 - m_2, \qquad b_2 = j_2 + m_2, \tag{E.36}$$

and each occupation of $\overline{e}_1$ introduces the factor $(-1)^{b_1}$.

The weight condition fixes two quantities that do not depend on a_3. From the first two components we obtain

$$m := m_1 + m_2 = \frac{W_1 - W_2}{2}, \tag{E.37}$$

while the third component gives

$$b_3 = a_3 + q - W_3. \tag{E.38}$$

Consequently,

$$j_1 = \frac{p - a_3}{2}, \qquad j_2 = \frac{W_3 - a_3}{2}, \tag{E.39}$$

and the difference

$$\Delta j := j_1 - j_2 = \frac{p - W_3}{2} \tag{E.40}$$

is also determined solely by $\mathbf{W}$. In contrast,

$$j_1 + j_2 = \frac{p + W_3}{2} - a_3 \tag{E.41}$$

depends on the block labeled by a_3.

For fixed a_3, the two doublets couple according to

$$j_1 \otimes j_2 = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} j. \tag{E.42}$$

We introduce the coupled states

$$|j, m; j_1, j_2; a_3, b_3\rangle = \sum_{m_1+m_2=m} (-1)^{b_1} C^{jm}_{j_1 m_1, j_2 m_2} \, |\mathbf{a}\rangle \, |\overline{\mathbf{b}}\rangle \,, \tag{E.43}$$

where $C^{jm}_{j_1 m_1, j_2 m_2}$ are Clebsch–Gordan coefficients with the phase convention used in [20].

For fixed $(\mathbf{W}, j)$, the contributions with different a_3 belong to orthogonal blocks of the ambient tensor product. Their combination in the traceless irrep (p, q) takes the form

$$|\mathbf{W}, j\rangle = \sum_{a_3=0}^{p} \sqrt{\frac{\mathcal{Q}(a_3|\mathbf{W}, j)}{\mathcal{N}(\mathbf{W}, j)}} \; |j, m; j_1, j_2; a_3, b_3\rangle \,, \tag{E.44}$$

where

$$\mathcal{Q}(a_3|\mathbf{W}, j) = \frac{(p+q)!}{a_3!\, b_3!\, (j_1 + j_2 - j)!\, (j_1 + j_2 + j + 1)!}, \tag{E.45}$$

and we understand $\mathcal{Q}(a_3|\mathbf{W}, j) = 0$ whenever one of the factorials does not correspond to a nonnegative integer. The normalization is

$$\mathcal{N}(\mathbf{W}, j) = \sum_{a_3=0}^{p} \mathcal{Q}(a_3|\mathbf{W}, j). \tag{E.46}$$

This combinatorial normalization fixes the basis vectors and must be distinguished from the frame-dependent projected norm $\mathcal{N}_{\Lambda,\boldsymbol{n}}(V)$.

To obtain a closed form, we define

$$u = \frac{p + W_3}{2}, \qquad v = \frac{p + 2q - W_3}{2}. \tag{E.47}$$

The sum in (E.46) can be evaluated using the Vandermonde identity,

$$\mathcal{N}(\mathbf{W}, j) = \frac{(p+q)!(p+q+1)!}{(u-j)!(u+j+1)!(v-j)!(v+j+1)!}. \tag{E.48}$$

Because the different a_3 blocks are orthogonal, (E.46) gives

$$\langle \mathbf{W}, j | \mathbf{W}, j \rangle = 1. \tag{E.49}$$

Moreover,

$$J^2 \, |\mathbf{W}, j\rangle = j(j+1) \, |\mathbf{W}, j\rangle \,, \tag{E.50}$$

so that

$$\langle \mathbf{W}, j | \mathbf{W}, j' \rangle = \delta_{jj'}. \tag{E.51}$$

The allowed vectors

$$\{ |\mathbf{W}, j\rangle : j_{\min} \le j \le j_{\max} \} \tag{E.52}$$

form an orthonormal basis of $\mathcal{H}_{\Lambda,\boldsymbol{n}}$. Since the $U(3) \downarrow U(2)$ branching is multiplicity-free, this basis agrees, up to phase conventions, with the Gelfand–Tsetlin basis above.

E.3 Wigner-D coordinates of the frame state

We now express the frame dependence of the projected state in the isospin basis $|\mathbf{W}, j\rangle$. Define

$$g(V) = \begin{pmatrix} \zeta_1 & \eta_2^* \\ \zeta_2 & -\eta_1^* \end{pmatrix}. \tag{E.53}$$

The orthogonality condition $\langle \eta | \zeta \rangle = 0$ implies

$$\det g = \zeta_3 \eta_3^*. \tag{E.54}$$

When $\det g \neq 0$, we define

$$\widetilde{g} = \frac{g}{\sqrt{\det g}}, \qquad \det \widetilde{g} = 1. \tag{E.55}$$

The polynomial expressions in g remain well defined when $\det g = 0$; the rescaled matrix $\widetilde{g}$ is only a local parametrization. When defined, $\widetilde{g} \in SL(2,\mathbb{C})$ but is not generally unitary. The Wigner-D elements are therefore understood through their polynomial continuation.

For a complex 2×2 matrix, the polynomial continuation of the Wigner elements can be written as [20, 21, 22]

$$D^j_{m,m'}(g) = \frac{1}{\sqrt{\binom{2j}{j+m}\binom{2j}{j+m'}}} \sum_k \frac{(2j)!}{A_{11}!A_{12}!A_{21}!A_{22}!} g_{11}^{A_{11}} g_{12}^{A_{12}} g_{21}^{A_{21}} g_{22}^{A_{22}}, \tag{E.56}$$

where

$$A(k) = \begin{pmatrix} j + m' - k & m - m' + k \\ k & j - m - k \end{pmatrix}, \tag{E.57}$$

and the sum runs over the values of k for which all exponents are nonnegative integers. Since this polynomial is homogeneous of degree $2j$,

$$D^j_{m,m'}(cg) = c^{2j} D^j_{m,m'}(g), \tag{E.58}$$

and, when $\det g \neq 0$,

$$D^j_{m,m'}(\widetilde{g}) = (\det g)^{-j} D^j_{m,m'}(g). \tag{E.59}$$

The sum over m_1, m_2 appearing in the projection can be reorganized exactly in this polynomial form. Defining

$$m' = j_1 - j_2 = \Delta j, \tag{E.60}$$

we obtain

$$\begin{aligned} \psi(j, m | j_1, j_2; \zeta, \eta) = \sqrt{\frac{2j+1}{(j_1 + j_2 - j)!(j_1 + j_2 + j + 1)!}} \\ \times (\det g)^{j_1 + j_2 - j} D^j_{m, j_1 - j_2}(g). \end{aligned} \tag{E.61}$$

To obtain this identity, one expands the polynomial (E.56) together with $(\det g)^{j_1+j_2-j}$ and collects terms with equal powers. Their coefficients reproduce the Racah expression for the Clebsch–Gordan coefficients [20, 21, 22]. This is an exact polynomial identity, not a semiclassical approximation.

When $\det g \neq 0$, the same expression can be written as

$$\psi(j,m|j_1,j_2;\zeta,\eta) = \sqrt{\frac{2j+1}{(j_1+j_2-j)!(j_1+j_2+j+1)!}} \times (\det g)^{j_1+j_2} D^j_{m,j_1-j_2}(\widetilde{g}). \tag{E.62}$$

Combining this result with (E.44), the full projection takes the form

$$P_{\mathbf{W}} \left|\Omega_{p,q}(V)\right\rangle = \sqrt{\frac{p!q!}{(p+q)!}} \sum_{j=j_{\min}}^{j_{\max}} \mathcal{B}_j(V) \left|\mathbf{W}, j\right\rangle, \tag{E.63}$$

with

$$\mathcal{B}_j(V) = \sqrt{(2j+1)\mathcal{N}(\mathbf{W},j)}\, \zeta_3^{u-j} (\eta_3^*)^{v-j} D^j_{m,\Delta j}(g(V)). \tag{E.64}$$

For every allowed j,

$$u - j \geq 0, \qquad v - j \geq 0, \tag{E.65}$$

so that $\mathcal{B}_j(V)$ is a polynomial expression and remains well defined even when $\det g = 0$.
In the region $\det g \neq 0$,

$$\mathcal{B}_j(V) = \zeta_3^u (\eta_3^*)^v A_j(V), \tag{E.66}$$

where

$$A_j(V) = \sqrt{(2j+1)\mathcal{N}(\mathbf{W},j)} D^j_{m,\Delta j}(\widetilde{g}(V)). \tag{E.67}$$

The factor relating $\mathcal{B}_j$ to A_j does not depend on j. Thus, wherever the projected state is nonzero, the two expressions give the same normalized coordinates, up to a common phase.

E.4 Three occupation configurations

We now consider the three configurations that will be used in the numerical analysis. In all cases we fix

$$N = 45, \qquad \Lambda = (33,9,3), \qquad (p,q,r) = (24,6,3), \tag{E.68}$$

and take the occupations

$$\boldsymbol{n}^{(1)} = (15,15,15), \qquad \boldsymbol{n}^{(2)} = (27,9,9), \qquad \boldsymbol{n}^{(3)} = (15,10,20). \tag{E.69}$$

The reduced weights $\mathbf{W} = \boldsymbol{n} - r\mathbf{1}$ are

$$\mathbf{W}^{(1)} = (12,12,12), \qquad \mathbf{W}^{(2)} = (24,6,6), \qquad \mathbf{W}^{(3)} = (12,7,17). \tag{E.70}$$

The labels determined by (E.37), (E.40), (E.24), and (E.25) are summarized in Table 1.

Configuration	$\mathbf{W}$	m	Δj	Allowed j
Balanced	$(12,12,12)$	0	6	$6,\ldots,12$
Partially unbalanced	$(24,6,6)$	9	9	$9,\ldots,15$
Asymmetric	$(12,7,17)$	$\frac{5}{2}$	$\frac{7}{2}$	$\frac{7}{2},\frac{9}{2},\ldots,\frac{19}{2}$

Table 1: Isospin labels for the three reference configurations. In all cases $N = 45$, $\Lambda = (33,9,3)$, and $m_\Lambda(\boldsymbol{n}) = 7$.

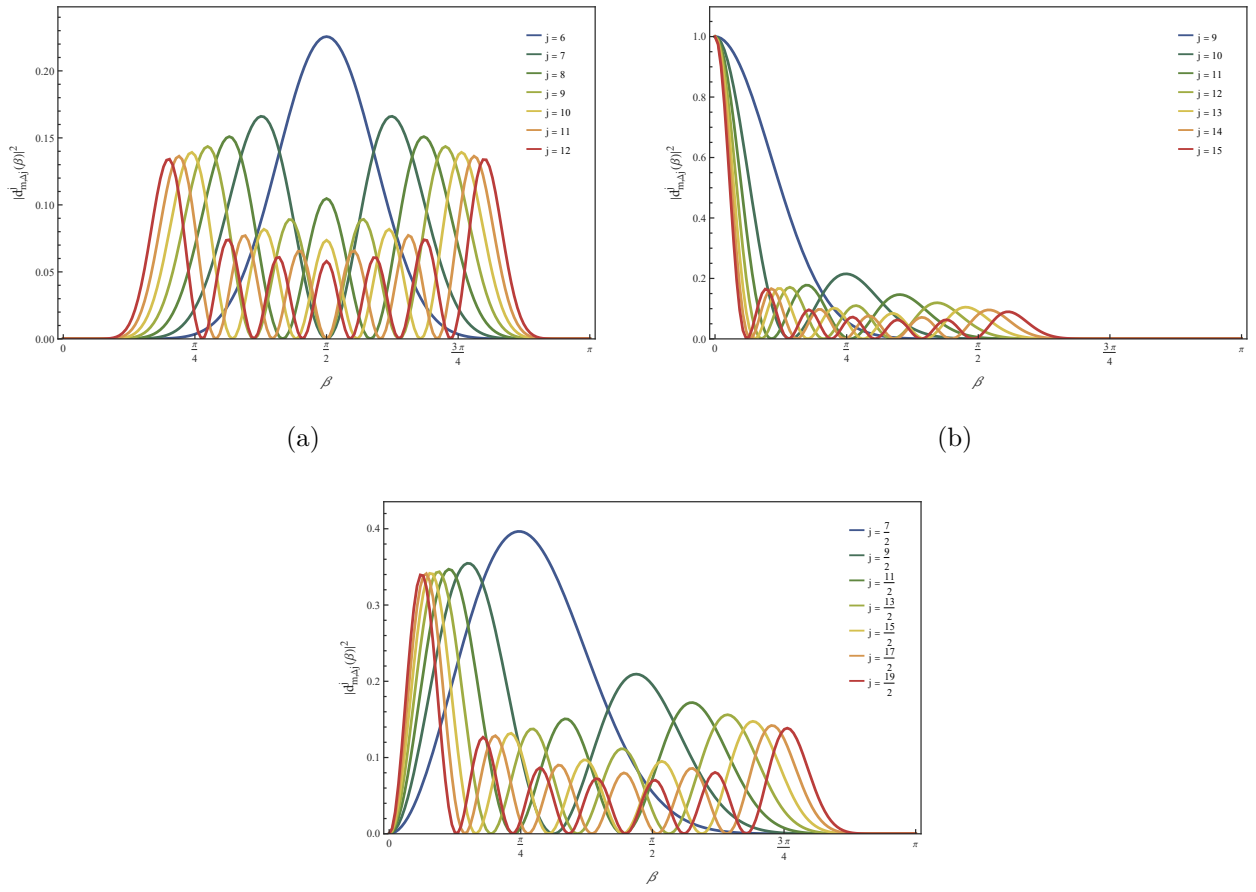
All three weights have multiplicity $m_\Lambda(\boldsymbol{n}) = 7$, but their isospin labels differ. Here $m = (W_1 - W_2)/2$ is the magnetic quantum number and should not be confused with the multiplicity $m_\Lambda(\boldsymbol{n})$. The balanced and partially unbalanced cases have integer j, whereas the asymmetric case has half-integer

j. Thus, equal weight multiplicities need not correspond to the same set of isospin channels; the labels m, Δj, and the interval $[j_{\min}, j_{\max}]$ distinguish these cases.

To illustrate the angular dependence of the Wigner elements in (E.67), we evaluate the reduced matrix $\widetilde{g}$ along a one-parameter family of $SU(2)$ rotations, parametrized by β. The resulting angular factor for each channel is

$$\left|d^{j}_{m,\Delta j}(\beta)\right|^2 . \tag{E.71}$$

Fig. 5 compares these functions for the three weights of Table 1. This is an illustrative angular section of the reduced Wigner variables, not a restriction of the full frame V to the embedded subgroup $SU(2)_{12}$. It does not impose the spectrum–diagonal compatibility condition of Appendix B, and these Wigner factors alone are not physical sector probabilities.



(a) (b)

(c)

Figure 5: Angular Wigner response $\left|d^{j}_{m,\Delta j}(\beta)\right|^2$ for the three reference configurations with $N = 45$ and $\Lambda = (33, 9, 3)$. (a) Balanced configuration: $m = 0$, $\Delta j = 6$, and $j = 6, \ldots, 12$. (b) Partially unbalanced configuration: $m = 9$, $\Delta j = 9$, and $j = 9, \ldots, 15$. (c) Asymmetric configuration: $m = 5/2$, $\Delta j = 7/2$, and $j = 7/2, 9/2, \ldots, 19/2$.

Although the three configurations have multiplicity seven, their channels exhibit different angular dependences. In the balanced configuration, $m = 0$ and the functions show a symmetric structure about $\beta = \pi/2$. In the partially unbalanced configuration, $m = \Delta j = 9$ and $d^{j}_{9,9}(0) = 1$ for all allowed values of j; as β increases, the different channels develop different oscillatory structures. In the asymmetric configuration, $m = 5/2 \neq 7/2 = \Delta j$, so that $d^{j}_{5/2,7/2}(0) = 0$ and the maxima occur at nonzero values of the angle.

These angular sections show that the multiplicity fixes the number of internal directions, but not the corresponding Wigner responses. The latter depend on $(j, m, \Delta j)$, while the factors $\mathcal{N}(\mathbf{W}, j)$ in (E.67)

set the relative combinatorial weights entering the projected-state coordinates.

E.5 Normalized state and orbital isospin distribution

For a frame with nonzero projected norm, define the normalized coordinates

$$c_j(V) = \frac{\mathcal{B}_j(V)}{\sqrt{\sum_k |\mathcal{B}_k(V)|^2}}. \tag{E.72}$$

Under the identification of the central factor, the projected pure state introduced in Appendix C can be written as

$$|\mathbf{W}, V\rangle \equiv |\boldsymbol{n}, V; \Lambda\rangle = \sum_{j=j_{\min}}^{j_{\max}} c_j(V)\, |\mathbf{W}, j\rangle\,. \tag{E.73}$$

When $\det g \neq 0$, we introduce

$$Z_{\mathrm{iso}}(\mathbf{W}, \widetilde{g}) = \sum_j (2j+1)\mathcal{N}(\mathbf{W}, j) \left| D^j_{m,\Delta j}(\widetilde{g}) \right|^2. \tag{E.74}$$

In this case,

$$c_j(V) = \frac{\sqrt{(2j+1)\mathcal{N}(\mathbf{W}, j)}\, D^j_{m,\Delta j}(\widetilde{g}(V))}{\sqrt{Z_{\mathrm{iso}}(\mathbf{W}, \widetilde{g}(V))}}. \tag{E.75}$$

The quantity Z_{iso} normalizes the isospin coordinates; it is not, by itself, the squared norm of the projected state. From (E.63),

$$\|P_{\mathbf{W}}\, |\Omega_{p,q}(V)\rangle\|^2 = \frac{p!q!}{(p+q)!} \sum_j |\mathcal{B}_j(V)|^2. \tag{E.76}$$

When $\det g \neq 0$,

$$\|P_{\mathbf{W}}\, |\Omega_{p,q}(V)\rangle\|^2 = \frac{p!q!}{(p+q)!} |\zeta_3|^{2u} |\eta_3|^{2v} Z_{\mathrm{iso}}(\mathbf{W}, \widetilde{g}). \tag{E.77}$$

The isospin distribution associated with the projected pure state is

$$P_{\mathrm{orb}}(j|\mathbf{W}, V) = |c_j(V)|^2. \tag{E.78}$$

In the region $\det g \neq 0$,

$$P_{\mathrm{orb}}(j|\mathbf{W}, V) = \frac{(2j+1)\mathcal{N}(\mathbf{W}, j) \left| D^j_{m,\Delta j}(\widetilde{g}(V)) \right|^2}{Z_{\mathrm{iso}}(\mathbf{W}, \widetilde{g}(V))}, \tag{E.79}$$

with

$$\sum_j P_{\mathrm{orb}}(j|\mathbf{W}, V) = 1. \tag{E.80}$$

For the angular section of $\widetilde{g}$ used in Fig. 5, consider a corresponding family of full frames $V(\beta)$ and define

$$P_{\mathrm{orb}}(j|\mathbf{W}, \beta) := P_{\mathrm{orb}}(j|\mathbf{W}, V(\beta)). \tag{E.81}$$

Where the projected norm is nonzero, this quantity describes the normalized weights of the isospin channels as β varies. Unlike $\left| d^j_{m,\Delta j}(\beta) \right|^2$, it includes the factors $(2j+1)\mathcal{N}(\mathbf{W}, j)$ and normalization over all channels. The angular family illustrates the projected-state coordinates; it need not consist of frames compatible with the prescribed spectrum and initial occupation. In particular, these curves are not the isospin populations of a fixed physical sector density. At any endpoint where all the channel amplitudes vanish, the normalized distribution is undefined; one-sided limits, rather than zero probability for every channel, are appropriate there.

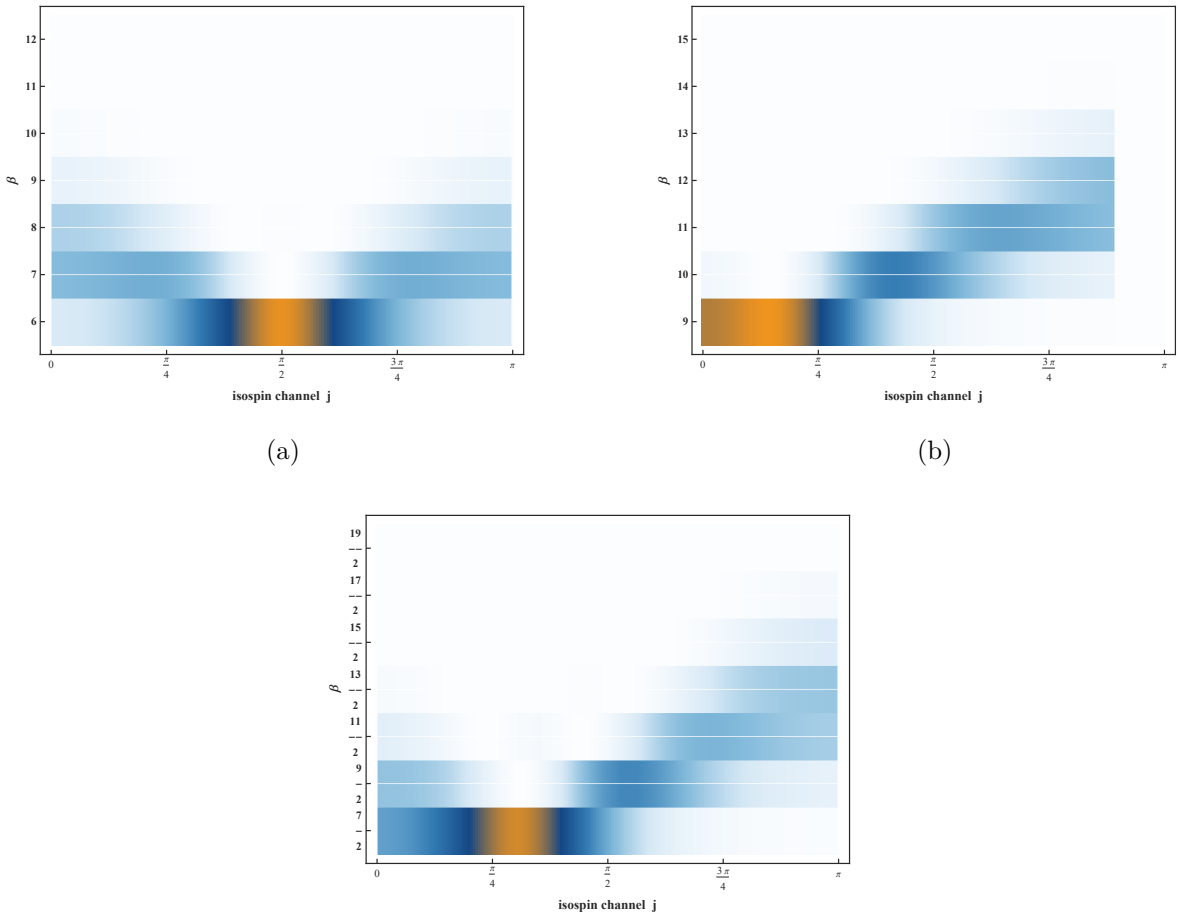


Figure 6: Orbital isospin distribution $P_{\mathrm{orb}}(j|\mathbf{W},\beta)$ for the three reference configurations with $N = 45$ and $\Lambda = (33, 9, 3)$. (a) Balanced configuration, $\mathbf{W} = (12, 12, 12)$, with $j = 6, \ldots, 12$. (b) Partially unbalanced configuration, $\mathbf{W} = (24, 6, 6)$, with $j = 9, \ldots, 15$. (c) Asymmetric configuration, $\mathbf{W} = (12, 7, 17)$, with $j = 7/2, 9/2, \ldots, 19/2$. The intensity gives the probability associated with each isospin channel as a function of β.

Fig. 6 displays the normalized isospin weights obtained from the Wigner angular sections after including their combinatorial factors. For the balanced configuration, Fig. 6(a), the distribution is symmetric about $\beta = \pi/2$. The $j = 6$ channel concentrates the largest probability around this value, while the higher channels acquire weight on both sides.

In the partially unbalanced configuration, Fig. 6(b), the distribution starts concentrated in the $j = 9$ channel and progressively shifts toward larger values of j as β increases. Although this case and the balanced case have the same multiplicity, the redistribution among their channels is different because they correspond to different values of m and Δj.

For the asymmetric configuration, Fig. 6(c), the basis is formed by half-integer channels and the probability is redistributed among $j = 7/2, \ldots, 19/2$ as β varies. The resulting structure reflects the labels $m = 5/2$ and $\Delta j = 7/2$ characteristic of this weight.

Figs. 5 and 6 show two complementary aspects of the decomposition in the isospin basis. The first describes the angular dependence of the Wigner elements associated with each channel, while the second incorporates the factors $(2j + 1)\mathcal{N}(\mathbf{W}, j)$ and the normalization over j, giving the orbital distribution $P_{\mathrm{orb}}(j|\mathbf{W}, V) = |c_j(V)|^2$ of the projected pure state $|\mathbf{W}, V\rangle$.

This distribution must be distinguished from the physical isospin populations $P_{\mathrm{phys}}(j) = [\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)]_{jj}$, defined in the same isospin basis for a sector with nonzero statistical weight. While $P_{\mathrm{orb}}(j|\mathbf{W}, V)$ is associated with a single pure state determined by the frame V, $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ describes the physical state of

the sector and may also contain coherences $[\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)]_{jj'}$ with $j \neq j'$. The isospin basis provides common coordinates for the pure orbital states and the physical density without identifying them. Neither these finite-N angular sections nor the concentration of irreducible labels establishes concentration of isospin populations as $N \to \infty$. In Appendix F, these coordinates are used to calculate projected-state overlaps and test whether families of frames span the weight space.

F Projected kernels and generation of the multiplicity space

This appendix studies the inner products of projected highest-weight orbit states and the subspaces they generate within a fixed weight space. We obtain an exact kernel from the exterior-power construction and express it in the isospin basis of Appendix E. For a finite family of frames, the Gram matrix $G = C^\dagger C$, formed from their isospin coordinates, determines the dimension of their linear span. We apply this criterion numerically to three weights of multiplicity seven and give an exact example in the adjoint representation $(1, 1)$, where two frames with the same $Q(V) = |V|^{\circ 2}$ yield orthogonal projected states. This distinction highlights phase information not determined by $Q(V)$.

F.1 Exact projected kernel

Let

$$V, W \in U(3) \tag{F.1}$$

be two frames and let

$$|\Omega_\Lambda(V)\rangle = D^\Lambda(V)\,|\Omega_\Lambda\rangle \tag{F.2}$$

be the highest-weight orbit vector defined in Appendix C. For a weight $\boldsymbol{n}$, we introduce the unnormalized projected kernel

$$\widetilde{K}_{\Lambda,\boldsymbol{n}}(V, W) := \langle\Omega_\Lambda(V)|\,P_{\Lambda,\boldsymbol{n}}\,|\Omega_\Lambda(W)\rangle\,. \tag{F.3}$$

In particular,

$$\widetilde{K}_{\Lambda,\boldsymbol{n}}(V, V) = \mathcal{N}_{\Lambda,\boldsymbol{n}}(V), \tag{F.4}$$

so the diagonal of the kernel gives the squared norm of the unnormalized projected state. Since $P_{\Lambda,\boldsymbol{n}}$ is an orthogonal projector,

$$\widetilde{K}_{\Lambda,\boldsymbol{n}}(W, V) = \overline{\widetilde{K}_{\Lambda,\boldsymbol{n}}(V, W)}, \qquad \widetilde{K}_{\Lambda,\boldsymbol{n}}(V, V) \geq 0. \tag{F.5}$$

The exterior-power construction evaluates $\widetilde{K}_{\Lambda,\boldsymbol{n}}(V, W)$ without requiring an explicit basis of $\mathcal{H}_{\Lambda,\boldsymbol{n}}$. Let

$$Z = \mathrm{diag}(z_1, z_2, z_3), \qquad z^{\mathbf{m}} = z_1^{m_1} z_2^{m_2} z_3^{m_3}. \tag{F.6}$$

The decomposition into weight spaces gives

$$\langle\Omega_\Lambda(V)|\,D^\Lambda(Z)\,|\Omega_\Lambda(W)\rangle = \sum_{\mathbf{m}} z^{\mathbf{m}}\,\langle\Omega_\Lambda(V)|\,P_{\Lambda,\mathbf{m}}\,|\Omega_\Lambda(W)\rangle\,, \tag{F.7}$$

and therefore

$$\widetilde{K}_{\Lambda,\boldsymbol{n}}(V, W) = [z^{\boldsymbol{n}}]\,\langle\Omega_\Lambda(V)|\,D^\Lambda(Z)\,|\Omega_\Lambda(W)\rangle\,. \tag{F.8}$$

We write

$$V_r = (v_1, \ldots, v_r), \qquad W_r = (w_1, \ldots, w_r), \tag{F.9}$$

and recall that $p_r = \Lambda_r - \Lambda_{r+1}$. The identity for exterior products gives

$$\langle v_1 \wedge \cdots \wedge v_r | \wedge^r Z | w_1 \wedge \cdots \wedge w_r\rangle = \det(V_r^\dagger Z W_r). \tag{F.10}$$

The different tensor factors multiply, so that

$$F_{V,W}(z) := \langle\Omega_\Lambda(V)|\,D^\Lambda(Z)\,|\Omega_\Lambda(W)\rangle = \prod_{r=1}^{3} \det(V_r^\dagger Z W_r)^{p_r}. \tag{F.11}$$

We thus obtain the exact expression

$$\widetilde{K}_{\Lambda,\boldsymbol{n}}(V, W) = [z^{\boldsymbol{n}}] \prod_{r=1}^{3} \det(V_r^\dagger Z W_r)^{p_r}. \tag{F.12}$$

The determinants can be expanded using Cauchy–Binet. We define

$$\Delta_I^{(r)}(V) := \det V_{I,[r]}, \qquad I \subset \{1,2,3\}, \qquad |I| = r. \tag{F.13}$$

Then

$$\det(V_r^\dagger Z W_r) = \sum_{\substack{I\subset\{1,2,3\}\\|I|=r}} \overline{\Delta_I^{(r)}(V)}\, \Delta_I^{(r)}(W) \prod_{i\in I} z_i. \tag{F.14}$$

The coefficient extraction in (F.12) can therefore be performed on a finite polynomial in z_1, z_2, z_3. Equivalently, the weight selection can be written through Fourier orthogonality,

$$[z^{\boldsymbol{n}}]F(z) = \int_{\mathbb{T}^3} \frac{\mathrm{d}^3\phi}{(2\pi)^3}\, \mathrm{e}^{-\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi}} F(\mathrm{e}^{\mathrm{i}\boldsymbol{\phi}}). \tag{F.15}$$

Coefficient extraction and the toroidal projector thus describe the same operation. Since all weights satisfy $m_1 + m_2 + m_3 = |\Lambda|$, one of the three phases is global and can be eliminated as in Appendix C.

F.2 Normalized kernel and Gram matrix

Whenever

$$\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = \widetilde{K}_{\Lambda,\boldsymbol{n}}(V,V) > 0, \tag{F.16}$$

the normalized projected state is

$$|\boldsymbol{n}, V; \Lambda\rangle = \frac{P_{\Lambda,\boldsymbol{n}}\,|\Omega_\Lambda(V)\rangle}{\sqrt{\widetilde{K}_{\Lambda,\boldsymbol{n}}(V,V)}}. \tag{F.17}$$

For two frames with nonzero projection, we define

$$K_{\Lambda,\boldsymbol{n}}(V,W) := \langle \boldsymbol{n}, V; \Lambda | \boldsymbol{n}, W; \Lambda\rangle\,, \tag{F.18}$$

so that

$$K_{\Lambda,\boldsymbol{n}}(V,W) = \frac{\widetilde{K}_{\Lambda,\boldsymbol{n}}(V,W)}{\sqrt{\widetilde{K}_{\Lambda,\boldsymbol{n}}(V,V)\widetilde{K}_{\Lambda,\boldsymbol{n}}(W,W)}}. \tag{F.19}$$

In particular,

$$K_{\Lambda,\boldsymbol{n}}(V,V) = 1. \tag{F.20}$$

The isospin basis constructed in Appendix E allows us to write

$$|\boldsymbol{n}, V; \Lambda\rangle = |\mathbf{W}, V\rangle = \sum_j c_j(V)\,|\mathbf{W}, j\rangle\,, \tag{F.21}$$

with

$$\sum_j |c_j(V)|^2 = 1. \tag{F.22}$$

Consequently,

$$K_{\Lambda,\boldsymbol{n}}(V,W) = \sum_j c_j(V)^* c_j(W). \tag{F.23}$$

In a fixed orthonormal isospin basis, the normalized kernel is the Hermitian product of the coordinate vectors

$$\mathbf{c}(V) = \begin{pmatrix} c_{j_{\min}}(V) \\ c_{j_{\min}+1}(V) \\ \vdots \\ c_{j_{\max}}(V) \end{pmatrix} \in \mathbb{C}^{m_\Lambda(\boldsymbol{n})}. \tag{F.24}$$

Equations (F.12) and (F.23) are therefore two representations of the same projected inner product: the first is evaluated directly from the frames, while the second expresses it in the weight-multiplicity space. Diagonal left and right rephasings, $V \mapsto D_L V D_R$, change the normalized projected state only by an overall phase. They therefore do not generate additional directions in the projective weight space.

We now fix a normalized one-particle spectrum $\boldsymbol{\lambda}$ and a normalized occupation

$$\overline{\boldsymbol{n}} = \frac{\boldsymbol{n}}{N}. \tag{F.25}$$

The frames compatible with these data satisfy

$$Q(V)\boldsymbol{\lambda} = \overline{\boldsymbol{n}}, \qquad Q_{ij}(V) = |V_{ij}|^2, \tag{F.26}$$

or, equivalently,

$$\operatorname{diag}\left[V[\boldsymbol{\lambda}]V^\dagger\right] = \overline{\boldsymbol{n}}. \tag{F.27}$$

The condition $Q(V)\boldsymbol{\lambda} = \overline{\boldsymbol{n}}$ specifies spectrum–diagonal compatibility for the prescribed spectrum $\boldsymbol{\lambda}$. Sectorial compatibility for the irrep Λ instead uses $Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}}$, where $\overline{\Lambda} = \Lambda/N$. These conditions coincide when $\boldsymbol{\lambda} = \overline{\Lambda}$, as in the examples below; otherwise they should not be identified.

We denote by

$$\mathcal{F}^*_{\Lambda,\boldsymbol{n}}(\overline{\boldsymbol{n}}, \boldsymbol{\lambda}) = \{V : Q(V)\boldsymbol{\lambda} = \overline{\boldsymbol{n}}, \quad \mathcal{N}_{\Lambda,\boldsymbol{n}}(V) > 0\} \tag{F.28}$$

the set of compatible frames whose projection onto the weight $\boldsymbol{n}$ does not vanish.

Now let

$$V_1, \ldots, V_K \in \mathcal{F}^*_{\Lambda,\boldsymbol{n}} \tag{F.29}$$

be a finite family and define

$$|\psi_a\rangle = |\boldsymbol{n}, V_a; \Lambda\rangle\,. \tag{F.30}$$

Its Gram matrix is

$$G_{ab} = \langle\psi_a|\psi_b\rangle = K_{\Lambda,\boldsymbol{n}}(V_a, V_b), \tag{F.31}$$

and satisfies

$$G^\dagger = G, \qquad G \succeq 0, \qquad G_{aa} = 1. \tag{F.32}$$

Collecting the coordinates of the states as columns,

$$C = \begin{pmatrix} | & | & & | \\ \mathbf{c}(V_1) & \mathbf{c}(V_2) & \cdots & \mathbf{c}(V_K) \\ | & | & & | \end{pmatrix} \in \mathbb{C}^{m_\Lambda(\boldsymbol{n})\times K}, \tag{F.33}$$

the identity (F.23) immediately gives

$$G = C^\dagger C. \tag{F.34}$$

The unnormalized Gram matrix can be used as well,

$$\widetilde{G}_{ab} = \widetilde{K}_{\Lambda,\boldsymbol{n}}(V_a, V_b). \tag{F.35}$$

If

$$D = \operatorname{diag}(\widetilde{G}_{11}, \ldots, \widetilde{G}_{KK}), \tag{F.36}$$

then

$$G = D^{-1/2}\widetilde{G}D^{-1/2}, \tag{F.37}$$

and, since D is invertible for the states considered,

$$\operatorname{rank} G = \operatorname{rank} \widetilde{G}. \tag{F.38}$$

F.3 Rank criterion and linear generation

The Gram matrix provides an exact criterion for determining the subspace generated by a finite family of projected states.

Proposition. Let $\{V_a\}_{a=1}^K \subset \mathcal{F}^*_{\Lambda,\boldsymbol{n}}$ and let G be the corresponding Gram matrix. Then

$$\operatorname{rank} G = \dim \operatorname{span} \{|\boldsymbol{n}, V_a; \Lambda\rangle\}_{a=1}^K \leq m_\Lambda(\boldsymbol{n}). \tag{F.39}$$

In particular,

$$\operatorname{rank} G = m_\Lambda(\boldsymbol{n}) \tag{F.40}$$

if and only if

$$\operatorname{span} \{|\boldsymbol{n}, V_a; \Lambda\rangle\}_{a=1}^K = \mathcal{H}_{\Lambda,\boldsymbol{n}}. \tag{F.41}$$

Proof. Define

$$T : \mathbb{C}^K \longrightarrow \mathcal{H}_{\Lambda,\boldsymbol{n}}, \qquad Te_a = |\psi_a\rangle\,. \tag{F.42}$$

Then

$$G = T^\dagger T. \tag{F.43}$$

Since

$$\ker(T^\dagger T) = \ker T, \tag{F.44}$$

we obtain

$$\operatorname{rank} G = \operatorname{rank} T. \tag{F.45}$$

The image of T is precisely the subspace generated by the states $|\psi_a\rangle$, which proves (F.39). The equivalence between (F.40) and (F.41) follows from $\dim \mathcal{H}_{\Lambda,\boldsymbol{n}} = m_\Lambda(\boldsymbol{n})$.

The same conclusion follows directly from (F.34),

$$\operatorname{rank} G = \operatorname{rank}(C^\dagger C) = \operatorname{rank} C. \tag{F.46}$$

Thus, the rank of C counts the linearly independent directions present in the selected family. The ranks of C and $G = C^\dagger C$ agree exactly, but finite-precision rank estimates depend on the chosen tolerances: the eigenvalues of G are the squared singular values of C. We therefore use the singular values of C as the primary numerical rank diagnostic.

The criterion concerns linear generation, which must be distinguished from two other properties. First,

$$\operatorname{rank} G = m_\Lambda(\boldsymbol{n}) \tag{F.47}$$

implies that the states considered linearly generate the full $\mathcal{H}_{\Lambda,\boldsymbol{n}}$, but not that every direction of this space can be obtained from a single frame. Linear generation of the space is not equivalent to covering all its directions through the map $V \mapsto |\boldsymbol{n}, V; \Lambda\rangle$, nor does it imply that a physical sector density $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ admits a positive mixture decomposition into these projected pure states.

Second, if a finite family gives

$$\operatorname{rank} G < m_\Lambda(\boldsymbol{n}), \tag{F.48}$$

one can only conclude that this family does not generate the full space. This result does not establish an obstruction for the total set of admissible frames.

Finally, generation and orthogonality are different conditions. If exactly $m_\Lambda(\boldsymbol{n})$ linearly independent states are selected, they form a basis, but for it to be orthonormal one further requires

$$G = I_{m_\Lambda(\boldsymbol{n})}. \tag{F.49}$$

F.4 Reference configurations

We now consider the same three configurations used in the previous appendices. In all cases we fix

$$N = 45, \qquad \Lambda = (33, 9, 3), \qquad \boldsymbol{\lambda} = \overline{\Lambda} = (11, 3, 1)/15, \tag{F.50}$$

and take

$$\boldsymbol{n}^{(1)} = (15, 15, 15), \qquad \boldsymbol{n}^{(2)} = (27, 9, 9), \qquad \boldsymbol{n}^{(3)} = (15, 10, 20). \tag{F.51}$$

The normalized occupations are

$$\overline{\boldsymbol{n}}^{(1)} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right), \qquad \overline{\boldsymbol{n}}^{(2)} = (3, 1, 1)/5, \qquad \overline{\boldsymbol{n}}^{(3)} = (3, 2, 4)/9. \tag{F.52}$$

Since $r = \Lambda_3 = 3$, the reduced weights are

$$\mathbf{W}^{(1)} = (12, 12, 12), \qquad \mathbf{W}^{(2)} = (24, 6, 6), \qquad \mathbf{W}^{(3)} = (12, 7, 17). \tag{F.53}$$

We represent the weight diagram using

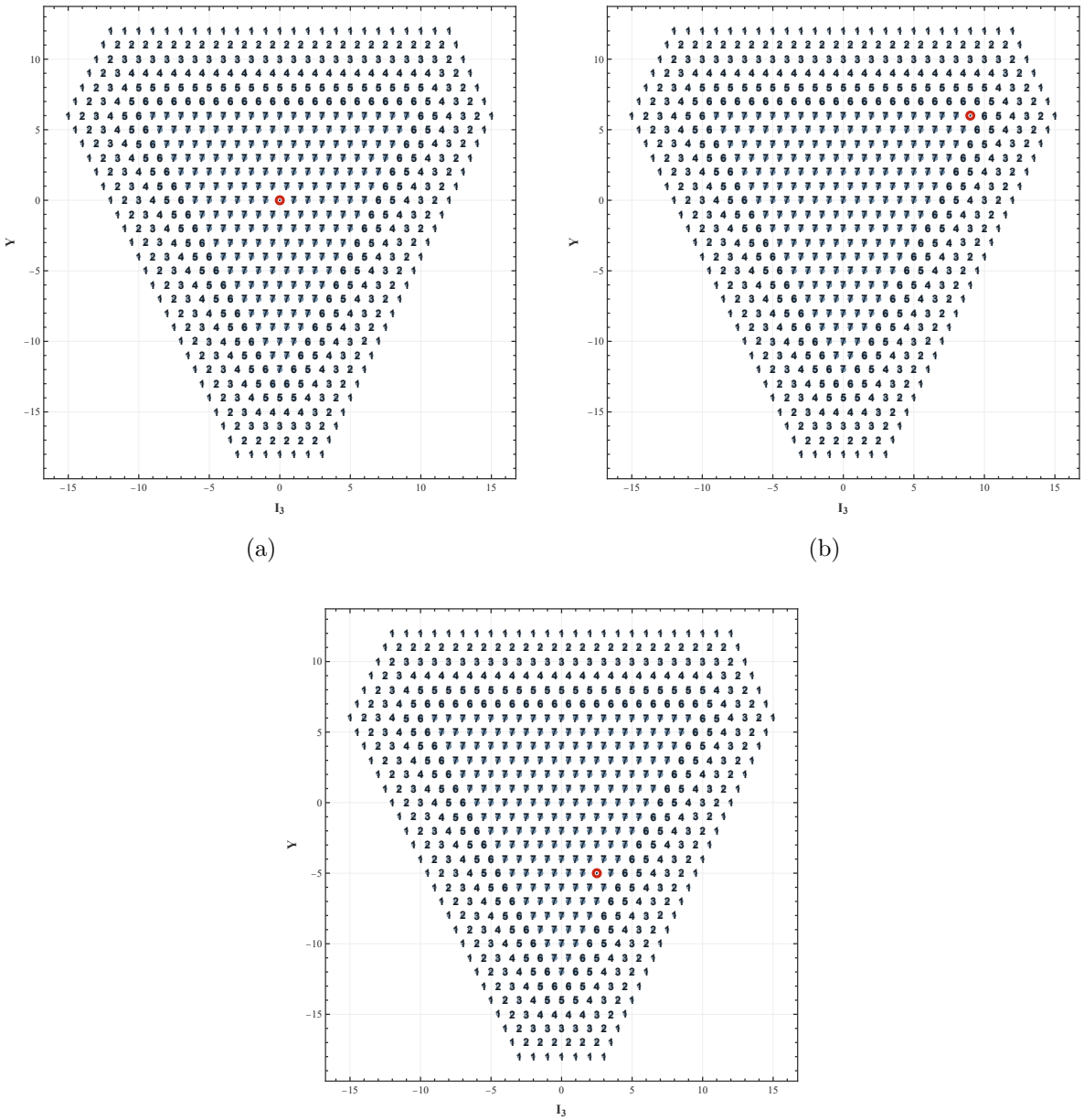
$$I_3 = \frac{W_1 - W_2}{2}, \qquad Y = \frac{W_1 + W_2 - 2W_3}{3}, \tag{F.54}$$

so the three selected weights are located at

$$(0, 0), \qquad (9, 6), \qquad \left(\frac{5}{2}, -5\right), \tag{F.55}$$

respectively.

Since the irrep $\Lambda = (33, 9, 3)$ is the same in all three cases, the complete multiplicity diagram is common to them. The different configurations, however, select different weights within that same diagram. For the rank calculations, the coordinates $\mathbf{c}(V_a)$ of the numerically generated sector-compatible frames are assembled into C. We estimate the rank from the singular values using the relative criterion $s_k > 10^{-8} s_{\max}$ and use $G = C^\dagger C$ as a consistency check. For each occupation, sampled families reach the seven-dimensional rank of the corresponding weight space. Fig. 7 identifies the three selected weights.



(a) (b)

(c)

Figure 7: Weight multiplicities for three initial occupations with $N = 45$, $\Lambda = (33, 9, 3)$, and $\boldsymbol{\lambda} = \overline{\Lambda} = (11, 3, 1)/15$. The highlighted points correspond to (a) $\mathbf{W} = (12, 12, 12)$, (b) $\mathbf{W} = (24, 6, 6)$, and (c) $\mathbf{W} = (12, 7, 17)$. Each selected weight has multiplicity $m_\Lambda(\boldsymbol{n}) = 7$.

The panels of Fig. 7 show that the three configurations select different weights of the same irrep. The balanced occupation selects the central weight $\mathbf{W} = (12, 12, 12)$, whereas the partially unbalanced and asymmetric occupations select $\mathbf{W} = (24, 6, 6)$ and $\mathbf{W} = (12, 7, 17)$, respectively. Despite their different positions in the diagram,

$$m_\Lambda(\boldsymbol{n}) = 7 \tag{F.56}$$

for all three. As shown in Appendix E, this same dimension is realized through different sets of isospin channels.

The observed rank $m_\Lambda(\boldsymbol{n}) = 7$ establishes full linear generation for the sampled families, not for every family of compatible frames. The cumulative rank depends on the selected frames and their order; it cannot be inferred from the weight diagram alone. Numerical full rank does not establish an asymptotic concentration of isospin populations.

F.5 Adjoint representation $(1,1)$

Finally, consider the adjoint irrep of $SU(3)$,

$$(p,q) = (1,1), \tag{F.57}$$

whose minimal polynomial representative of $U(3)$ is [1, 2]

$$\Lambda = (2,1,0). \tag{F.58}$$

The central weight is

$$\mathbf{W} = (1,1,1), \qquad m_{(2,1,0)}(1,1,1) = 2. \tag{F.59}$$

The multiplicity can be seen directly in the fundamental–antifundamental formulation. The central-weight sector is formed by diagonal tensors

$$X = \sum_{i=1}^{3} c_i \left|i\right\rangle \left|\bar{i}\right\rangle, \qquad c_1 + c_2 + c_3 = 0, \tag{F.60}$$

and therefore has dimension two.

Let

$$B = (\alpha, v_2, \gamma) \in SU(3). \tag{F.61}$$

The corresponding orbit vector can be written as

$$\alpha \otimes \overline{\gamma}. \tag{F.62}$$

The orthogonality between the first and third columns,

$$\gamma^{\dagger}\alpha = \sum_{i=1}^{3} \alpha_i \gamma_i^{*} = 0, \tag{F.63}$$

guarantees that this tensor belongs to the traceless subspace. Its projection onto the central weight retains only the diagonal components,

$$P_0^{SU(3)}\left(\alpha \otimes \overline{\gamma}\right) = \sum_{i=1}^{3} \alpha_i \gamma_i^{*} \left|i\right\rangle \left|\bar{i}\right\rangle. \tag{F.64}$$

Now consider the Fourier matrix

$$F_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}, \qquad \omega = \mathrm{e}^{2\pi \mathrm{i}/3}. \tag{F.65}$$

A global phase can be removed to obtain a representative in $SU(3)$ without modifying the following results.

Taking the first column as α and the third as γ, the normalized projected state is determined by

$$\left|\psi_1\right\rangle = \frac{1}{\sqrt{3}} \left(1, \omega, \omega^2\right). \tag{F.66}$$

If we interchange the second and third columns of F_3, we obtain a second frame with

$$\left|\psi_2\right\rangle = \frac{1}{\sqrt{3}} \left(1, \omega^2, \omega\right). \tag{F.67}$$

The identity

$$1 + \omega + \omega^2 = 0 \tag{F.68}$$

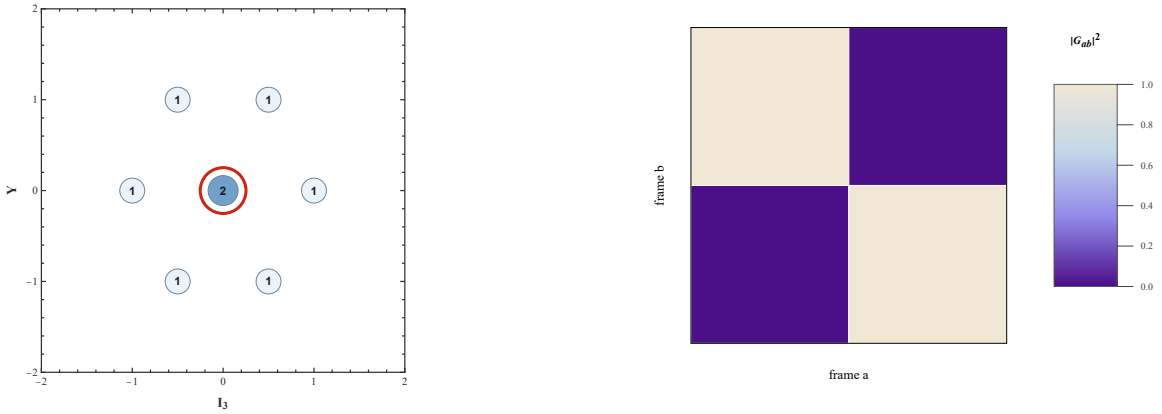
implies

$$\langle \psi_1 | \psi_2 \rangle = 0, \qquad \|\psi_1\| = \|\psi_2\| = 1. \tag{F.69}$$

The two states therefore form an orthonormal basis of the two-dimensional central-weight space. However, both frames have the same unistochastic matrix. Since all entries of F_3 have modulus $1/\sqrt{3}$,

$$Q(F_3) = |F_3|^{\circ 2} = \frac{1}{3}\mathbf{1}\mathbf{1}^T. \tag{F.70}$$

Permuting its second and third columns does not modify this matrix.



(a) (b)

Figure 8: Adjoint representation for $\Lambda = (2, 1, 0)$ and $\mathbf{W} = (1, 1, 1)$. (a) Weight diagram in the (I_3, Y) plane. The highlighted central weight has multiplicity $m_{(2,1,0)}(1, 1, 1) = 2$. (b) Fidelity matrix $|G_{ab}|^2 = |\langle \psi_a | \psi_b \rangle|^2$ for the two projected states obtained from the Fourier frame F_3 and from the frame obtained by interchanging its second and third columns.

Fig. 8(a) identifies the central weight of multiplicity two, while Fig. 8(b) shows

$$\left(|G_{ab}|^2\right)_{a,b=1}^{2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{F.71}$$

in agreement with (F.69). The two states therefore form an orthonormal basis of the central-weight space, not of the full adjoint representation.

Although the frames have the same $Q(V)$ and satisfy the same spectrum–diagonal condition, their projected states are orthogonal. Thus the squared moduli of the frame entries do not determine the projected state: relative phases can distinguish independent directions. Orthogonality alone does not establish different isospin populations $|c_j(V)|^2$ for the two states. This case makes explicit a general property of the construction developed in this appendix: the projected kernels retain information about the frame that is not contained in $Q(V)$, and the Gram matrix allows one to determine how that information is reflected in the inner products and in the dimension of the generated subspace.

This analysis concerns projected pure states $|\boldsymbol{n}, V; \Lambda\rangle$ and their linear relations. The generally mixed physical density $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ instead encodes partial distinguishability in the conditional Born rule and must not be identified with a single frame state. The following appendix studies its populations, coherences, and contribution to physical detection probabilities.

G Physical multiplicity density and detection probabilities

This appendix gives an exact description of the physical state and detection probabilities within each irreducible sector, using the isospin basis of Appendix E. We distinguish the pure state obtained by projecting a highest-weight orbit from the physical density associated with the internal-state Gram matrix Γ [3, 4, 7]. Both act on the same weight space, but they need not coincide. We express the physical density in terms of the auxiliary single-particle matrix ρ and identify the highest-weight contribution in a decomposition induced by the spectral resolution of $D^{\Lambda}(\rho)$. For $\Gamma > 0$, the density has full rank on any nonempty weight space, whereas a pure frame state determines a rank-one projector. We then derive the conditioned detection probabilities and the statistical weights of the irreducible sectors, and compare the resulting distributions in the three reference configurations.

G.1 Physical density in the multiplicity space

We start from the effective single-particle matrix

$$\rho = [\overline{\boldsymbol{n}}]^{1/2}\Gamma[\overline{\boldsymbol{n}}]^{1/2}, \qquad \Gamma \succeq 0, \qquad \Gamma_{ii} = 1. \tag{G.1}$$

It follows from Appendix A that, for an irrep Λ and an initial weight $\boldsymbol{n}$, the physical density on the weight-multiplicity space $\mathcal{H}_{\Lambda,\boldsymbol{n}}$, distinct from the Schur–Weyl multiplicity factor $\mathcal{M}_{\Lambda}$, is

$$\sigma_{\Lambda,\boldsymbol{n}}(\Gamma) = \frac{P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(\Gamma) P_{\Lambda,\boldsymbol{n}}}{\operatorname{Tr}\left[P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(\Gamma) P_{\Lambda,\boldsymbol{n}}\right]}, \tag{G.2}$$

provided that the denominator is nonzero. This operator acts on

$$\mathcal{H}_{\Lambda,\boldsymbol{n}}, \qquad \dim \mathcal{H}_{\Lambda,\boldsymbol{n}} = m_{\Lambda}(\boldsymbol{n}), \tag{G.3}$$

is positive semidefinite, and has unit trace.

The same density can be evaluated directly from ρ. Since

$$\rho = S\Gamma S, \qquad S = [\overline{\boldsymbol{n}}]^{1/2}, \tag{G.4}$$

the action of $D^{\Lambda}(S)$ on $\mathcal{H}_{\Lambda,\boldsymbol{n}}$ is multiplication by $\prod_i (\overline{n}_i)^{n_i/2}$. The corresponding positive scalar factors cancel after normalization, giving

$$\sigma_{\Lambda,\boldsymbol{n}}(\Gamma) = \frac{P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(\rho) P_{\Lambda,\boldsymbol{n}}}{\operatorname{Tr}\left[P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(\rho) P_{\Lambda,\boldsymbol{n}}\right]}. \tag{G.5}$$

If any component of $\overline{\boldsymbol{n}}$ vanishes, the construction is restricted to the modes with nonzero initial occupation, as discussed in Appendix A.

G.2 Pure frame state and decomposition of the density

We diagonalize the effective single-particle matrix as

$$\rho = V[\boldsymbol{\lambda}]V^{\dagger}, \qquad [\boldsymbol{\lambda}] = \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3). \tag{G.6}$$

The condition

$$\operatorname{diag}\rho = \overline{\boldsymbol{n}} \tag{G.7}$$

is equivalent to

$$Q(V)\boldsymbol{\lambda} = \overline{\boldsymbol{n}}, \qquad Q_{ij}(V) = |V_{ij}|^2. \tag{G.8}$$

A compatible frame then determines

$$\Gamma(V) = [\overline{\boldsymbol{n}}]^{-1/2} V[\boldsymbol{\lambda}]V^{\dagger}[\overline{\boldsymbol{n}}]^{-1/2}, \tag{G.9}$$

whenever the inverse is defined on the modes under consideration.
The same frame also defines a normalized projected pure state, provided $\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) > 0$:

$$|\boldsymbol{n}, V; \Lambda\rangle = \frac{P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(V)\,|\Omega_{\Lambda}\rangle}{\sqrt{\mathcal{N}_{\Lambda,\boldsymbol{n}}(V)}}. \tag{G.10}$$

We thus have two different maps,

$$V \longmapsto \Gamma(V), \qquad V \longmapsto |\boldsymbol{n}, V; \Lambda\rangle\,, \tag{G.11}$$

and the existence of both does not imply that

$$\sigma_{\Lambda,\boldsymbol{n}}(\Gamma(V)) = |\boldsymbol{n}, V; \Lambda\rangle\,\langle \boldsymbol{n}, V; \Lambda|\,. \tag{G.12}$$

In general,

$$\sigma_{\Lambda,\boldsymbol{n}}(\Gamma(V)) \neq |\boldsymbol{n}, V; \Lambda\rangle\,\langle \boldsymbol{n}, V; \Lambda|\,. \tag{G.13}$$

The difference can be seen directly from the spectral resolution of $D^{\Lambda}(\rho)$. For a vector

$$|\Lambda, \mathbf{m}, a\rangle \in \mathcal{H}_{\Lambda,\mathbf{m}}, \tag{G.14}$$

the diagonal matrix $[\boldsymbol{\lambda}]$ acts as [1, 2]

$$D^{\Lambda}([\boldsymbol{\lambda}])\,|\Lambda, \mathbf{m}, a\rangle = \boldsymbol{\lambda}^{\mathbf{m}}\,|\Lambda, \mathbf{m}, a\rangle\,, \qquad \boldsymbol{\lambda}^{\mathbf{m}} = \prod_i \lambda_i^{m_i}. \tag{G.15}$$

Therefore,

$$D^{\Lambda}([\boldsymbol{\lambda}]) = \sum_{\mathbf{m}} \boldsymbol{\lambda}^{\mathbf{m}} P_{\Lambda,\mathbf{m}}. \tag{G.16}$$

Substituting this resolution into $D^{\Lambda}(\rho)$,

$$P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(\rho) P_{\Lambda,\boldsymbol{n}} = \sum_{\mathbf{m}} \boldsymbol{\lambda}^{\mathbf{m}} P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(V) P_{\Lambda,\mathbf{m}} D^{\Lambda}(V^{\dagger}) P_{\Lambda,\boldsymbol{n}}. \tag{G.17}$$

Let

$$P_{\Lambda,\mathbf{m}} = \sum_{a=1}^{m_{\Lambda}(\mathbf{m})} |\Lambda, \mathbf{m}, a\rangle\,\langle \Lambda, \mathbf{m}, a|\,, \tag{G.18}$$

and define

$$|\widetilde{\psi}_{\mathbf{m},a}(V)\rangle := P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(V)\,|\Lambda, \mathbf{m}, a\rangle\,. \tag{G.19}$$

Then

$$P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(\rho) P_{\Lambda,\boldsymbol{n}} = \sum_{\mathbf{m},a} \boldsymbol{\lambda}^{\mathbf{m}}\,|\widetilde{\psi}_{\mathbf{m},a}(V)\rangle\,\langle \widetilde{\psi}_{\mathbf{m},a}(V)|\,. \tag{G.20}$$

The physical density is thus obtained by normalizing a positive sum of contributions from the weight spaces of the representation. This sum is not, in general, the spectral decomposition of the restricted density. It also does not express the density as a positive mixture of highest-weight projected frame states alone. A given contribution may vanish if its spectral coefficient or projected vector is zero. The term associated with the highest weight $\mathbf{m} = \Lambda$ is

$$\boldsymbol{\lambda}^{\Lambda}\,|\widetilde{\psi}_{\mathrm{hw}}(V)\rangle\,\langle \widetilde{\psi}_{\mathrm{hw}}(V)|\,, \qquad |\widetilde{\psi}_{\mathrm{hw}}(V)\rangle = P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(V)\,|\Omega_{\Lambda}\rangle\,. \tag{G.21}$$

Normalizing this vector recovers precisely $|\boldsymbol{n}, V; \Lambda\rangle$. The pure frame state therefore corresponds to the highest-weight contribution within (G.20), but does not in general represent the full physical density. Nor does this exact decomposition establish asymptotic dominance of the highest-weight term.

This observation also clarifies the role of possible degeneracies of ρ. The same matrix ρ may admit

different choices of eigenvectors within a degenerate subspace and, consequently, different pure states associated with those choices. The physical density, by contrast, depends on ρ or, equivalently, on Γ, and not on a particular choice of eigenbasis.

G.3 Physical density in the isospin basis

For

$$\Lambda = (p+q+r, q+r, r), \qquad \mathbf{W} = \boldsymbol{n} - r\mathbf{1}, \tag{G.22}$$

we use the orthonormal basis

$$\{|\mathbf{W}, j\rangle\}_{j \in J_{\mathbf{W}}} \tag{G.23}$$

constructed in Appendix E. We write $\sigma_{\mathbf{W}}(\Gamma) \equiv \sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ with $\mathbf{W} = \boldsymbol{n} - r\mathbf{1}$ when the sector density is defined. In this basis,

$$[\sigma_{\mathbf{W}}(\Gamma)]_{jj'} = \frac{\langle \mathbf{W}, j|\, D^{(p,q)}_{\mathrm{pol}}(\Gamma)\, |\mathbf{W}, j'\rangle}{\sum_{k \in J_{\mathbf{W}}} \langle \mathbf{W}, k|\, D^{(p,q)}_{\mathrm{pol}}(\Gamma)\, |\mathbf{W}, k\rangle}. \tag{G.24}$$

The diagonal defines the physical populations

$$P_{\mathrm{phys}}(j|\mathbf{W}, \Gamma) = [\sigma_{\mathbf{W}}(\Gamma)]_{jj}, \tag{G.25}$$

with

$$P_{\mathrm{phys}}(j|\mathbf{W}, \Gamma) \geq 0, \qquad \sum_j P_{\mathrm{phys}}(j|\mathbf{W}, \Gamma) = 1. \tag{G.26}$$

The decomposition (G.20) provides an equivalent expression. Defining

$$D_{\mathbf{W}j,\mathbf{m}a}(V) := \langle \mathbf{W}, j|\, D^{\Lambda}(V)\, |\Lambda, \mathbf{m}, a\rangle\,, \tag{G.27}$$

the unnormalized block is

$$\widetilde{\sigma}_{jj'}(V) = \sum_{\mathbf{m},a} \boldsymbol{\lambda}^{\mathbf{m}} D_{\mathbf{W}j,\mathbf{m}a}(V) D_{\mathbf{W}j',\mathbf{m}a}(V)^*, \tag{G.28}$$

and

$$\sigma_{jj'}(V) = \frac{\widetilde{\sigma}_{jj'}(V)}{\sum_k \widetilde{\sigma}_{kk}(V)}. \tag{G.29}$$

For the highest-weight term, we define the matrix element

$$\mathcal{A}^{\mathrm{hw}}_j(V) := \langle \mathbf{W}, j|\, D^{\Lambda}(V)\, |\Omega_\Lambda\rangle\,, \tag{G.30}$$

whose contribution to the unnormalized density is

$$\boldsymbol{\lambda}^{\Lambda} \mathcal{A}^{\mathrm{hw}}_j(V) \mathcal{A}^{\mathrm{hw}}_{j'}(V)^*. \tag{G.31}$$

This notation distinguishes the matrix element from the coefficient $\mathcal{B}_j(V)$ of Appendix E, where

$$\mathcal{A}^{\mathrm{hw}}_j(V) = \sqrt{\frac{p!q!}{(p+q)!}}\, \mathcal{B}_j(V). \tag{G.32}$$

Either set of amplitudes determines the normalized coordinates $c_j(V)$ of the projected pure state.

Proposition. If $\Gamma > 0$ and the sector $\mathcal{H}_{\Lambda,\boldsymbol{n}}$ is nonempty, then

$$\operatorname{rank} \sigma_{\Lambda,\boldsymbol{n}}(\Gamma) = m_\Lambda(\boldsymbol{n}). \tag{G.33}$$

Proof. Since $\Gamma > 0$, there exists an invertible matrix A such that

$$\Gamma = A^\dagger A. \tag{G.34}$$

The representation satisfies

$$D^\Lambda(\Gamma) = D^\Lambda(A)^\dagger D^\Lambda(A) > 0. \tag{G.35}$$

Its restriction to any nonzero subspace is therefore positive definite. After trace normalization, the density $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$ retains the same rank and, consequently, has rank $m_\Lambda(\boldsymbol{n})$. □

For $\Gamma > 0$ and $m_\Lambda(\boldsymbol{n}) > 1$, the proposition implies that the physical density cannot coincide with a rank-one projector,

$$\sigma_{\Lambda,\boldsymbol{n}}(\Gamma(V)) \neq |\mathbf{W}, V\rangle \langle \mathbf{W}, V| . \tag{G.36}$$

The difference is also reflected in the isospin populations. The pure state determines

$$P_{\text{orb}}(j|\mathbf{W}, V) = |c_j(V)|^2, \tag{G.37}$$

whereas the physical density determines

$$P_{\text{phys}}(j|\mathbf{W}, \Gamma) = \langle \mathbf{W}, j| \, \sigma_{\mathbf{W}}(\Gamma) \, |\mathbf{W}, j\rangle . \tag{G.38}$$

In general,

$$P_{\text{orb}} \neq P_{\text{phys}}. \tag{G.39}$$

In addition to its diagonal populations, $\sigma_{\mathbf{W}}$ may contain coherences $\sigma_{jj'}$ for $j \neq j'$, subject to

$$|\sigma_{jj'}|^2 \leq \sigma_{jj}\sigma_{j'j'}. \tag{G.40}$$

Consequently, the comparison between P_{orb} and P_{phys} does not exhaust the difference between the pure state and the physical density.

The rank criterion of Appendix F allows a family of projected states to be used as a generally nonorthogonal basis of the multiplicity space. In that case, any operator $\sigma_{\Lambda,\boldsymbol{n}}$ can be expanded in that basis and its dual basis. This representation is algebraically exact, but its coefficients are neither probabilities nor weights of a physical mixture of frame states.

G.4 Transitions between weights and the Born rule

Let U be the interferometer. Conditioned on an irrep Λ with nonzero statistical weight, the probability of detecting the occupation $\boldsymbol{n}'$ is [7]

$$P(\boldsymbol{n}'|\boldsymbol{n}, \Lambda, U, \Gamma) = \text{Tr}\left[P_{\Lambda,\boldsymbol{n}'} D^\Lambda(U) \sigma_{\Lambda,\boldsymbol{n}}(\Gamma) D^\Lambda(U^\dagger)\right]. \tag{G.41}$$

In $SU(3)$ we define

$$\mathbf{W} = \boldsymbol{n} - r\mathbf{1}, \qquad \mathbf{W}' = \boldsymbol{n}' - r\mathbf{1}, \tag{G.42}$$

and use the bases

$$\{|\mathbf{W}, j\rangle\}_{j \in J_{\mathbf{W}}}, \qquad \{|\mathbf{W}', J\rangle\}_{J \in J_{\mathbf{W}'}}. \tag{G.43}$$

The transition block between the two weights is

$$\mathcal{D}^{(p,q)}_{Jj}(\mathbf{W}', \mathbf{W}; U) := \langle \mathbf{W}', J| \, D^{(p,q)}(U) \, |\mathbf{W}, j\rangle . \tag{G.44}$$

The Born rule then takes the form

$$P(\mathbf{W}'|\mathbf{W}, (p,q), U, \Gamma) = \sum_J \sum_{j,j'} \mathcal{D}^{(p,q)}_{Jj} \sigma_{jj'} \mathcal{D}^{(p,q)*}_{Jj'}. \tag{G.45}$$

The sum over J appears because detection resolves the final weight, but not the internal multiplicity index.

Defining the rectangular matrix

$$T^{\mathbf{W}'\leftarrow\mathbf{W}}(U) = \left[\mathcal{D}^{(p,q)}_{Jj}(\mathbf{W}', \mathbf{W}; U)\right]_{Jj}, \tag{G.46}$$

we obtain the compact expression for the sector-conditioned physical probability (not yet the full distribution)

$$P_{\text{phys}}(\mathbf{W}') = \text{Tr}\left[T^{\mathbf{W}'\leftarrow\mathbf{W}}\sigma_{\mathbf{W}}T^{\mathbf{W}'\leftarrow\mathbf{W}\dagger}\right]. \tag{G.47}$$

If we replace the physical density by the projector associated with a pure frame state,

$$\sigma_{\text{orb}} = |\mathbf{W}, V\rangle\langle\mathbf{W}, V|, \tag{G.48}$$

then

$$[\sigma_{\text{orb}}]_{jj'} = c_j(V)c_{j'}(V)^*, \tag{G.49}$$

and

$$P_{\text{orb}}(\mathbf{W}') = \|T^{\mathbf{W}'\leftarrow\mathbf{W}}(U)\mathbf{c}(V)\|^2. \tag{G.50}$$

Equations (G.47) and (G.50) use the same transition block. The difference between the two probabilities originates exclusively from the internal object propagated through the interferometer.

G.5 Total probability and statistical weight of the irrep

The previous probability is conditioned on a fixed irrep. To obtain the complete physical distribution, we include the statistical weight of each sector (which is not contained in the normalized density),

$$P(\Lambda|\boldsymbol{n}, \Gamma) = \frac{d_{[\Lambda]}}{C_{\boldsymbol{n}}}\,\text{Tr}\left[P_{\Lambda,\boldsymbol{n}}D^{\Lambda}(\Gamma)P_{\Lambda,\boldsymbol{n}}\right]. \tag{G.51}$$

The total output distribution is

$$P(\boldsymbol{n}'|\boldsymbol{n}, U, \Gamma) = \sum_{\Lambda} P(\Lambda|\boldsymbol{n}, \Gamma)P(\boldsymbol{n}'|\boldsymbol{n}, \Lambda, U, \Gamma). \tag{G.52}$$

This expression separates two different levels: the weight with which each irrep participates in the initial state and the internal dynamics within the selected sector.

G.6 Numerical results

We now consider the three reference configurations used in the previous appendices. In all cases we fix

$$\boldsymbol{\lambda} = \frac{1}{15}(11, 3, 1), \qquad N = 45, \qquad \Lambda = (33, 9, 3), \tag{G.53}$$

so that

$$(p, q, r) = (24, 6, 3). \tag{G.54}$$

The normalized occupations are

$$\overline{\boldsymbol{n}}^{(1)} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right), \qquad \overline{\boldsymbol{n}}^{(2)} = \frac{1}{5}(3, 1, 1), \qquad \overline{\boldsymbol{n}}^{(3)} = \frac{1}{9}(3, 2, 4), \tag{G.55}$$

corresponding to

$$\boldsymbol{n}^{(1)} = (15, 15, 15), \qquad \boldsymbol{n}^{(2)} = (27, 9, 9), \qquad \boldsymbol{n}^{(3)} = (15, 10, 20). \tag{G.56}$$

The reduced weights are

$$\mathbf{W}^{(1)} = (12, 12, 12), \qquad \mathbf{W}^{(2)} = (24, 6, 6), \qquad \mathbf{W}^{(3)} = (12, 7, 17). \tag{G.57}$$

In all three cases,

$$m_\Lambda(\boldsymbol{n}^{(k)}) = 7, \qquad k = 1, 2, 3. \tag{G.58}$$

The numerical illustrations use the exact expressions derived above. The supplementary code provides routines for constructing $\sigma_{\mathbf{W}}$ and the transition blocks $T^{\mathbf{W}'\leftarrow\mathbf{W}}(U)$, computing conditioned probabilities and sector weights, and summing the irreducible-sector contributions. The figures use the associated frame banks and exported datasets; reproducing each panel requires identifying its frame and preparation data.

When comparing the physical density with a projected pure state, we use in each configuration the frame labeled V_1 in the corresponding bank. This label denotes the first selected frame in each bank, not the same unitary matrix across configurations. The spectrum $\boldsymbol{\lambda}$, N, the irrep and, for the output probabilities, the interferometer $U = F_3$ remain fixed; the initial occupation and its associated compatible frame change.

G.6.1 Physical density and isospin distribution

For each configuration, $\sigma_{\mathbf{W}}$ is a seven-dimensional matrix in the corresponding isospin basis. Its diagonal elements determine

$$P_{\mathrm{phys}}(j) = \sigma_{jj}, \tag{G.59}$$

whereas the pure state associated with V_1 determines

$$P_{\mathrm{orb}}(j) = |c_j(V_1)|^2. \tag{G.60}$$

We define

$$\Delta P_j = P_{\mathrm{phys}}(j) - P_{\mathrm{orb}}(j). \tag{G.61}$$

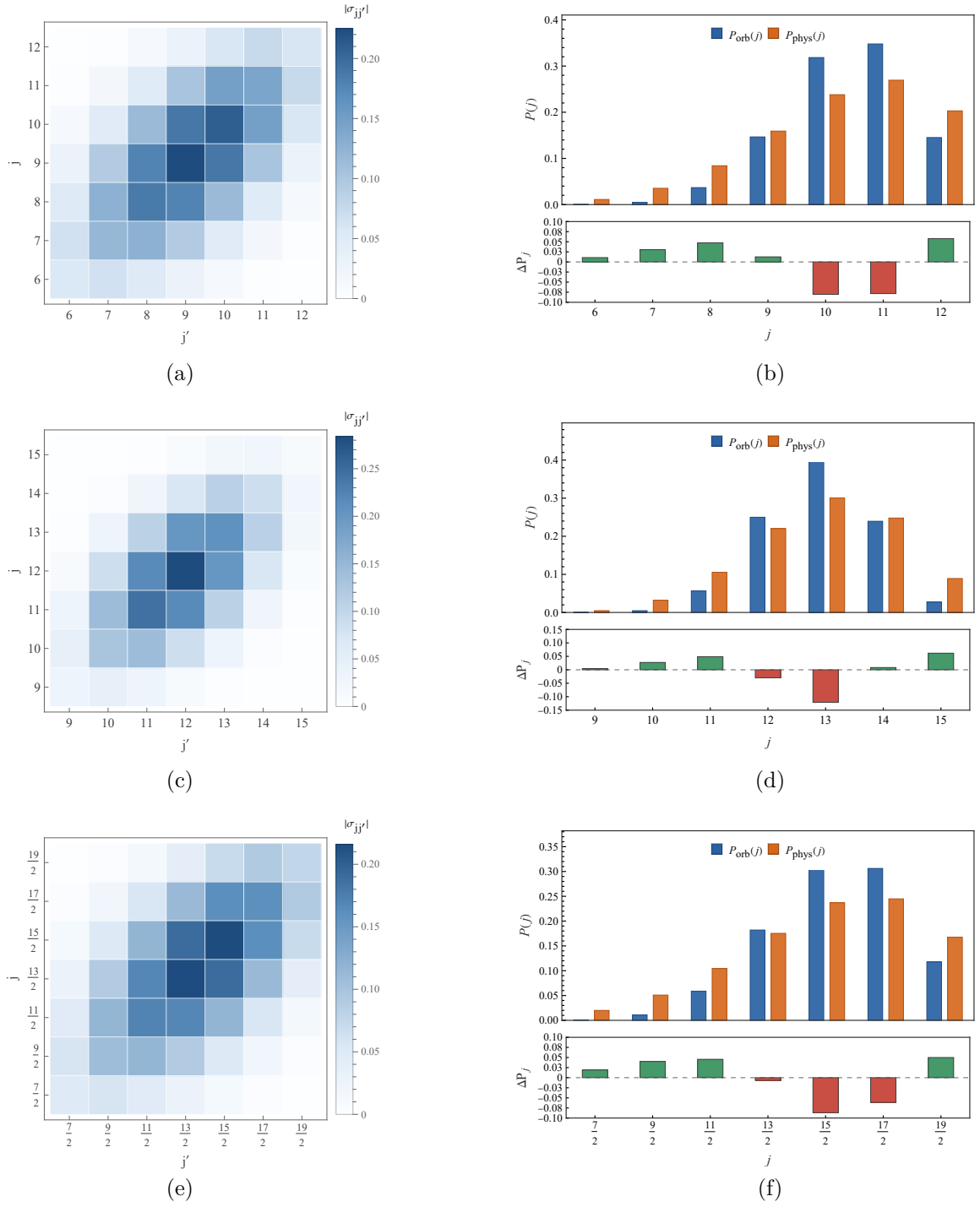


Figure 9: Physical density and isospin distributions for three initial occupations, with $\boldsymbol{\lambda} = (0.7333, 0.2000, 0.0667)$, $N = 45$, $\Lambda = (33, 9, 3)$, and the reference frame V_1 of each configuration. (a,b) Balanced configuration $\overline{\boldsymbol{n}} = (1/3, 1/3, 1/3)$. (c,d) Partially unbalanced configuration $\overline{\boldsymbol{n}} = (0.6, 0.2, 0.2)$. (e,f) Asymmetric configuration $\overline{\boldsymbol{n}} = (0.3333, 0.2222, 0.4444)$. The left column shows the moduli $|\sigma_{jj'}|$ of the physical density matrix in the isospin basis. The right column compares $P_{\mathrm{orb}}(j)$ and $P_{\mathrm{phys}}(j)$ and shows $\Delta P_j = P_{\mathrm{phys}}(j) - P_{\mathrm{orb}}(j)$.

Fig. 9 shows, for all three configurations, nonzero off-diagonal elements of $\sigma_{\mathbf{W}}$. The physical density therefore contains coherences between different isospin states and is not characterized solely by the populations $P_{\mathrm{phys}}(j)$.

Parts (b), (d), and (f) further show that these populations differ from those of the projected pure state. In the balanced case, the orbital distribution is concentrated mainly around the central channels, and the physical density transfers part of this weight toward other values of j. An analogous behavior

appears in the partially unbalanced configuration, although over the isospin interval corresponding to $\mathbf{W} = (24, 6, 6)$. For the asymmetric weight, the same distinction appears in a basis formed by half-integer channels.

In all three cases, the physical density differs from the pure-state projector both in its populations and in its off-diagonal matrix elements. The plots establish the presence of coherences but do not isolate their individual contribution to the detection probabilities; doing so would require separating the diagonal and off-diagonal terms in the Born rule.

G.6.2 Detection probabilities after the interferometer

We now consider how this internal difference manifests itself after the Fourier interferometer $U = F_3$. For a final occupation $\boldsymbol{n}'$, the pure state determines

$$P_{\mathrm{orb}}(\boldsymbol{n}') = \|T^{\mathbf{W}'\leftarrow\mathbf{W}}(F_3)\mathbf{c}(V_1)\|^2, \tag{G.62}$$

whereas the physical density determines

$$P_{\mathrm{phys}}(\boldsymbol{n}') = \mathrm{Tr}\left[T^{\mathbf{W}'\leftarrow\mathbf{W}}(F_3)\sigma_{\mathbf{W}}T^{\mathbf{W}'\leftarrow\mathbf{W}}(F_3)^\dagger\right]. \tag{G.63}$$

We define

$$\Delta P(\boldsymbol{n}') = P_{\mathrm{phys}}(\boldsymbol{n}') - P_{\mathrm{orb}}(\boldsymbol{n}'). \tag{G.64}$$

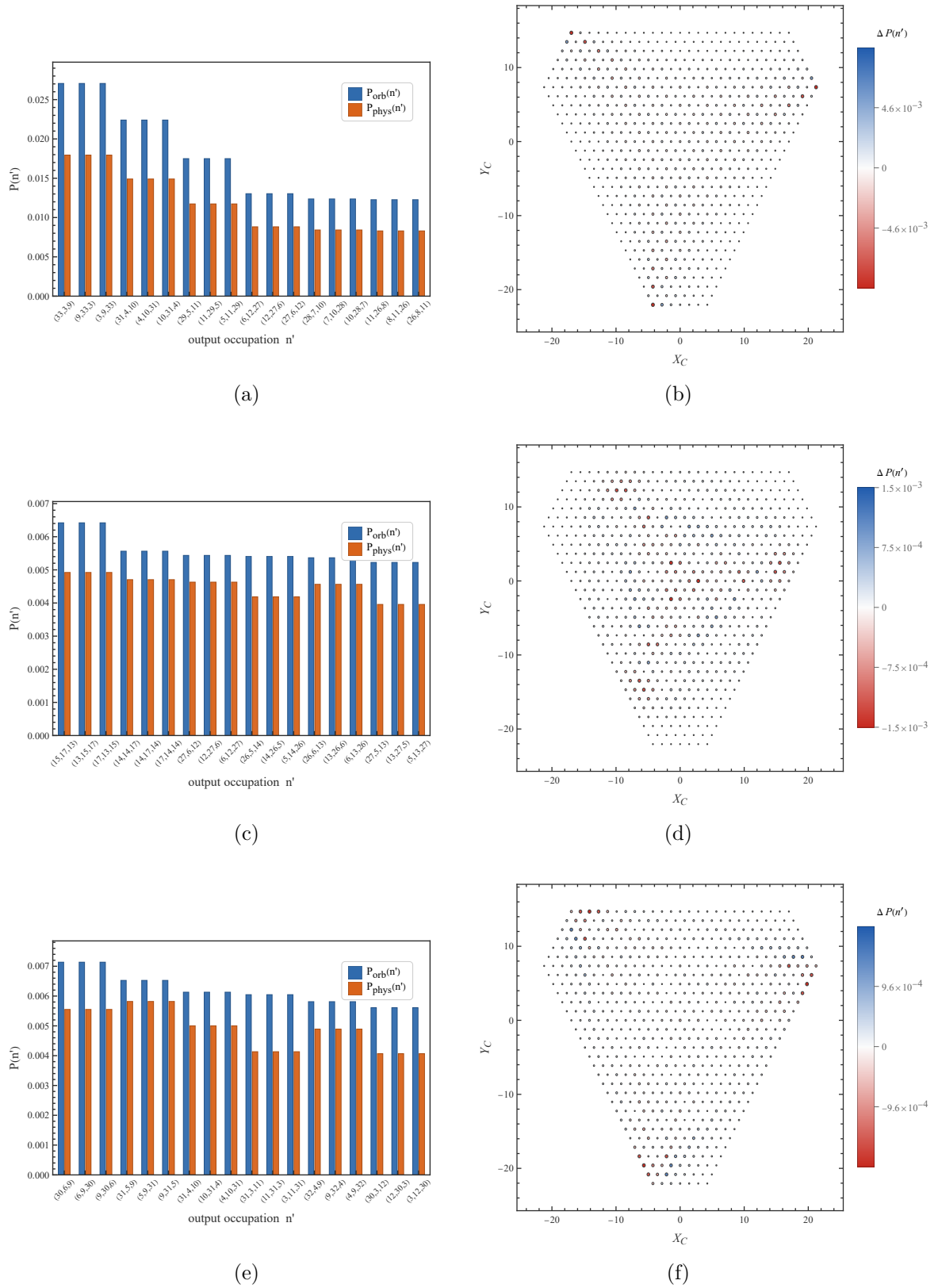


Figure 10: Detection probabilities after the Fourier interferometer $U = F_3$ for three initial occupations, with $\boldsymbol{\lambda} = (0.7333, 0.2000, 0.0667)$, $N = 45$, $\Lambda = (33, 9, 3)$, and the reference frame V_1 of each configuration. (a,b) Balanced configuration $\overline{\boldsymbol{n}} = (1/3, 1/3, 1/3)$. (c,d) Partially unbalanced configuration $\overline{\boldsymbol{n}} = (0.6, 0.2, 0.2)$. (e,f) Asymmetric configuration $\overline{\boldsymbol{n}} = (0.3333, 0.2222, 0.4444)$. The left column compares $P_{\text{orb}}(\boldsymbol{n}')$ and $P_{\text{phys}}(\boldsymbol{n}')$. The right column shows $\Delta P(\boldsymbol{n}') = P_{\text{phys}}(\boldsymbol{n}') - P_{\text{orb}}(\boldsymbol{n}')$ over the output-weight diagram.

Fig. 10 shows that the internal difference between the two descriptions remains after the interferometric transformation. Several occupations with high orbital probability lose weight when the physical density is used, whereas others receive positive corrections.

This redistribution does not alter the normalization. Both P_{orb} and P_{phys} sum to one over the complete set of final weights, so the positive and negative corrections of $\Delta P(\boldsymbol{n}')$ compensate globally. The maps in parts (b), (d), and (f) show that this compensation is distributed over multiple output weights.

For the configurations shown, the maximum scales are approximately

$$|\Delta P| \simeq 4.6 \times 10^{-3}, \qquad 1.5 \times 10^{-3}, \qquad 9.6 \times 10^{-4}, \tag{G.65}$$

for the balanced, partially unbalanced, and asymmetric cases, respectively. This sequence pertains only to the three configurations studied and does not establish a general dependence between occupation imbalance and the magnitude of the correction.

The spatial structure of ΔP is more informative than this comparison of scales. According to (G.63), the physical probability depends on the full matrix $\sigma_{\mathbf{W}}$. Therefore, both its populations and its coherences may contribute to the output statistics. The interferometric evolution thereby makes differences between the two internal states visible in the detection probabilities. The plots do not, by themselves, separate the effects of populations from those of coherences.

G.6.3 Sectorial contribution and total output distribution

Up to this point we have considered the dynamics conditioned on $\Lambda = (33, 9, 3)$. To incorporate the statistical weight of this irrep, we define

$$P_\Lambda(\boldsymbol{n}') := P(\Lambda|\boldsymbol{n}, \Gamma) P(\boldsymbol{n}'|\boldsymbol{n}, \Lambda, U, \Gamma) = P(\Lambda, \boldsymbol{n}'). \tag{G.66}$$

The total distribution is obtained by summing over all sectors,

$$P_T(\boldsymbol{n}') := \sum_\Lambda P_\Lambda(\boldsymbol{n}') = P(\boldsymbol{n}'|\boldsymbol{n}, U, \Gamma). \tag{G.67}$$

The quantity $P_\Lambda(\boldsymbol{n}')$ is therefore a joint probability and should not be confused with the conditioned probability $P(\boldsymbol{n}'|\boldsymbol{n}, \Lambda, U, \Gamma)$.

Figs. 11–13 compare this sectorial contribution with the total distribution for the three configurations.

Parts (a,b) of Figs. 11–13 show that the contribution of the selected irrep is restricted to the weights of $\Lambda = (33, 9, 3)$. Although the representation is common to all three cases, the distribution $P_\Lambda(\boldsymbol{n}')$ changes with the initial occupation and with the physical density compatible with it.

Parts (c,d) correspond to a different quantity. After summing over all irreps, the distribution extends over the full simplex of output occupations. Therefore, $P_T(\boldsymbol{n}')$ is not a simple rescaling of $P_\Lambda(\boldsymbol{n}')$: the other sectors contribute probability outside the diagram of the selected irrep and also modify the distribution within the simplex.

The distributions exhibit different patterns for the three preparations. The balanced case has a comparatively symmetric appearance, whereas the other two cases show different spatial structures. The equality of two components of the initial occupation in the second configuration does not by itself imply an exact symmetry of Γ or of the selected frame. These observations concern the numerical examples shown, rather than a general ordering of the preparations by asymmetry.

Taken together, the three numerical comparisons follow the same sequence:

$$\sigma_{\Lambda,\boldsymbol{n}} \longrightarrow P(\boldsymbol{n}'|\boldsymbol{n}, \Lambda, U, \Gamma) \longrightarrow P_\Lambda(\boldsymbol{n}') \longrightarrow P_T(\boldsymbol{n}'). \tag{G.68}$$

The first object describes the internal structure of the multiplicity space; the second, the dynamics conditioned on the irrep; the third incorporates the statistical weight of the sector; and the last collects the contributions from all irreducible representations.

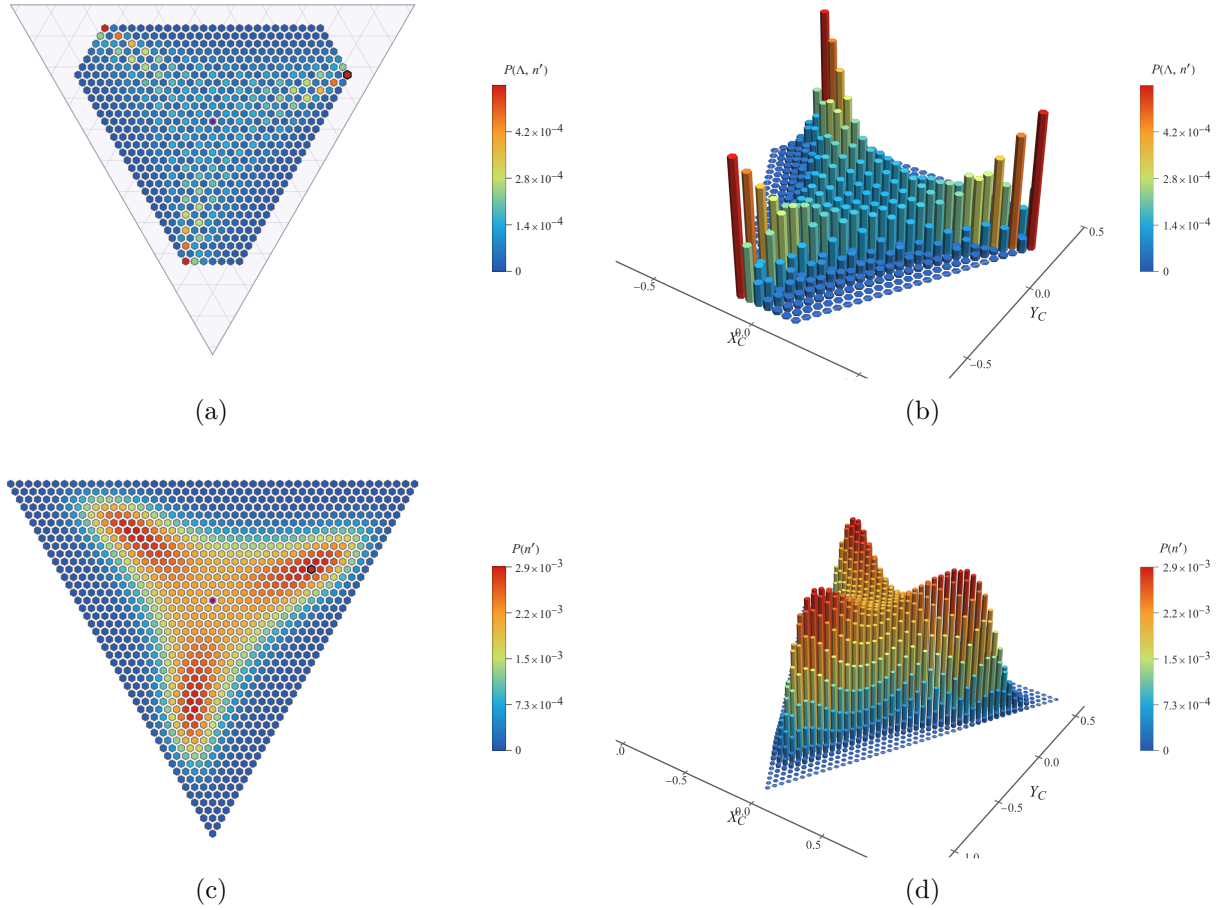


Figure 11: Sectorial contribution and total physical output distribution for the balanced configuration $\overline{\boldsymbol{n}} = (1/3, 1/3, 1/3)$, with $\boldsymbol{\lambda} = (0.7333, 0.2000, 0.0667)$, $N = 45$, and $\Lambda = (33, 9, 3)$. (a) Joint sectorial probability $P(\Lambda, \boldsymbol{n}') = P_\Lambda(\boldsymbol{n}')$. (b) Three-dimensional representation of the same sectorial contribution. (c) Total physical output distribution $P_T(\boldsymbol{n}') = \sum_\Lambda P(\Lambda, \boldsymbol{n}')$. (d) Three-dimensional representation of the total distribution. The color scale is determined independently for the sectorial and total distributions.

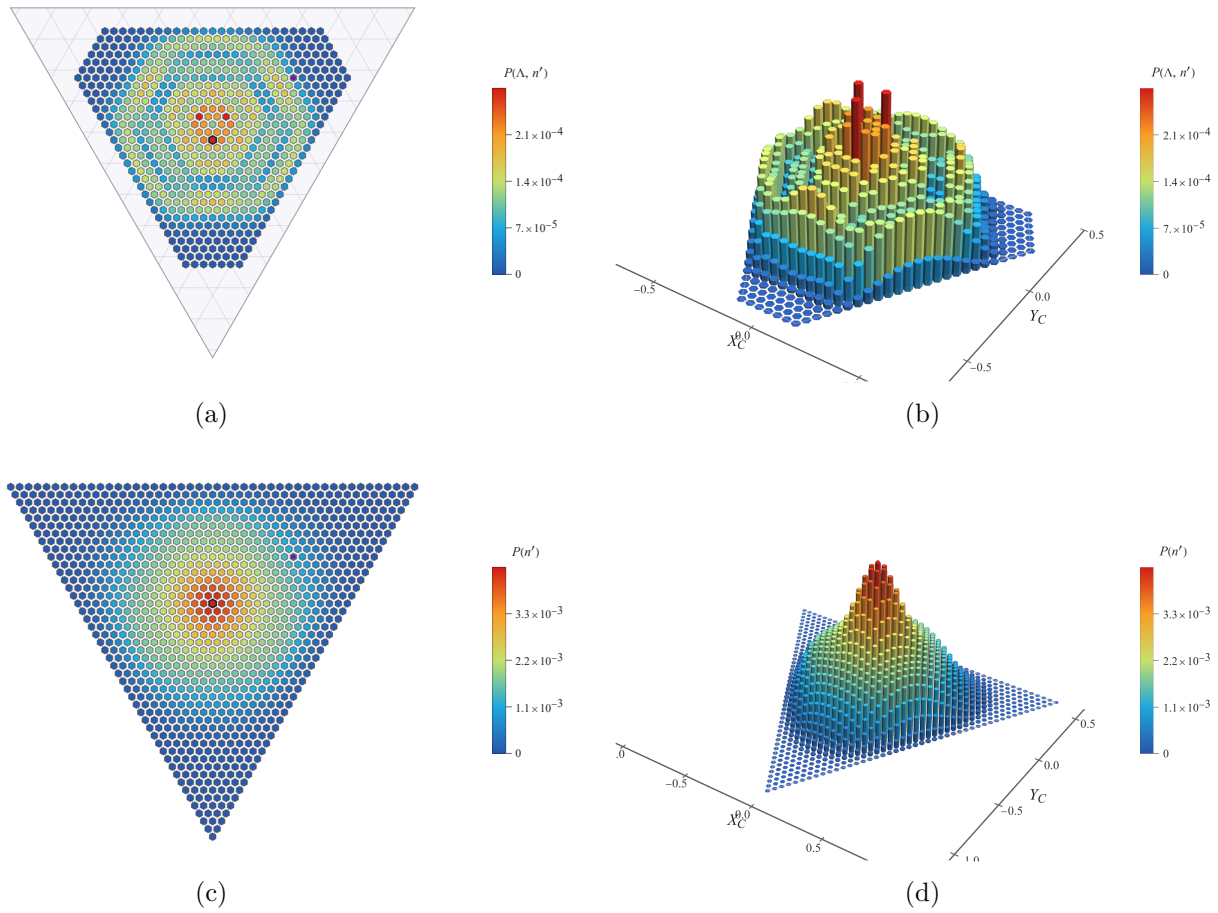


Figure 12: Sectorial contribution and total physical output distribution for the partially unbalanced configuration $\overline{\boldsymbol{n}} = (0.6, 0.2, 0.2)$, with $\boldsymbol{\lambda} = (0.7333, 0.2000, 0.0667)$, $N = 45$, and $\Lambda = (33, 9, 3)$. (a) Joint sectorial probability $P(\Lambda, \boldsymbol{n}') = P_\Lambda(\boldsymbol{n}')$. (b) Three-dimensional representation of the same sectorial contribution. (c) Total physical output distribution $P_T(\boldsymbol{n}') = \sum_\Lambda P(\Lambda, \boldsymbol{n}')$. (d) Three-dimensional representation of the total distribution. The color scale is determined independently for the sectorial and total distributions.

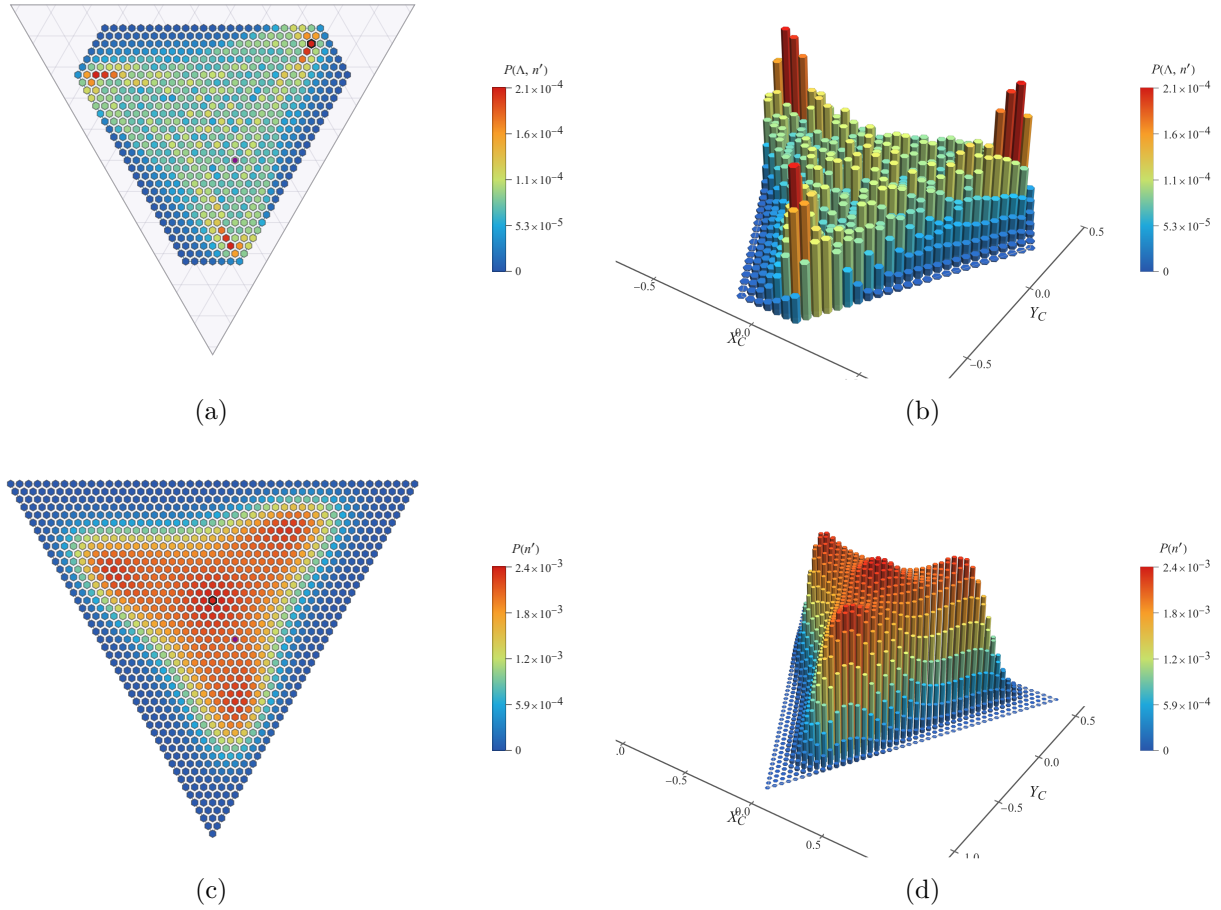


Figure 13: Sectorial contribution and total physical output distribution for the asymmetric configuration $\overline{\boldsymbol{n}} = (0.3333, 0.2222, 0.4444)$, with $\boldsymbol{\lambda} = (0.7333, 0.2000, 0.0667)$, $N = 45$, and $\Lambda = (33, 9, 3)$. (a) Joint sectorial probability $P(\Lambda, \boldsymbol{n}') = P_\Lambda(\boldsymbol{n}')$. (b) Three-dimensional representation of the same sectorial contribution. (c) Total physical output distribution $P_T(\boldsymbol{n}') = \sum_\Lambda P(\Lambda, \boldsymbol{n}')$. (d) Three-dimensional representation of the total distribution. The color scale is determined independently for the sectorial and total distributions.

G.6.4 Quantum–classical comparison of the total physical distribution

The preceding results distinguish the detection probability conditioned on an irreducible sector from the complete physical distribution, obtained by summing over all Schur–Weyl sectors. We now compare this distribution with the classical map introduced in the main candidacy document. The comparison is performed for four initial occupations at fixed photon number and interferometer, allowing us to examine the dependence of the quantum probabilities and classical images on the preparation.

For a two-port interferometer with partially indistinguishable photons, a many-photon asymptotic quantum–classical correspondence was studied in [23]. Here the numerical comparison concerns three modes and the complete physical distribution; it does not by itself establish an asymptotic limit.

We consider $M = 3$, $N = 45$, the Fourier interferometer $U = F_3$, and a common normalized spectrum

$$\boldsymbol{\lambda} = \frac{1}{15}(11, 3, 1). \tag{G.69}$$

The reference partition $\Lambda_0 = (33, 9, 3)$ has normalized shape $\overline{\Lambda}_0 = \boldsymbol{\lambda}$. It serves to construct frames compatible with each initial occupation; it does not restrict the physical probability to the sector Λ_0.

For each occupation $\boldsymbol{n}$, we consider unitary frames V_a satisfying the spectrum–diagonal condition of Appendix B. These frames define the effective single-particle matrices and their associated Gram matrices through

$$\begin{aligned} Q(V_a)\boldsymbol{\lambda} = \overline{\boldsymbol{n}}, \qquad \rho_a = V_a[\boldsymbol{\lambda}]V_a^\dagger, \\ \Gamma_a = [\overline{\boldsymbol{n}}]^{-1/2}\rho_a[\overline{\boldsymbol{n}}]^{-1/2}. \end{aligned} \tag{G.70}$$

All occupations considered here have strictly positive components. Hence, Γ_a is positive semidefinite with unit diagonal and defines a physically admissible set of internal-state overlaps. Although the spectrum and initial occupation are fixed within each family, different compatible frames need not produce the same ρ_a or Γ_a.

For each frame, the complete physical detection probability and the classical map are evaluated from the same preparation data:

$$\begin{aligned} P_a(\boldsymbol{n}') &= \sum_{\Lambda'} P(\Lambda' \mid \boldsymbol{n}, \Gamma_a) P(\boldsymbol{n}' \mid \boldsymbol{n}, \Lambda', F_3, \Gamma_a), \\ C_a(\boldsymbol{\phi}) &= \operatorname{diag}\Big(F_3 t_{\boldsymbol{\phi}} \rho_a t_{\boldsymbol{\phi}}^\dagger F_3^\dagger\Big). \end{aligned} \tag{G.71}$$

The first expression includes all populated irreducible sectors, without resolving the internal states. The second maps the two relative phases of $\mathbb{T}^2$ to normalized output intensities. Its image, $\mathcal{R}_a = C_a(\mathbb{T}^2)$, is the classically allowed region for that preparation.

In the figures below, the upper plot of each pair represents the physical probabilities on the discrete output-occupation lattice using heights and colors. The lower plot places the same probabilities on the simplex of normalized output occupations, $\overline{\boldsymbol{n}}' = \boldsymbol{n}'/N$, together with black points obtained by sampling the classical map. These points indicate accessible output intensities, rather than quantum detection events. They sample the image $\mathcal{R}_a$; the corresponding classical density is instead given by the pushforward of the uniform phase-torus measure.

By the Schur–Horn theorem, $\mathcal{R}_a \subseteq \Pi(\boldsymbol{\lambda})$ for every compatible frame, where $\Pi(\boldsymbol{\lambda})$ is the spectral permutohedron. Thus, all the preparations share a common spectral bound, while their classical images and physical probabilities can vary with the frame. Diagonal rephasings of a frame can change Γ_a without changing these observables; differences between such frames should not be counted as independent physical effects. The comparison at $N = 45$ illustrates the geometric correspondence between these quantities. It does not establish asymptotic convergence: exponential suppression away from the classical image and convergence to the induced classical density require the conditions stated in the corresponding asymptotic results [7].

Balanced input. For $\boldsymbol{n} = (15, 15, 15)$, Fig. 14 compares nine compatible frames. The occupation and spectrum are the same in every panel, but the resulting classical images and output probabilities need not coincide. Their differences reflect the freedom in the internal-state overlaps that remains under the spectrum–diagonal constraint.

Asymmetric input. Figure 15 presents the corresponding comparison for $\boldsymbol{n} = (15, 10, 20)$, or $\overline{\boldsymbol{n}} = (3, 2, 4)/9$. Again, nine compatible frames are considered at the same spectrum and interferometer. The variations among the panels show that the unequal occupation does not uniquely fix the effective single-particle matrix or the geometry of its classical image.

Intermediate asymmetry. For $\boldsymbol{n} = (24, 15, 6)$, Fig. 16 displays nine further preparations. Both the occupation and the admissible Gram matrices differ from those of the preceding families. Consequently, changes between families cannot be attributed to the initial occupation alone; the internal-state overlaps also enter the physical probabilities and the classical map.

More pronounced asymmetry. Finally, Fig. 17 shows the occupation $\boldsymbol{n} = (27, 9, 9)$. Its last two components are equal, but this equality does not impose the same symmetry on Γ_a. As in the other cases, compatible frames with a common spectrum and diagonal can generate distinct output patterns and classical-map images.

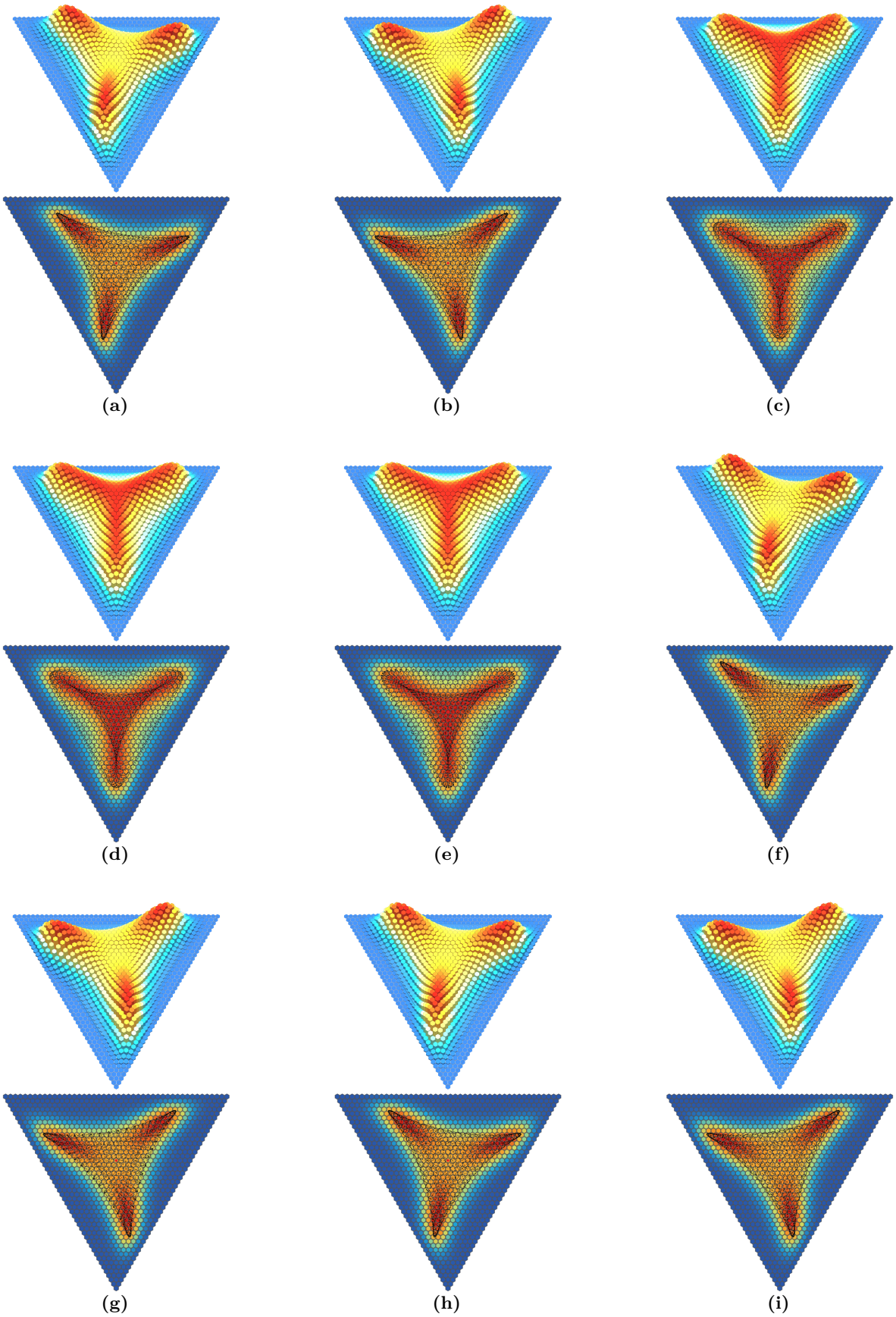


Figure 14: Quantum–classical comparison for nine compatible frames with $N = 45$, $\boldsymbol{n} = (15, 15, 15)$, $\boldsymbol{\lambda} = (11, 3, 1)/15$, and $U = F_3$. In each panel, the upper plot shows the physical output probabilities, while the lower plot superimposes them on the sampled classical-map image (black points). The frames share the same spectrum and input occupation but may correspond to different Gram matrices.

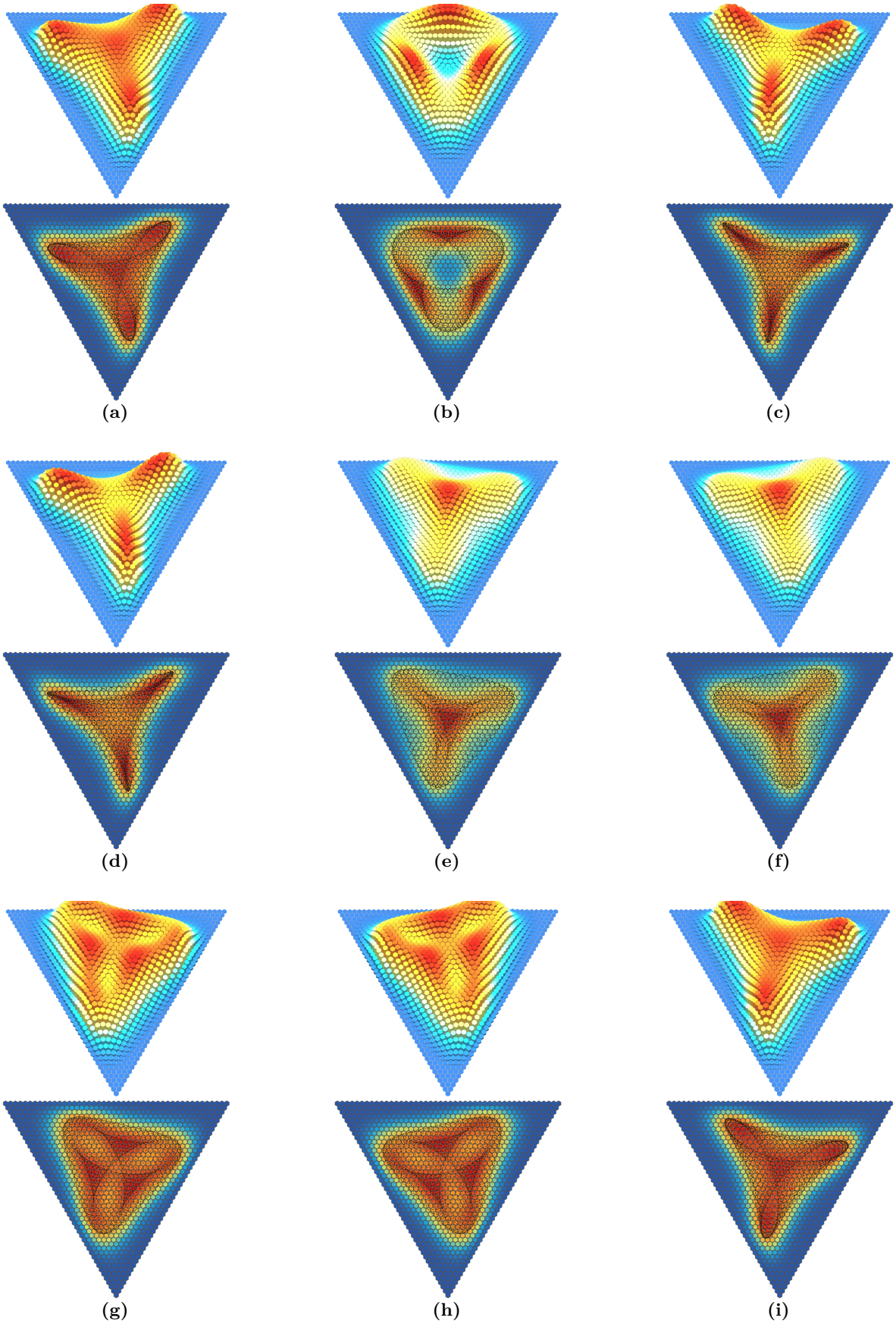


Figure 15: Quantum–classical comparison for nine compatible frames with $N = 45$, $\boldsymbol{n} = (15, 10, 20)$, $\overline{\boldsymbol{n}} = (3, 2, 4)/9$, $\boldsymbol{\lambda} = (11, 3, 1)/15$, and $U = F_3$. In each panel, the upper plot shows the physical output probabilities, while the lower plot superimposes them on the sampled classical-map image (black points). The frames share the same spectrum and input occupation but may correspond to different Gram matrices.

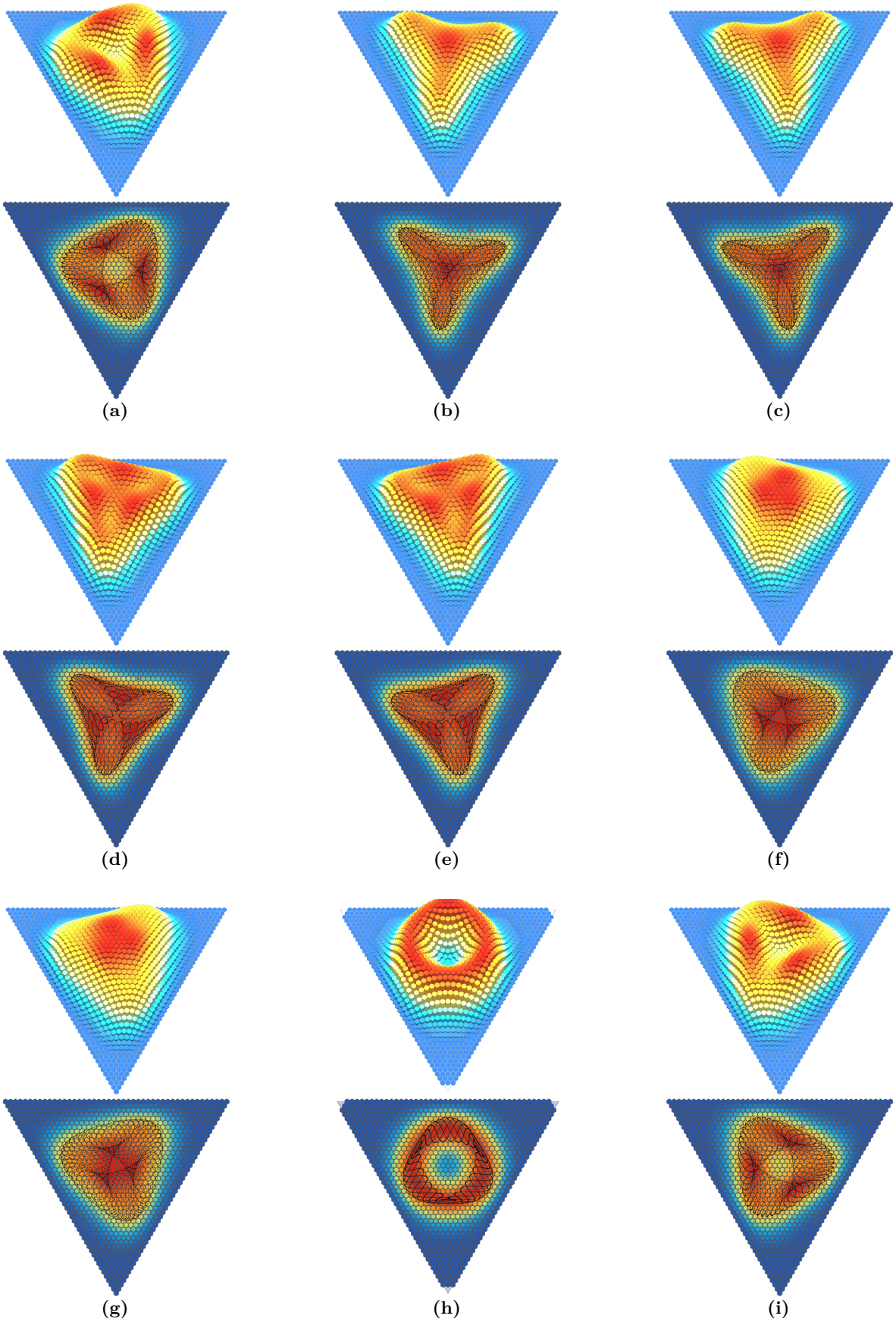


Figure 16: Quantum–classical comparison for nine compatible frames with $N = 45$, $\boldsymbol{n} = (24, 15, 6)$, $\boldsymbol{\lambda} = (11, 3, 1)/15$, and $U = F_3$. In each panel, the upper plot shows the physical output probabilities, while the lower plot superimposes them on the sampled classical-map image (black points). The frames share the same spectrum and input occupation but may correspond to different Gram matrices.

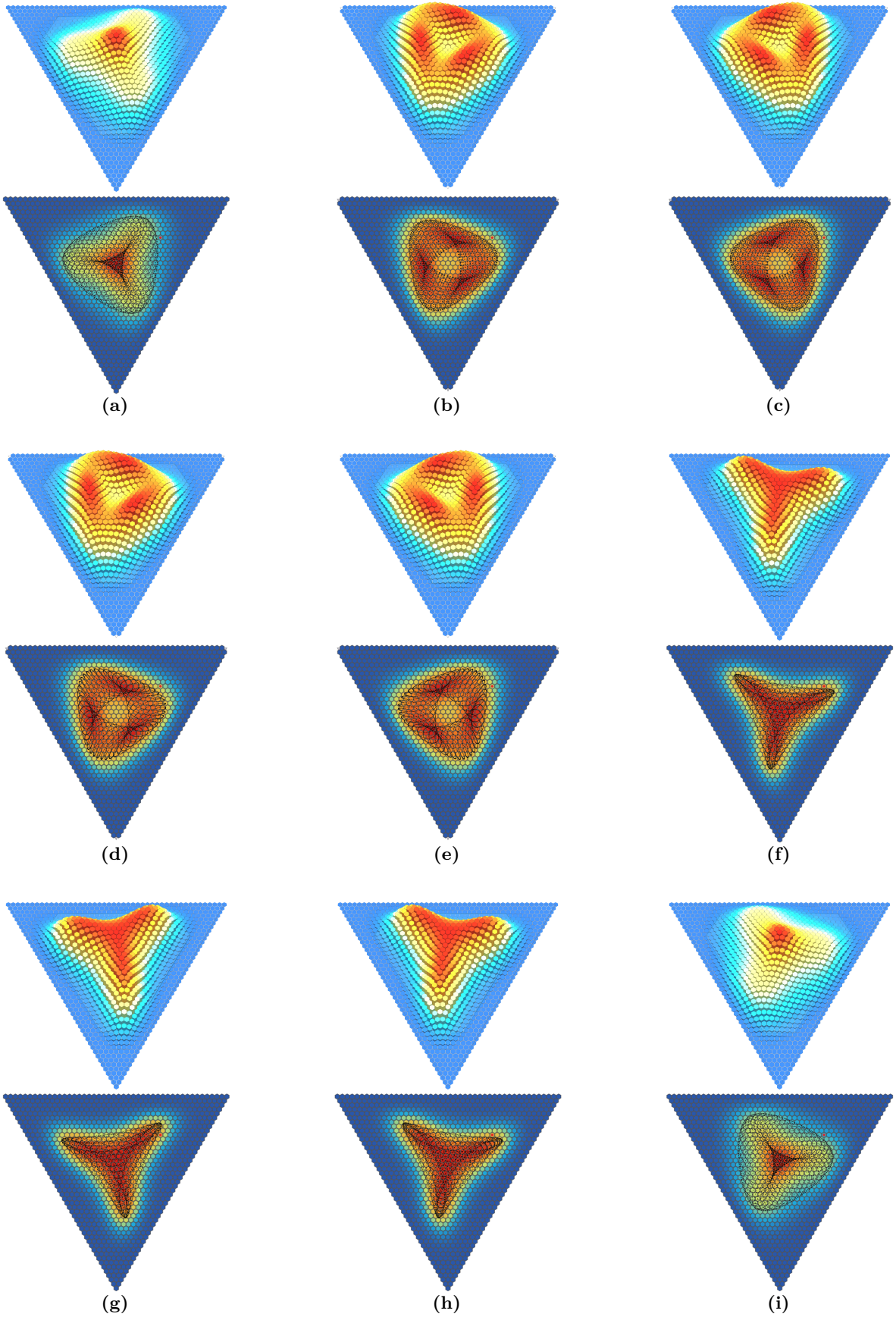


Figure 17: Quantum–classical comparison for nine compatible frames with $N = 45$, $\boldsymbol{n} = (27, 9, 9)$, $\boldsymbol{\lambda} = (11, 3, 1)/15$, and $U = F_3$. In each panel, the upper plot shows the physical output probabilities, while the lower plot superimposes them on the sampled classical-map image (black points). The frames share the same spectrum and input occupation but may correspond to different Gram matrices.

The four comparisons illustrate that the common spectral permutohedron alone need not determine the image of the classical map or the complete physical output distribution. Both quantities depend on the effective matrix specified by the preparation, while the correspondence between them is illustrated here at finite N. In particular, the plots compare the full Schur–Weyl sum with the classical map of the same ρ_a; they should not be interpreted as probabilities associated with the reference irrep alone.

G.7 Details of the algebraic evaluation

We close the appendix by summarizing the expressions used to evaluate numerically the physical density and the transition blocks. These formulas introduce no new physical objects; they provide an explicit evaluation of the previous expressions in the isospin basis.

G.7.1 Polynomial action for general matrices

For $A \in M_3(\mathbb{C})$, the action on $\wedge^2\mathbb{C}^3$ is identified with the cofactor matrix [1, 2],

$$\mathcal{J}(\wedge^2 A)\mathcal{J}^{-1} = \mathrm{Cof}(A), \tag{G.72}$$

where $\mathcal{J}$ is the isomorphism between $\wedge^2\mathbb{C}^3$ and the antifundamental representation defined in Appendix D. If A is invertible,

$$\mathrm{Cof}(A) = \det(A)A^{-T}, \tag{G.73}$$

whereas for $U \in SU(3)$,

$$\mathrm{Cof}(U) = \overline{U}. \tag{G.74}$$

The cofactor matrix, rather than entrywise complex conjugation, gives the polynomial action on the second exterior power for a general matrix, including ρ and Γ.

For

$$\Lambda = (p+q+r, q+r, r), \tag{G.75}$$

the full action can be written as

$$D^{\Lambda}(A) = (\det A)^r D^{(p,q)}_{\mathrm{pol}}(A). \tag{G.76}$$

When the sector has nonzero statistical weight, the scalar factor cancels upon normalization of the sector density, but it remains relevant to the statistical weight $P(\Lambda \mid \boldsymbol{n}, \Gamma)$:

$$\sigma(A) = \frac{P_{\mathbf{W}} D^{(p,q)}_{\mathrm{pol}}(A) P_{\mathbf{W}}}{\mathrm{Tr}\left[P_{\mathbf{W}} D^{(p,q)}_{\mathrm{pol}}(A) P_{\mathbf{W}}\right]}. \tag{G.77}$$

G.7.2 Operator blocks in the isospin basis

Let

$$S^{(N)}_{\mathbf{a},\mathbf{b}}(A) = \langle \mathbf{a}| \,\mathrm{Sym}^N(A)\, |\mathbf{b}\rangle\,. \tag{G.78}$$

These elements can be evaluated by coefficient extraction,

$$S^{(N)}_{\mathbf{a},\mathbf{b}}(A) = \sqrt{\frac{\prod_i a_i!}{\prod_j b_j!}}\,[x^{\mathbf{a}}] \prod_{j=1}^{3} \left(\sum_i A_{ij} x_i\right)^{b_j}. \tag{G.79}$$

In

$$\mathrm{Sym}^p(\mathbb{C}^3) \otimes \mathrm{Sym}^q(\overline{\mathbb{C}^3}), \tag{G.80}$$

the states compatible with a weight $\mathbf{W}$ satisfy

$$|\mathbf{k}| = p, \qquad |\mathbf{l}| = q, \qquad \mathbf{k} - \mathbf{l} = \mathbf{W} - q\mathbf{1}. \tag{G.81}$$

Let $B^{(\mathbf{W})}$ be the matrix whose columns contain the coordinates of the orthonormal basis $\{|\mathbf{W}, j\rangle\}$ in these tensor-product states. By construction,

$$(B^{(\mathbf{W})})^\dagger B^{(\mathbf{W})} = I. \tag{G.82}$$

We define the operator block in the tensor-product space by

$$[O_{\mathbf{W}}(A)]_{st} = S^{(p)}_{\mathbf{k}_s,\mathbf{k}_t}(A) S^{(q)}_{\mathbf{l}_s,\mathbf{l}_t}(\mathrm{Cof}\, A). \tag{G.83}$$

Its restriction to the irrep (p, q) and the weight $\mathbf{W}$ is

$$M_{\mathbf{W}}(A) = (B^{(\mathbf{W})})^\dagger O_{\mathbf{W}}(A) B^{(\mathbf{W})}. \tag{G.84}$$

Taking $A = \rho$,

$$\sigma_{\mathbf{W}} = \frac{M_{\mathbf{W}}(\rho)}{\mathrm{Tr}\, M_{\mathbf{W}}(\rho)}. \tag{G.85}$$

The same density is obtained from Γ,

$$\frac{M_{\mathbf{W}}(\rho)}{\mathrm{Tr}\, M_{\mathbf{W}}(\rho)} = \frac{M_{\mathbf{W}}(\Gamma)}{\mathrm{Tr}\, M_{\mathbf{W}}(\Gamma)}, \tag{G.86}$$

within sectors with nonzero occupations.

For a nonzero sector, the evaluation must satisfy

$$\sigma^\dagger = \sigma, \qquad \mathrm{Tr}\, \sigma = 1, \qquad \sigma \succeq 0, \tag{G.87}$$

together with the numerical equality between the constructions obtained from ρ and from Γ.

G.7.3 Transition blocks

For two weights $\mathbf{W}$ and $\mathbf{W}'$, we define the tensor-product block

$$[O_{\mathbf{W}',\mathbf{W}}(U)]_{s's} = S^{(p)}_{\mathbf{k}_{s'},\mathbf{k}_s}(U) S^{(q)}_{\mathbf{l}_{s'},\mathbf{l}_s}(\overline{U}), \tag{G.88}$$

where U is taken in $SU(3)$ and $\mathrm{Cof}(U) = \overline{U}$. For a general $U \in U(3)$, one may remove its determinant phase and use an $SU(3)$ representative: the resulting change in $D^\Lambda(U)$ is an overall phase and does not affect detection probabilities. In particular, this convention can be used for the Fourier interferometer F_3. The transformation to the isospin bases gives

$$T^{\mathbf{W}'\leftarrow\mathbf{W}}(U) = (B^{(\mathbf{W}')})^\dagger O_{\mathbf{W}',\mathbf{W}}(U) B^{(\mathbf{W})}. \tag{G.89}$$

This block coincides with $\mathcal{D}^{(p,q)}$ introduced in (G.44) and directly gives

$$P_{\mathrm{phys}}(\mathbf{W}') = \mathrm{Tr}\left[T^{\mathbf{W}'\leftarrow\mathbf{W}}(U)\sigma_{\mathbf{W}} T^{\mathbf{W}'\leftarrow\mathbf{W}}(U)^\dagger\right], \tag{G.90}$$

or, for a pure frame state,

$$P_{\mathrm{orb}}(\mathbf{W}') = \|T^{\mathbf{W}'\leftarrow\mathbf{W}}(U)\mathbf{c}(V)\|^2. \tag{G.91}$$

The unitarity of $D^{(p,q)}(U)$ and the resolution of the identity in weight spaces guarantee

$$\sum_{\mathbf{W}'} P_{\mathrm{phys}}(\mathbf{W}') = 1, \tag{G.92}$$

and the same normalization for P_{orb}.

H Stationary condition and Hessian of the toroidal integral

This appendix studies the toroidal integral for the squared norm of a projected highest-weight orbit state. Starting from the exact representation in Appendix C, we show that $Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}}$ is equivalent to stationarity at the origin of the relative phases. We compute the Hessian of the intensive action, its positive-semidefinite Laplacian structure, and the null mode associated with the global phase. In three modes, we obtain explicit expressions for the reduced Hessian and its determinant. These statements are exact. Under additional regularity and dominance assumptions, a local Gaussian approximation gives an algebraic contribution to the projected norm. This conditional analysis is relevant to the normalization estimates needed in the sectorial semiclassical construction, following methods analogous to those of [7].

H.1 Toroidal integral and intensive action

Let Λ be a partition of N and

$$|\Omega_\Lambda(V)\rangle = D^\Lambda(V)\,|\Omega_\Lambda\rangle \tag{H.1}$$

the highest-weight orbit vector constructed in Appendix C. The norm of its projection onto the weight $\boldsymbol{n}$ is

$$\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = \|P_{\Lambda,\boldsymbol{n}} D^\Lambda(V)\,|\Omega_\Lambda\rangle\|^2. \tag{H.2}$$

Using the toroidal representation of the projector,

$$P_{\Lambda,\boldsymbol{n}} = \int_{[0,2\pi]^M} \frac{\mathrm{d}^M\boldsymbol{\phi}}{(2\pi)^M}\, \mathrm{e}^{-\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi}} D^\Lambda(t_{\boldsymbol{\phi}}), \qquad t_{\boldsymbol{\phi}} = \mathrm{diag}(\mathrm{e}^{\mathrm{i}\phi_1},\dots,\mathrm{e}^{\mathrm{i}\phi_M}), \tag{H.3}$$

we obtain

$$\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = \int_{[0,2\pi]^M} \frac{\mathrm{d}^M\boldsymbol{\phi}}{(2\pi)^M}\, \mathrm{e}^{-\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi}}\, \langle\Omega_\Lambda(V)|\, D^\Lambda(t_{\boldsymbol{\phi}})\,|\Omega_\Lambda(V)\rangle\,. \tag{H.4}$$

Let V_r be the matrix formed by the first r columns of V and define

$$A_r(\boldsymbol{\phi}) = V_r^\dagger t_{\boldsymbol{\phi}} V_r. \tag{H.5}$$

The construction through exterior powers gives

$$\langle\Omega_\Lambda(V)|\, D^\Lambda(t_{\boldsymbol{\phi}})\,|\Omega_\Lambda(V)\rangle = \prod_{r=1}^{M} \det A_r(\boldsymbol{\phi})^{p_r}, \tag{H.6}$$

where

$$p_r = \Lambda_r - \Lambda_{r+1}, \qquad \Lambda_{M+1} = 0. \tag{H.7}$$

Therefore,

$$\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = \int_{[0,2\pi]^M} \frac{\mathrm{d}^M\boldsymbol{\phi}}{(2\pi)^M}\, \exp\left[S_N(\boldsymbol{\phi})\right], \tag{H.8}$$

with

$$S_N(\boldsymbol{\phi}) = -\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi} + \sum_{r=1}^{M} p_r \log\det A_r(\boldsymbol{\phi}). \tag{H.9}$$

The logarithmic expression defines a local branch of the action near $\boldsymbol{\phi} = \mathbf{0}$, where

$$A_r(\mathbf{0}) = V_r^\dagger V_r = I_r, \tag{H.10}$$

and the logarithm is analytic in a sufficiently small neighborhood of that point. The integral itself is defined globally by the product of determinants, including at phases where one of them vanishes.

In the semiclassical regime we consider partitions whose normalized shape is

$$\overline{\Lambda} = \frac{\Lambda}{N}, \qquad \overline{\boldsymbol{n}} = \frac{\boldsymbol{n}}{N}, \qquad \bar{p}_r = \frac{p_r}{N} = \overline{\Lambda}_r - \overline{\Lambda}_{r+1}, \tag{H.11}$$

with $\overline{\Lambda}_{M+1} = 0$. Along an asymptotic sequence, the normalized occupations and partition shapes are assumed to approach fixed limits. In the numerical examples considered here, $\overline{\Lambda}$ coincides with the physical spectrum $\boldsymbol{\lambda}$ used to select the compatible frames; this equality is not required by the general construction.

The exact exponent can be written as

$$S_N(\boldsymbol{\phi}) = N\, s(\boldsymbol{\phi}), \tag{H.12}$$

where

$$s(\boldsymbol{\phi}) = -\mathrm{i}\overline{\boldsymbol{n}} \cdot \boldsymbol{\phi} + \sum_{r=1}^{M} \bar{p}_r \log\det A_r(\boldsymbol{\phi}). \tag{H.13}$$

The intensive action depends on N through the normalized data along an asymptotic sequence. In particular,

$$s(\mathbf{0}) = 0. \tag{H.14}$$

By Cauchy–Binet, each determinant in the integrand is a sum of phases with nonnegative coefficients whose sum is one. Hence $|\det A_r(\boldsymbol{\phi})| \le 1$ for real phases, and the modulus of the integrand cannot exceed its value at the origin. This does not ensure that the origin is the only point of maximal modulus or that its local contribution controls the full integral.

H.2 Stationary condition

The first variation of the action follows from

$$\partial_k \log\det A_r = \mathrm{Tr}\left(A_r^{-1}\partial_k A_r\right). \tag{H.15}$$

At the origin,

$$\partial_k t_{\boldsymbol{\phi}}|_{\mathbf{0}} = \mathrm{i}E_k, \qquad E_k = |k\rangle\langle k|, \tag{H.16}$$

and therefore

$$\partial_k A_r|_{\mathbf{0}} = \mathrm{i}V_r^\dagger E_k V_r. \tag{H.17}$$

Defining

$$X_k^{(r)} := V_r^\dagger E_k V_r, \tag{H.18}$$

we obtain

$$\begin{aligned}\partial_k \log\det A_r|_{\mathbf{0}} &= \mathrm{i}\,\mathrm{Tr}\, X_k^{(r)} \\ &= \mathrm{i}\sum_{a=1}^{r} |V_{ka}|^2.\end{aligned} \tag{H.19}$$

Thus,

$$\left.\frac{\partial s}{\partial \phi_k}\right|_{\mathbf{0}} = -\mathrm{i}\overline{n}_k + \mathrm{i}\sum_{r=1}^{M} \bar{p}_r \sum_{a=1}^{r} |V_{ka}|^2. \tag{H.20}$$

Interchanging the order of the sums and using

$$\sum_{r=a}^{M} \bar{p}_r = \overline{\Lambda}_a, \tag{H.21}$$

we obtain

$$\sum_{r=1}^{M} \bar{p}_r \sum_{a=1}^{r} |V_{ka}|^2 = \sum_{a=1}^{M} |V_{ka}|^2 \overline{\Lambda}_a. \tag{H.22}$$

With

$$Q(V)_{ka} = |V_{ka}|^2, \tag{H.23}$$

the first variation takes the compact form

$$\nabla s(\mathbf{0}) = \mathrm{i}\left[Q(V)\overline{\Lambda} - \overline{\boldsymbol{n}}\right]. \tag{H.24}$$

Proposition. For the local action defined above, the origin is stationary if and only if the frame satisfies the sectorial spectrum–diagonal condition,

$$Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}} \qquad \Longleftrightarrow \qquad \nabla s(\mathbf{0}) = 0. \tag{H.25}$$

The equivalence follows directly from (H.24). Moreover,

$$\sum_k \overline{n}_k = \sum_k [Q(V)\overline{\Lambda}]_k = 1, \tag{H.26}$$

so only $M-1$ of the stationarity conditions are independent. This redundancy reflects the global phase, which cancels in the Fourier projection and leaves only $M-1$ independent phases.

H.3 Hessian and positivity

To compute the second variation we use the identity

$$\partial_l \partial_k \log\det A = \mathrm{Tr}\left(A^{-1}A_{kl} - A^{-1}A_l A^{-1}A_k\right), \tag{H.27}$$

where

$$A_k = \partial_k A, \qquad A_{kl} = \partial_l \partial_k A. \tag{H.28}$$

For the diagonal torus,

$$\partial_l \partial_k t_\phi|_{\mathbf{0}} = -\delta_{kl} E_k. \tag{H.29}$$

Consequently,

$$A_{r,k}(\mathbf{0}) = \mathrm{i}X_k^{(r)}, \qquad A_{r,kl}(\mathbf{0}) = -\delta_{kl}X_k^{(r)}, \tag{H.30}$$

and, since $A_r(\mathbf{0}) = I_r$,

$$\partial_l \partial_k \log\det A_r|_{\mathbf{0}} = -\delta_{kl}\,\mathrm{Tr}\,X_k^{(r)} + \mathrm{Tr}\left(X_l^{(r)}X_k^{(r)}\right). \tag{H.31}$$

We introduce the projector

$$P_r = V_r V_r^\dagger = \sum_{a=1}^{r} |v_a\rangle \langle v_a|. \tag{H.32}$$

Then

$$\mathrm{Tr}\,X_k^{(r)} = (P_r)_{kk}, \tag{H.33}$$

and

$$\mathrm{Tr}\left(X_l^{(r)}X_k^{(r)}\right) = |(P_r)_{kl}|^2. \tag{H.34}$$

Therefore,

$$\partial_l \partial_k s|_{\mathbf{0}} = \sum_{r=1}^{M} \bar{p}_r \left[-\delta_{kl}(P_r)_{kk} + |(P_r)_{kl}|^2\right]. \tag{H.35}$$

We define

$$\mathbb{H} := -\nabla^2 s(\mathbf{0}). \tag{H.36}$$

Its components are

$$\mathbb{H}_{kl} = \sum_{r=1}^{M} \bar{p}_r \left[\delta_{kl} (P_r)_{kk} - |(P_r)_{kl}|^2 \right]. \tag{H.37}$$

The matrix $\mathbb{H}$ is the negative Hessian of the intensive action; for the original exponent,

$$-\nabla^2 S_N(\mathbf{0}) = N\mathbb{H}. \tag{H.38}$$

The positivity of $\mathbb{H}$ follows from $P_r^2 = P_r$, which implies

$$(P_r)_{kk} = \sum_{l=1}^{M} |(P_r)_{kl}|^2. \tag{H.39}$$

For any real vector $x = (x_1, \ldots, x_M)^T$,

$$\begin{aligned} x^T \mathbb{H} x &= \frac{1}{2} \sum_{r=1}^{M} \bar{p}_r \sum_{k,l=1}^{M} |(P_r)_{kl}|^2 (x_k - x_l)^2 \\ &\geq 0. \end{aligned} \tag{H.40}$$

Consequently,

$$\mathbb{H} \succeq 0. \tag{H.41}$$

The same identity expresses $\mathbb{H}$ as a weighted graph Laplacian. For $k \neq l$, we define

$$\omega_{kl} = \sum_{r=1}^{M} \bar{p}_r |(P_r)_{kl}|^2 = \omega_{lk} \geq 0. \tag{H.42}$$

Then

$$\mathbb{H}_{kl} = -\omega_{kl}, \qquad k \neq l, \tag{H.43}$$

while

$$\mathbb{H}_{kk} = \sum_{l \neq k} \omega_{kl}. \tag{H.44}$$

In particular,

$$\mathbb{H}\mathbf{1} = 0, \qquad \mathbf{1} = (1, \ldots, 1)^T. \tag{H.45}$$

The vector $\mathbf{1}$ represents a common shift of all phases. Additional null modes occur when the weighted graph is disconnected; otherwise, the Hessian is positive definite after the global phase has been removed.

H.4 Reduced form for three modes

For $M = 3$,

$$\bar{p}_1 = \overline{\Lambda}_1 - \overline{\Lambda}_2, \qquad \bar{p}_2 = \overline{\Lambda}_2 - \overline{\Lambda}_3, \qquad \bar{p}_3 = \overline{\Lambda}_3. \tag{H.46}$$

Let

$$q_{ka} = |V_{ka}|^2. \tag{H.47}$$

For $r = 1$,

$$P_1 = v_1 v_1^\dagger, \tag{H.48}$$

so that, for $k \neq l$,

$$|(P_1)_{kl}|^2 = q_{k1} q_{l1}. \tag{H.49}$$

For $r = 2$, the unitarity of V gives

$$P_2 = I - v_3 v_3^\dagger, \tag{H.50}$$

and therefore

$$|(P_2)_{kl}|^2 = q_{k3} q_{l3}, \qquad k \neq l. \tag{H.51}$$

Finally,

$$P_3 = I, \tag{H.52}$$

so the central component $\bar{p}_3$ does not contribute to the Hessian.

For $k < l$, we then define

$$\omega_{kl} = \bar{p}_1 q_{k1} q_{l1} + \bar{p}_2 q_{k3} q_{l3}. \tag{H.53}$$

The intensive Hessian is

$$\mathbb{H} = \begin{pmatrix} \omega_{12} + \omega_{13} & -\omega_{12} & -\omega_{13} \\ -\omega_{12} & \omega_{12} + \omega_{23} & -\omega_{23} \\ -\omega_{13} & -\omega_{23} & \omega_{13} + \omega_{23} \end{pmatrix}. \tag{H.54}$$

The identity

$$\mathbb{H} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 0 \tag{H.55}$$

makes explicit the redundancy associated with the global phase. We fix, for example, $\phi_1 = 0$ and use as independent variables

$$\boldsymbol{\varphi} = (\phi_2, \phi_3)^T. \tag{H.56}$$

The reduced Hessian is then

$$\mathbb{H}_{\mathrm{red}} = \begin{pmatrix} \omega_{12} + \omega_{23} & -\omega_{23} \\ -\omega_{23} & \omega_{13} + \omega_{23} \end{pmatrix}. \tag{H.57}$$

Its determinant is

$$\begin{aligned} \det \mathbb{H}_{\mathrm{red}} &= (\omega_{12} + \omega_{23})(\omega_{13} + \omega_{23}) - \omega_{23}^2 \\ &= \omega_{12}\omega_{13} + \omega_{12}\omega_{23} + \omega_{13}\omega_{23}. \end{aligned} \tag{H.58}$$

The three principal cofactors of $\mathbb{H}$ have the same value, which is also the weighted spanning-tree sum of this three-vertex Laplacian. The condition

$$\det \mathbb{H}_{\mathrm{red}} > 0 \tag{H.59}$$

is equivalent to requiring that, after eliminating the global phase, no additional null mode remains. The stationary point is then nondegenerate on the relative-phase space, although this alone does not establish that it dominates the full toroidal integral.

H.5 Regular Gaussian approximation

The preceding identities are exact. We now consider the local Gaussian approximation to the projection integral, following the stationary-phase and Laplace methods used in [7]. Related asymptotic studies of Wigner functions for $SU(2)$ and selected $SU(3)$ representations provide further representation-theoretic context [24, 25]. The calculation below concerns the projected norm rather than a general asymptotic formula for representation matrix elements.

Suppose that

$$Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}}, \tag{H.60}$$

so that the origin is stationary, and suppose that the reduced Hessian is positive definite, with eigenvalues bounded away from zero along the asymptotic sequence. We also assume the data vary

sufficiently regularly for the local expansion to be uniform. After eliminating the global phase, let

$$\boldsymbol{\varphi} \in \mathbb{R}^d, \qquad d = M - 1, \tag{H.61}$$

be the relative-phase coordinates. Since

$$s(\mathbf{0}) = 0, \qquad \nabla s(\mathbf{0}) = 0, \tag{H.62}$$

the local expansion is

$$s(\boldsymbol{\varphi}) = -\frac{1}{2}\boldsymbol{\varphi}^T \mathbb{H}_{\mathrm{red}} \boldsymbol{\varphi} + O(\|\boldsymbol{\varphi}\|^3). \tag{H.63}$$

Consequently,

$$\exp[S_N(\boldsymbol{\varphi})] = \exp[N s(\boldsymbol{\varphi})] \simeq \exp\left[-\frac{N}{2}\boldsymbol{\varphi}^T \mathbb{H}_{\mathrm{red}} \boldsymbol{\varphi}\right]. \tag{H.64}$$

On the scale $\|\boldsymbol{\varphi}\| = O(N^{-1/2})$, the leading local contribution is determined by the Gaussian integral

$$\int_{\mathbb{R}^d} \exp\left[-\frac{N}{2} x^T H x\right] \mathrm{d}^d x = (2\pi)^{d/2} N^{-d/2} (\det H)^{-1/2}. \tag{H.65}$$

Denoting by $\mathcal{N}^{(0)}_{\Lambda,\boldsymbol{n}}(V)$ the contribution from a sufficiently small neighborhood of the origin in the reduced torus, and using the measure factor $(2\pi)^{-d}$, we obtain

$$\mathcal{N}^{(0)}_{\Lambda,\boldsymbol{n}}(V) \sim \frac{1}{(2\pi N)^{d/2}\sqrt{\det \mathbb{H}_{\mathrm{red}}}}. \tag{H.66}$$

If this neighborhood supplies the unique leading contribution to the reduced toroidal integral, with the remaining contributions controlled, the local formula also gives the leading asymptotic behavior of the full squared norm,

$$\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) \sim \frac{1}{(2\pi N)^{(M-1)/2}\sqrt{\det \mathbb{H}_{\mathrm{red}}}}. \tag{H.67}$$

For three modes,

$$\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) \sim \frac{1}{2\pi N\sqrt{\det \mathbb{H}_{\mathrm{red}}}}. \tag{H.68}$$

Under these assumptions, the leading dependence on N is algebraic. In particular,

$$-\frac{1}{N}\log \mathcal{N}_{\Lambda,\boldsymbol{n}}(V) \longrightarrow 0, \qquad N \longrightarrow \infty. \tag{H.69}$$

Thus the squared norm does not introduce an exponential rate when the regular contribution from the origin controls the integral.

If other stationary points contribute at the same order, their contributions must be included before drawing a conclusion about the full norm. The zero-rate statement (H.69) then requires that the total leading contribution remain nonzero. If

$$\det \mathbb{H}_{\mathrm{red}} = 0, \tag{H.70}$$

the regular Gaussian expansion ceases to be valid and higher-order terms must be retained.

H.6 Numerical verification of the local expansion

We consider the balanced configuration used throughout the manuscript,

$$N = 45, \qquad \Lambda = (33, 9, 3), \qquad \overline{\Lambda} = \boldsymbol{\lambda} = \frac{1}{15}(11, 3, 1), \qquad \overline{\boldsymbol{n}} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right). \tag{H.71}$$

We use the frame labeled V_1 in the corresponding numerical bank. The construction of this frame imposes

$$Q(V_1)\overline{\Lambda} = \overline{\boldsymbol{n}}, \tag{H.72}$$

and, according to (H.25), the origin

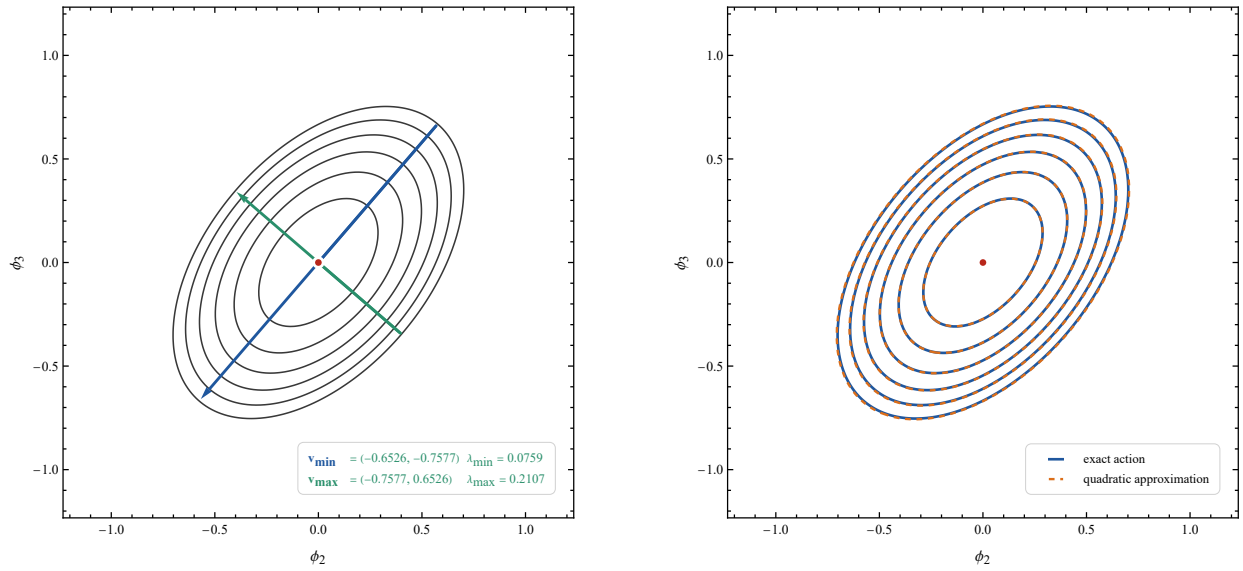
$$(\phi_2, \phi_3) = (0, 0) \tag{H.73}$$

is a stationary point of the reduced integral.

To compare the exact action with its quadratic expansion we define

$$\Delta s(\boldsymbol{\varphi}) = \Re s(\boldsymbol{\varphi}) - \Re s(\mathbf{0}), \qquad \Delta s_{\mathrm{quad}}(\boldsymbol{\varphi}) = -\frac{1}{2}\boldsymbol{\varphi}^T \mathbb{H}_{\mathrm{red}} \boldsymbol{\varphi}. \tag{H.74}$$

The numerical illustration uses the exact toroidal action and the reduced Hessian for the same reference frame V_1. Diagonalizing the Hessian, we compare the real part of the exact action with its quadratic expansion near the origin in Fig. 18.



(a) (b)

Figure 18: Local expansion around the stationary point for the balanced configuration $\overline{\boldsymbol{n}} = (1/3, 1/3, 1/3)$, with $\overline{\Lambda} = \boldsymbol{\lambda} = (0.7333, 0.2000, 0.0667)$, $N = 45$, $\Lambda = (33, 9, 3)$, and the reference frame V_1. (a) Level curves of the real part of the exact local action and the principal directions determined by $\mathbb{H}_{\mathrm{red}}$. (b) Comparison between the real part of the exact action and the quadratic approximation $-\frac{1}{2}\boldsymbol{\varphi}^T \mathbb{H}_{\mathrm{red}} \boldsymbol{\varphi}$.

Fig. 18(a) shows that the eigenvector directions of $\mathbb{H}_{\mathrm{red}}$ determine the characteristic directions of the local variation of $\Re s$. The smaller eigenvalue corresponds to the direction along which $\Re s(\boldsymbol{\varphi})$ decreases more slowly, whereas the larger eigenvalue corresponds to a faster decrease.

The comparison in Fig. 18(b) shows agreement between Δs and Δs_{quad} near the origin, as expected from the exact Hessian. This comparison checks the local quadratic behavior for the selected frame; it does not, by itself, establish the large-N asymptotics of the complete projection integral. When $\det \mathbb{H}_{\mathrm{red}} > 0$, the origin is nondegenerate in the relative phases and (H.68) gives its leading Gaussian contribution under the regularity assumptions stated above.

The analysis thus establishes a direct connection between the sectorial spectrum–diagonal condition and the local structure of the toroidal integral: $Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}}$ is exactly equivalent to stationarity of the origin, whereas $\mathbb{H}_{\mathrm{red}}$ determines the quadratic variation around that point. If this regular contribution determines the leading behavior of the full norm, its dependence on N is algebraic and introduces

no additional exponential rate. This conditional result separates the normalization of the initial projected state from the stationary conditions that determine the sectorial classical region after the interferometer, as discussed in the next appendix.

I Derivation of the sectorial classical map

In this appendix we derive the classical map associated with the propagation of a projected highest-weight orbit state within a fixed irreducible representation. Starting from an exact expansion of its detection probability, we identify an exact relative-entropy penalty in the moduli of the toroidal coefficients. The vanishing of this penalty leads to the sectorial map $\boldsymbol{\nu}' = Q(Ut_{\boldsymbol{\phi}}V)\overline{\Lambda}$. For a frame satisfying the initial spectrum–diagonal constraint, the corresponding points also satisfy the stationarity condition of the phase. We then establish an exponential bound outside the image of the map, provided that the initial projected norm is not exponentially small. The map identifies where the modulus penalty can vanish, but does not determine the full probability within its image: the stationary-point structure, prefactors, and possible interference must still be analyzed. The distinction between a projected pure-state probability and the physical sectorial density of Appendix G is maintained throughout [7].

I.1 Sectorial probability and exact expansion

Consider an irrep

$$\Lambda = (\Lambda_1, \ldots, \Lambda_M), \qquad \Lambda_1 \geq \cdots \geq \Lambda_M \geq 0, \qquad |\Lambda| = N. \tag{I.1}$$

The column multiplicities are

$$p_k = \Lambda_k - \Lambda_{k+1}, \qquad \Lambda_{M+1} = 0, \tag{I.2}$$

and satisfy

$$\sum_{k=1}^{M} k\, p_k = N. \tag{I.3}$$

We introduce the normalized quantities

$$\overline{\Lambda}_a = \frac{\Lambda_a}{N}, \qquad \alpha_k = \frac{p_k}{N}, \tag{I.4}$$

so that

$$\sum_{k=a}^{M} \alpha_k = \overline{\Lambda}_a. \tag{I.5}$$

The limit $N \to \infty$ is taken with M fixed, along sequences of partitions and occupations for which the normalized quantities converge. At each finite N, however, $\alpha_k = p_k/N$ is the exact ratio used below. The index N is suppressed when no confusion is possible.

As in Appendix C, the highest-weight vector can be written as

$$|\Omega_\Lambda\rangle = \bigotimes_{k=1}^{M} |\Omega_k\rangle^{\otimes p_k}, \qquad |\Omega_k\rangle = e_1 \wedge \cdots \wedge e_k. \tag{I.6}$$

For

$$V = (v_1, \ldots, v_M) \in U(M), \tag{I.7}$$

its orbit is

$$D^\Lambda(V)\,|\Omega_\Lambda\rangle = \bigotimes_{k=1}^{M} (v_1 \wedge \cdots \wedge v_k)^{\otimes p_k}. \tag{I.8}$$

Let

$$\boldsymbol{n} = (n_1, \ldots, n_M), \qquad \sum_i n_i = N, \tag{I.9}$$

be the initial weight. The toroidal representation of the projector is

$$P_{\Lambda,\boldsymbol{n}} = \int_{\mathbb{T}^M} \frac{\mathrm{d}^M \boldsymbol{\phi}}{(2\pi)^M} \, \mathrm{e}^{-\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi}} D^{\Lambda}(t_{\boldsymbol{\phi}}), \qquad t_{\boldsymbol{\phi}} = \mathrm{diag}(\mathrm{e}^{\mathrm{i}\phi_1}, \ldots, \mathrm{e}^{\mathrm{i}\phi_M}). \tag{I.10}$$

Since all weights of the irrep sum to N, one combination of the phases is global. We eliminate this degree of freedom and write

$$\mathrm{d}\mu(\boldsymbol{\phi}) = \frac{\mathrm{d}^{M-1}\phi}{(2\pi)^{M-1}} \tag{I.11}$$

for the normalized measure over the torus of relative phases.

The unnormalized projected state is

$$|\widetilde{\boldsymbol{n}}, V\rangle = P_{\Lambda,\boldsymbol{n}} D^{\Lambda}(V) \, |\Omega_{\Lambda}\rangle \, , \tag{I.12}$$

with

$$\mathcal{N}_{\Lambda,\boldsymbol{n}}(V) = \langle \widetilde{\boldsymbol{n}}, V | \widetilde{\boldsymbol{n}}, V \rangle \, . \tag{I.13}$$

Whenever this norm is nonzero,

$$|\boldsymbol{n}, V; \Lambda\rangle = \frac{|\widetilde{\boldsymbol{n}}, V\rangle}{\sqrt{\mathcal{N}_{\Lambda,\boldsymbol{n}}(V)}}. \tag{I.14}$$

After applying the interferometer U, we define

$$g_{\boldsymbol{\phi}} = U t_{\boldsymbol{\phi}} V, \tag{I.15}$$

and obtain

$$D^{\Lambda}(U) \, |\widetilde{\boldsymbol{n}}, V\rangle = \int \mathrm{d}\mu(\boldsymbol{\phi}) \, \mathrm{e}^{-\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi}} D^{\Lambda}(g_{\boldsymbol{\phi}}) \, |\Omega_{\Lambda}\rangle \, . \tag{I.16}$$

Consider now the rank-k block. If $g_1, \ldots, g_k$ are the first k columns of $g_{\boldsymbol{\phi}}$,

$$(\wedge^k g_{\boldsymbol{\phi}}) \, |\Omega_k\rangle = \sum_{\substack{I \subset \{1,\ldots,M\} \\ |I|=k}} A_I^{(k)}(\boldsymbol{\phi}) \, |I\rangle \, , \tag{I.17}$$

where

$$A_I^{(k)}(\boldsymbol{\phi}) = \det(g_{\boldsymbol{\phi}})_{I,[k]}. \tag{I.18}$$

The unitarity of $g_{\boldsymbol{\phi}}$ implies

$$\sum_{|I|=k} \left| A_I^{(k)}(\boldsymbol{\phi}) \right|^2 = 1. \tag{I.19}$$

We therefore define

$$q_I^{(k)}(\boldsymbol{\phi}) = \left| A_I^{(k)}(\boldsymbol{\phi}) \right|^2, \tag{I.20}$$

which defines a normalized distribution over subsets I of cardinality k.

The block contains p_k copies. Let $m_I^{(k)}$ be the number of copies occupying the exterior state $|I\rangle$. Then

$$\sum_{|I|=k} m_I^{(k)} = p_k. \tag{I.21}$$

The number of sequences associated with the set $m^{(k)} = \{m_I^{(k)}\}_{|I|=k}$ is

$$N_{m^{(k)}} = \frac{p_k!}{\prod_I m_I^{(k)}!}, \tag{I.22}$$

and the corresponding normalized symmetrized state can be written as

$$|m^{(k)}\rangle = \frac{1}{\sqrt{N_{m^{(k)}}}} \sum_{\text{sequences of type } m^{(k)}} |I_1\rangle \otimes \cdots \otimes |I_{p_k}\rangle \,. \tag{I.23}$$

States corresponding to distinct type vectors are orthogonal. The associated coefficient is

$$C_{m^{(k)}}(\boldsymbol{\phi}) = \sqrt{\frac{p_k!}{\prod_I m_I^{(k)}!}} \prod_I \left[A_I^{(k)}(\boldsymbol{\phi})\right]^{m_I^{(k)}} . \tag{I.24}$$

For all blocks we write

$$\mathbf{m} = \{m_I^{(k)}\}_{k,I}, \qquad |\mathbf{m}\rangle = \bigotimes_k |m^{(k)}\rangle \,, \tag{I.25}$$

and

$$C_{\mathbf{m}}(\boldsymbol{\phi}) = \prod_k \sqrt{\frac{p_k!}{\prod_{|I|=k} m_I^{(k)}!}} \prod_{|I|=k} \left[A_I^{(k)}(\boldsymbol{\phi})\right]^{m_I^{(k)}} . \tag{I.26}$$

The weight of the exterior state $|I\rangle$ is

$$w(I) = \sum_{i \in I} e_i, \tag{I.27}$$

so that the weight associated with $\mathbf{m}$ has components

$$w_i(\mathbf{m}) = \sum_k \sum_{I \ni i} m_I^{(k)}. \tag{I.28}$$

A table $\mathbf{m}$ therefore contributes to the final weight $\boldsymbol{n}' = (n'_1, \ldots, n'_M)$ if

$$n'_i = \sum_k \sum_{I \ni i} m_I^{(k)}, \qquad i = 1, \ldots, M. \tag{I.29}$$

We denote by $\mathcal{T}_\Lambda(\boldsymbol{n}')$ the set of tables satisfying (I.21) and (I.29).
The output probability associated with the projected pure state can be defined as

$$P_V^{(\Lambda)}(\boldsymbol{n}'|\boldsymbol{n}, U) = \|P_{\Lambda,\boldsymbol{n}'} D^\Lambda(U) \,|\boldsymbol{n}, V; \Lambda\rangle\|^2. \tag{I.30}$$

The orthogonality of the states $|\mathbf{m}\rangle$ leads to

$$P_V^{(\Lambda)}(\boldsymbol{n}'|\boldsymbol{n}, U) = \frac{1}{\mathcal{N}_{\Lambda,\boldsymbol{n}}(V)} \sum_{\mathbf{m} \in \mathcal{T}_\Lambda(\boldsymbol{n}')} \left| \int \mathrm{d}\mu(\boldsymbol{\phi})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{n}\cdot\boldsymbol{\phi}} C_{\mathbf{m}}(\boldsymbol{\phi}) \right|^2 . \tag{I.31}$$

This expression is exact for the projected pure state. It is not the physical sectorial probability obtained by propagating $\sigma_{\Lambda,\boldsymbol{n}}(\Gamma)$, which was derived in Appendix G. The bounds below concern this projected pure-state probability and cannot be transferred to physical sectorial probabilities without additional arguments.

I.2 Macroscopic limit of the modulus and phase

In a region where the relevant minors do not vanish, we write

$$A_I^{(k)}(\boldsymbol{\phi}) = \sqrt{q_I^{(k)}(\boldsymbol{\phi})}\, \mathrm{e}^{\mathrm{i}\theta_I^{(k)}(\boldsymbol{\phi})}, \tag{I.32}$$

with phases $\theta_I^{(k)}(\boldsymbol{\phi})$ that are locally continuous.

For every block with $p_k > 0$, define its type frequencies by

$$f_I^{(k)} = \frac{m_I^{(k)}}{p_k}, \qquad \sum_{|I|=k} f_I^{(k)} = 1. \tag{I.33}$$

We continue to use the exact ratios $\alpha_k = p_k/N$. Blocks with $p_k = o(N)$ may affect subleading terms but have no contribution to the limiting rate. Let

$$D_k = \binom{M}{k} \tag{I.34}$$

denote the number of subsets of cardinality k. When $q_I^{(k)}(\boldsymbol{\phi}) > 0$ for every component with positive frequency, the polar decomposition is well defined on the support of the type.

The coefficient modulus can be factored exactly. Indeed,

$$|C_{\mathbf{m}}(\boldsymbol{\phi})|^2 = \prod_{k:p_k>0} \left[\frac{p_k!}{\prod_{|I|=k} m_I^{(k)}!} \prod_{|I|=k} [q_I^{(k)}(\boldsymbol{\phi})]^{m_I^{(k)}} \right]. \tag{I.35}$$

With

$$D_{\mathrm{KL}}(f\|q) = \sum_I f_I \log \frac{f_I}{q_I}, \tag{I.36}$$

where the standard conventions are used for zero components [26], define

$$\mathscr{D}_{\mathbf{f}}(\boldsymbol{\phi}) = \sum_k \alpha_k D_{\mathrm{KL}}\Big(f^{(k)} \Big\| q^{(k)}(\boldsymbol{\phi})\Big). \tag{I.37}$$

The exact prefactor, independent of $\boldsymbol{\phi}$, is

$$\mathcal{B}_N(\mathbf{f})^2 := \prod_{k:p_k>0} \left[\frac{p_k!}{\prod_{|I|=k} m_I^{(k)}!} \prod_{|I|=k} (f_I^{(k)})^{m_I^{(k)}} \right]. \tag{I.38}$$

Here $0^0 = 1$, and the prefactor is the product of the probabilities of the observed type vectors under their own empirical frequencies; hence $0 \le \mathcal{B}_N(\mathbf{f}) \le 1$. It follows that

$$|C_{\mathbf{m}}(\boldsymbol{\phi})| = \mathcal{B}_N(\mathbf{f}) \exp\left[-\frac{N}{2}\mathscr{D}_{\mathbf{f}}(\boldsymbol{\phi})\right]. \tag{I.39}$$

This identity is exact, including on the boundaries of the frequency simplices. If $q_I^{(k)} = 0$ and $f_I^{(k)} > 0$, both sides vanish, with $D_{\mathrm{KL}} = +\infty$.

Stirling's formula is needed only to approximate the prefactor. For a block with $p_k \to \infty$ and frequencies bounded away from zero, it gives

$$\begin{aligned} &\sqrt{\frac{p_k!}{\prod_I m_I^{(k)}!}} \prod_I (f_I^{(k)})^{m_I^{(k)}/2} \\ &\quad = (2\pi p_k)^{-(D_k-1)/4} \prod_I (f_I^{(k)})^{-1/4} \left[1 + O(p_k^{-1})\right]. \end{aligned} \tag{I.40}$$

For frequencies approaching a boundary, this prefactor approximation requires modification; the exact exponential factorization remains valid. In particular, no Stirling approximation is needed for the suppression estimate derived below.

The phase of the coefficient also separates exactly wherever the relevant minors are nonzero. With

$$S_{\mathbf{m}}(\boldsymbol{\phi}) = \sum_{k,I} m_I^{(k)} \theta_I^{(k)}(\boldsymbol{\phi}), \tag{I.41}$$

we can write

$$-\boldsymbol{n} \cdot \boldsymbol{\phi} + S_{\mathbf{m}}(\boldsymbol{\phi}) = N \Phi_{\mathbf{f}}(\boldsymbol{\phi}), \tag{I.42}$$

where

$$\Phi_{\mathbf{f}}(\boldsymbol{\phi}) = -\overline{\boldsymbol{n}} \cdot \boldsymbol{\phi} + \sum_{k,I} \alpha_k f_I^{(k)} \theta_I^{(k)}(\boldsymbol{\phi}). \tag{I.43}$$

The relative entropy controls the modulus, whereas $\Phi_{\mathbf{f}}$ controls the phase oscillations. Their asymptotic analysis requires different conditions.

I.3 Zero set of the relative-entropy penalty

Since

$$D_{\mathrm{KL}}\left(f^{(k)} \middle\| q^{(k)}\right) \geq 0, \tag{I.44}$$

the relative-entropy penalty in an individual coefficient modulus can vanish only when, for each macroscopic block,

$$f_I^{(k)} = q_I^{(k)}(\boldsymbol{\phi}) = \left|A_I^{(k)}(\boldsymbol{\phi})\right|^2. \tag{I.45}$$

In terms of the discrete occupations,

$$m_I^{(k)} = p_k q_I^{(k)}(\boldsymbol{\phi}) + o(N). \tag{I.46}$$

Here the $o(N)$ term accommodates integer type vectors along an asymptotic sequence; exact equality of the frequencies at finite N need not be possible. The vanishing of this coefficient-level penalty is not sufficient to establish a nonzero leading detection probability, because the toroidal integral also contains oscillatory phases.

Let

$$\boldsymbol{\nu}' = \frac{\boldsymbol{n}'}{N} \tag{I.47}$$

be the normalized final occupation. Dividing (I.29) by N and using (I.45), we obtain

$$\nu_i' = \sum_k \alpha_k \sum_{I \ni i} \left|A_I^{(k)}(\boldsymbol{\phi})\right|^2. \tag{I.48}$$

This is the macroscopic restriction on the final weight when the limiting relative-entropy penalty vanishes.

I.4 Derivation of the sectorial classical map

The expression (I.48) can be reduced to a single-particle formula. To this end, we use

$$\sum_{I \ni i} \left|A_I^{(k)}\right|^2 = \sum_{a=1}^{k} |g_{ia}|^2. \tag{I.49}$$

Indeed, for the normalized exterior state

$$|\chi_k\rangle = g_1 \wedge \cdots \wedge g_k = \sum_I A_I^{(k)} |I\rangle, \tag{I.50}$$

the number operator of mode i satisfies

$$\hat{n}_i \left|I\right\rangle = \mathbf{1}_{i\in I} \left|I\right\rangle . \tag{I.51}$$

Therefore,

$$\left\langle \chi_k\right| \hat{n}_i \left|\chi_k\right\rangle = \sum_{I\ni i} |A_I^{(k)}|^2. \tag{I.52}$$

On the other hand, the one-particle matrix associated with $|\chi_k\rangle$ is

$$\gamma_k = \sum_{a=1}^{k} \left|g_a\right\rangle \left\langle g_a\right| , \tag{I.53}$$

and

$$\left\langle \chi_k\right| \hat{n}_i \left|\chi_k\right\rangle = \left\langle i\right| \gamma_k \left|i\right\rangle = \sum_{a=1}^{k} |g_{ia}|^2, \tag{I.54}$$

which proves (I.49).
Substituting this identity into (I.48),

$$\begin{aligned} \nu_i' &= \sum_k \alpha_k \sum_{a=1}^{k} |(g_{\boldsymbol{\phi}})_{ia}|^2 \\ &= \sum_{a=1}^{M} |(g_{\boldsymbol{\phi}})_{ia}|^2 \sum_{k=a}^{M} \alpha_k \\ &= \sum_{a=1}^{M} \overline{\Lambda}_a |(g_{\boldsymbol{\phi}})_{ia}|^2, \end{aligned} \tag{I.55}$$

where we used (I.5). With

$$Q(g)_{ia} = |g_{ia}|^2 \tag{I.56}$$

and $g_{\boldsymbol{\phi}} = Ut_{\boldsymbol{\phi}}V$, we obtain the central result [7]

$$\boldsymbol{\nu}' = Q(Ut_{\boldsymbol{\phi}}V)\overline{\Lambda}. \tag{I.57}$$

We then define the sectorial classical map

$$\mathcal{C}_{\overline{\Lambda},V,U} : \mathbb{T}^{M-1} \longrightarrow \Delta_{M-1}, \qquad \mathcal{C}_{\overline{\Lambda},V,U}(\boldsymbol{\phi}) = Q(Ut_{\boldsymbol{\phi}}V)\overline{\Lambda}. \tag{I.58}$$

Equivalently,

$$\mathcal{C}_{\overline{\Lambda},V,U}(\boldsymbol{\phi}) = \operatorname{diag}\left[Ut_{\boldsymbol{\phi}}V[\overline{\Lambda}]V^{\dagger}t_{\boldsymbol{\phi}}^{\dagger}U^{\dagger}\right]. \tag{I.59}$$

The map is defined for any unitary frame V; the initial compatibility condition $Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}}$ is needed only when relating its zero-penalty points to phase stationarity below.
The matrix inside the diagonal is unitarily equivalent to $[\overline{\Lambda}]$. By the Schur–Horn theorem [16, 17],

$$\mathcal{C}_{\overline{\Lambda},V,U}(\boldsymbol{\phi}) \prec \overline{\Lambda} \tag{I.60}$$

for every $\boldsymbol{\phi}$. If we define the spectral region [27]

$$\mathcal{S}_{\overline{\Lambda}} = \left\{\boldsymbol{\nu}' \in \Delta_{M-1} : \boldsymbol{\nu}' \prec \overline{\Lambda}\right\}, \tag{I.61}$$

then

$$\mathcal{R}_{\overline{\Lambda},V,U} \subseteq \mathcal{S}_{\overline{\Lambda}} \subseteq \Delta_{M-1}, \tag{I.62}$$

where

$$\mathcal{R}_{\overline{\Lambda},V,U} := \mathcal{C}_{\overline{\Lambda},V,U}\left(\mathbb{T}^{M-1}\right) \tag{I.63}$$

is the image of the relative-phase torus.

The spectral region provides the convex constraint imposed by the spectrum, whereas the toroidal image can be a proper subset and also depends on V and U. Neither the inclusion nor the image alone specifies the classical pushforward density, which requires the phase-torus measure and, at regular points, the Jacobian of the map.

I.5 Stationary condition of the phase

Consider a point $\boldsymbol{\phi}_*$ for which (I.45) is satisfied. In the following local differentiation we assume that all minors in the macroscopic blocks are nonzero at this point. Write

$$f_I^{(k)} = q_I^{(k)}(\boldsymbol{\phi}_*). \tag{I.64}$$

Differentiating the relative divergence while keeping $\mathbf{f}$ fixed,

$$\partial_{\phi_j} D_{\mathrm{KL}}(f\|q) = -\sum_I f_I \frac{\partial_{\phi_j} q_I}{q_I}. \tag{I.65}$$

At $f = q$,

$$\partial_{\phi_j} D_{\mathrm{KL}}(f\|q)\Big|_{f=q} = -\sum_I \partial_{\phi_j} q_I = 0, \tag{I.66}$$

because of the normalization of q.

To study the phase, we write

$$|\chi_k(\boldsymbol{\phi})\rangle = \sum_I \sqrt{q_I^{(k)}(\boldsymbol{\phi})}\, \mathrm{e}^{\mathrm{i}\theta_I^{(k)}(\boldsymbol{\phi})}\, |I\rangle\,. \tag{I.67}$$

It follows that

$$\mathrm{Im}\,\langle \chi_k|\partial_{\phi_j}\chi_k\rangle = \sum_I q_I^{(k)} \partial_{\phi_j}\theta_I^{(k)}. \tag{I.68}$$

On the other hand,

$$|\chi_k(\boldsymbol{\phi})\rangle = \wedge^k(Ut_{\boldsymbol{\phi}}V)\,|\Omega_k\rangle\,. \tag{I.69}$$

Since U is independent of $\boldsymbol{\phi}$,

$$\mathrm{Im}\,\langle \chi_k|\partial_{\phi_j}\chi_k\rangle = \sum_{a=1}^{k} |V_{ja}|^2. \tag{I.70}$$

Comparing (I.68) and (I.70),

$$\sum_I q_I^{(k)} \partial_{\phi_j}\theta_I^{(k)} = \sum_{a=1}^{k} |V_{ja}|^2. \tag{I.71}$$

Evaluating now the derivative of $\Phi_{\mathbf{f}}$ at $\boldsymbol{\phi}_*$,

$$\begin{aligned}
\partial_{\phi_j}\Phi_{\mathbf{f}}\Big|_{\boldsymbol{\phi}_*} &= -\overline{n}_j + \sum_k \alpha_k \sum_I q_I^{(k)}(\boldsymbol{\phi}_*)\partial_{\phi_j}\theta_I^{(k)}(\boldsymbol{\phi}_*) \\
&= -\overline{n}_j + \sum_k \alpha_k \sum_{a=1}^{k} |V_{ja}|^2 \\
&= -\overline{n}_j + \sum_a \overline{\Lambda}_a |V_{ja}|^2.
\end{aligned} \tag{I.72}$$

In vector form,

$$\nabla_{\boldsymbol{\phi}}\Phi_{\mathbf{f}}(\boldsymbol{\phi}_*) = Q(V)\overline{\Lambda} - \overline{\boldsymbol{n}}. \tag{I.73}$$

Consequently, if the initial frame satisfies

$$Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}}, \tag{I.74}$$

every regular point $\boldsymbol{\phi}_*$ at which the exponential penalty in the modulus disappears also satisfies

$$\nabla_{\boldsymbol{\phi}} \Phi_{\mathbf{f}}(\boldsymbol{\phi}_*) = 0. \tag{I.75}$$

This is the same spectrum–diagonal condition that, in Appendix H, determined the stationarity of the normalization integral. The output map and the stationary condition are thus linked by the same compatible frame.

I.6 Classical region and exponential suppression

For a fixed phase $\boldsymbol{\phi}$ and a normalized final occupation $\boldsymbol{\nu}'$, we define

$$\mathcal{I}_{\boldsymbol{\phi}}(\boldsymbol{\nu}') = \inf_{\{f^{(k)}\}} \sum_k \alpha_k D_{\mathrm{KL}} \left(f^{(k)} \middle\| q^{(k)}(\boldsymbol{\phi}) \right), \tag{I.76}$$

where the infimum is taken over distributions satisfying

$$\sum_I f_I^{(k)} = 1 \tag{I.77}$$

and

$$\nu_i' = \sum_k \alpha_k \sum_{I \ni i} f_I^{(k)}. \tag{I.78}$$

The infimum equals $+\infty$ if the constraint set is empty. Since each relative entropy is nonnegative,

$$\mathcal{I}_{\boldsymbol{\phi}}(\boldsymbol{\nu}') = 0 \qquad \Longleftrightarrow \qquad \boldsymbol{\nu}' = \mathcal{C}_{\overline{\Lambda},V,U}(\boldsymbol{\phi}). \tag{I.79}$$

Indeed, a zero value requires $f^{(k)} = q^{(k)}(\boldsymbol{\phi})$ for every block with $\alpha_k > 0$. Minimizing over the relative-phase torus,

$$\mathcal{I}(\boldsymbol{\nu}') = \inf_{\boldsymbol{\phi} \in \mathbb{T}^{M-1}} \mathcal{I}_{\boldsymbol{\phi}}(\boldsymbol{\nu}'), \tag{I.80}$$

we find

$$\mathcal{I}(\boldsymbol{\nu}') = 0 \qquad \Longleftrightarrow \qquad \boldsymbol{\nu}' \in \mathcal{R}_{\overline{\Lambda},V,U}. \tag{I.81}$$

This equivalence concerns the infimum of the relative-entropy penalty in the coefficient moduli, not the full detection probability or its exact large-deviation rate.

The positivity of the rate can be made quantitative without using Stirling's approximation. Pinsker's inequality implies, for each mode i and each block,

$$\left| \sum_{I \ni i} (f_I^{(k)} - q_I^{(k)}) \right| \le \frac{1}{2} \| f^{(k)} - q^{(k)} \|_1 \le \sqrt{\frac{1}{2} D_{\mathrm{KL}}(f^{(k)} \| q^{(k)})}. \tag{I.82}$$

Using Cauchy–Schwarz and $\sum_k \alpha_k \le 1$, we obtain

$$\mathscr{D}_{\mathbf{f}}(\boldsymbol{\phi}) \ge 2 \left\| \boldsymbol{\nu}' - \mathcal{C}_{\overline{\Lambda},V,U}(\boldsymbol{\phi}) \right\|_\infty^2. \tag{I.83}$$

For an asymptotic sequence, the normalized partition, frame, interferometer, and hence the image $\mathcal{R}_{\overline{\Lambda},V,U}$ may depend on N; these indices are suppressed for readability. Consequently, if

$$\delta_N := \mathrm{dist}_\infty \left(\frac{\boldsymbol{n}_N'}{N}, \mathcal{R}_{\overline{\Lambda},V,U} \right), \tag{I.84}$$

then each coefficient satisfying the output-weight constraint obeys

$$\sup_{\boldsymbol{\phi}} |C_{\mathbf{m}}(\boldsymbol{\phi})|^2 \leq \exp(-2N\delta_N^2). \tag{I.85}$$

The number of type vectors is at most

$$B_{M,N} := \prod_{k=1}^{M} (p_k + 1)^{\binom{M}{k}}, \tag{I.86}$$

which is polynomial in N when M is fixed. Bounding the phase integral in (I.31) by the supremum of its integrand gives the finite-N estimate

$$P_V^{(\Lambda)}(\boldsymbol{n}'_N|\boldsymbol{n}, U) \leq \frac{B_{M,N}}{\mathcal{N}_{\Lambda,\boldsymbol{n}}(V)} \exp(-2N\delta_N^2), \tag{I.87}$$

whenever the initial projected norm is positive. This finite-N bound provides an upper estimate but need not give the sharp exponential decay rate; in particular, it supplies no matching lower bound.

Proposition. Consider a sequence with fixed M, positive projected norms, and

$$\liminf_{N\to\infty} \delta_N > 0, \qquad -\frac{1}{N} \log \mathcal{N}_{\Lambda,\boldsymbol{n}}(V) \longrightarrow 0. \tag{I.88}$$

Then there exists $c > 0$ such that

$$P_V^{(\Lambda)}(\boldsymbol{n}'_N|\boldsymbol{n}, U) \leq \mathrm{poly}(N)\, \mathrm{e}^{-Nc} \tag{I.89}$$

for all sufficiently large N.

The second hypothesis states that the inverse initial normalization grows subexponentially. The regular Gaussian regime of Appendix H, when its global hypotheses are satisfied, provides one sufficient mechanism for this property. In contrast, the bound (I.87) itself does not require regularity of the output map or strictly positive minor probabilities.

Inside $\mathcal{R}_{\overline{\Lambda},V,U}$ the modulus penalty can vanish. For compatible frames, its regular zero-rate points also satisfy the phase-stationarity condition (I.73). These two observations do not establish a nonzero leading probability. The result still depends on the prefactors, stationary-point geometry, and possible interference. The rate associated with the individual coefficient moduli need not equal the true exponential rate of the full probability, which can be altered by interference or exact zeros.

I.7 Explicit form for three modes

For $M = 3$, we write

$$\Lambda = (p + q + r, q + r, r), \tag{I.90}$$

so that

$$p_1 = p, \qquad p_2 = q, \qquad p_3 = r, \qquad N = p + 2q + 3r. \tag{I.91}$$

Let

$$g = (x, y, z) \in U(3) \tag{I.92}$$

be the columns of $g = Ut_{\boldsymbol{\phi}}V$. The relevant minors are

$$A_i^{(1)} = x_i, \qquad A_{ij}^{(2)} = x_i y_j - x_j y_i, \tag{I.93}$$

whereas

$$|A_{123}^{(3)}|^2 = 1. \tag{I.94}$$

The first component of (I.48) is

$$\nu_1' = \frac{p}{N}|x_1|^2 + \frac{q}{N}\left(|A_{12}^{(2)}|^2 + |A_{13}^{(2)}|^2\right) + \frac{r}{N}. \tag{I.95}$$

The identity (I.49) gives

$$|A_{12}^{(2)}|^2 + |A_{13}^{(2)}|^2 = |x_1|^2 + |y_1|^2. \tag{I.96}$$

Since

$$|x_1|^2 + |y_1|^2 + |z_1|^2 = 1, \tag{I.97}$$

it follows that

$$\nu_1' = \frac{p+q+r}{N}|x_1|^2 + \frac{q+r}{N}|y_1|^2 + \frac{r}{N}|z_1|^2. \tag{I.98}$$

This expression is precisely the first component of

$$Q(g)\frac{1}{N}\begin{pmatrix} p+q+r \\ q+r \\ r \end{pmatrix}. \tag{I.99}$$

Repeating the argument for the other two components,

$$\boldsymbol{\nu'} = Q(g)\overline{\Lambda}, \qquad \overline{\Lambda} = \frac{1}{N}\begin{pmatrix} p+q+r \\ q+r \\ r \end{pmatrix}. \tag{I.100}$$

The general form of the map is thus recovered directly for $SU(3)$.

I.8 Expression in terms of the effective single-particle matrix

The preceding derivation uses the normalized shape of the irrep,

$$\overline{\Lambda} = \frac{\Lambda}{N}. \tag{I.101}$$

Let

$$\boldsymbol{\lambda} = \operatorname{spec}\rho_{\overline{\boldsymbol{n}},\Gamma} \tag{I.102}$$

denote the physical spectrum of the effective single-particle matrix. When

$$\overline{\Lambda} = \boldsymbol{\lambda} \tag{I.103}$$

and V is a corresponding eigenbasis,

$$\rho_{\overline{\boldsymbol{n}},\Gamma} = [\overline{\boldsymbol{n}}]^{1/2}\Gamma[\overline{\boldsymbol{n}}]^{1/2} = V[\boldsymbol{\lambda}]V^\dagger = V[\overline{\Lambda}]V^\dagger, \tag{I.104}$$

the expression (I.59) can be written directly as

$$\mathcal{C}_{\overline{\Lambda},V,U}(\boldsymbol{\phi}) = \operatorname{diag}\left[Ut_{\boldsymbol{\phi}}\rho_{\overline{\boldsymbol{n}},\Gamma}t_{\boldsymbol{\phi}}^\dagger U^\dagger\right]. \tag{I.105}$$

This identity is exact under the identification (I.104). In particular, when $\rho_{\overline{\boldsymbol{n}},\Gamma}$ is fixed, the map does not depend on a particular eigenbasis within a degenerate spectral subspace. This equality of maps does not imply equality between the projected pure-state probability and the physical sectorial probability. The suppression statement above concerns only the former.

For the three reference configurations used in this manuscript, the spectral identification is exact:

$$\overline{\Lambda} = \frac{\Lambda}{N} = \frac{(33,9,3)}{45} = \frac{1}{15}(11,3,1) = \boldsymbol{\lambda}, \tag{I.106}$$

so the identification with the effective single-particle matrix is direct.

I.9 Maps for the reference configurations

Finally, we consider the same three occupations used in the previous appendices,

$$\overline{\boldsymbol{n}}^{(1)} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right), \qquad \overline{\boldsymbol{n}}^{(2)} = \frac{1}{5}(3,1,1), \qquad \overline{\boldsymbol{n}}^{(3)} = \frac{1}{9}(3,2,4), \tag{I.107}$$

with

$$\overline{\Lambda} = \boldsymbol{\lambda} = \frac{1}{15}(11,3,1), \qquad N = 45, \qquad \Lambda = (33,9,3), \qquad U = F_3. \tag{I.108}$$

For each occupation, the figures display twelve numerically selected compatible frames,

$$Q(V_a)\overline{\Lambda} = \overline{\boldsymbol{n}}, \qquad a = 1, \ldots, 12, \tag{I.109}$$

and display sampled images of $\mathcal{C}_{\overline{\Lambda},V_a,F_3}$. For each frame, the relative torus $\mathbb{T}^2$ is sampled at 2500 phase pairs using a deterministic quasi-uniform sequence. The black points are sampled classical intensities; they do not directly represent the induced classical density or the quantum probabilities. Figs. 19–21 show these images within the simplex of normalized occupations.

The three figures share the same spectral region $\mathcal{S}_{\overline{\Lambda}}$, since $\overline{\Lambda}$ remains fixed. This region must be distinguished from the image of the map: the light-blue region contains all diagonals compatible with the spectrum, whereas the black points represent only

$$\mathcal{R}_{\overline{\Lambda},V_a,F_3} = \mathcal{C}_{\overline{\Lambda},V_a,F_3}(\mathbb{T}^2) \tag{I.110}$$

for the corresponding frame. In general, there is no reason for this image to cover the full spectral region.

Within the same configuration, the frames V_a satisfy the same initial condition

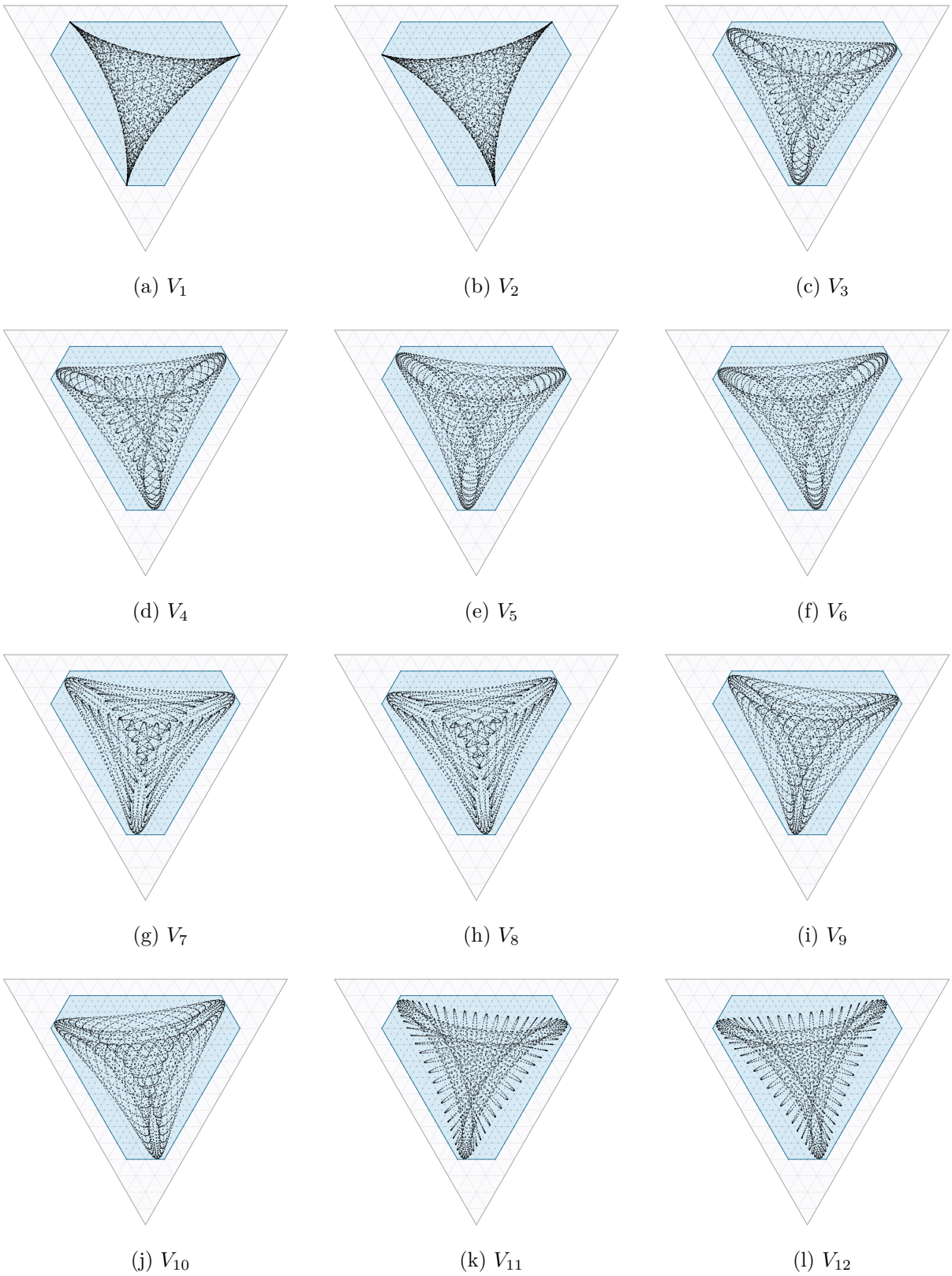
$$Q(V_a)\overline{\Lambda} = \overline{\boldsymbol{n}}, \tag{I.111}$$

but this constraint does not completely determine the unitary matrix. The output map depends on $Ut_{\boldsymbol{\phi}}V_a$ and, therefore, different compatible frames may produce different toroidal images. Frames differing only by left or right diagonal rephasings, however, have the same image: right rephasings leave $V[\overline{\Lambda}]V^\dagger$ unchanged, and left rephasings can be absorbed into the relative torus phases. The spectrum–diagonal condition determines the initial stationary point, but not the complete geometry of the sectorial image after the interferometer.

When moving from the balanced configuration to the partially unbalanced and asymmetric configurations, the occupation $\overline{\boldsymbol{n}}$ changes and, with it, the set of compatible frames. The spectrum, the normalized irrep, and the interferometer remain fixed. The three families thus allow two distinct structures to be separated: the common spectral region, determined solely by $\overline{\Lambda}$, and the toroidal images, which are additionally determined by the choice of frame and by U.

The map $\mathcal{C}_{\overline{\Lambda},V,U}$ is an intensive quantity: once $\overline{\Lambda}$, V, and U are fixed, it does not depend explicitly on N. The total number of particles, by contrast, determines the discretization of the quantum occupations within the simplex and controls the regime in which the asymptotic description becomes valid.

At regular points, the local image of the torus is determined by the Jacobian of $\mathcal{C}_{\overline{\Lambda},V,U}$. Where this Jacobian loses rank, the regular pushforward-density description requires a separate local analysis. Such singularities should not be identified with degeneracy of the normalization Hessian in Appendix H without an additional argument. Boundary frequencies and degenerate stationary points also require care in any more detailed approximation. None of these regularity issues changes the finite-N distance bound (I.87), which does not rely on a Gaussian approximation.



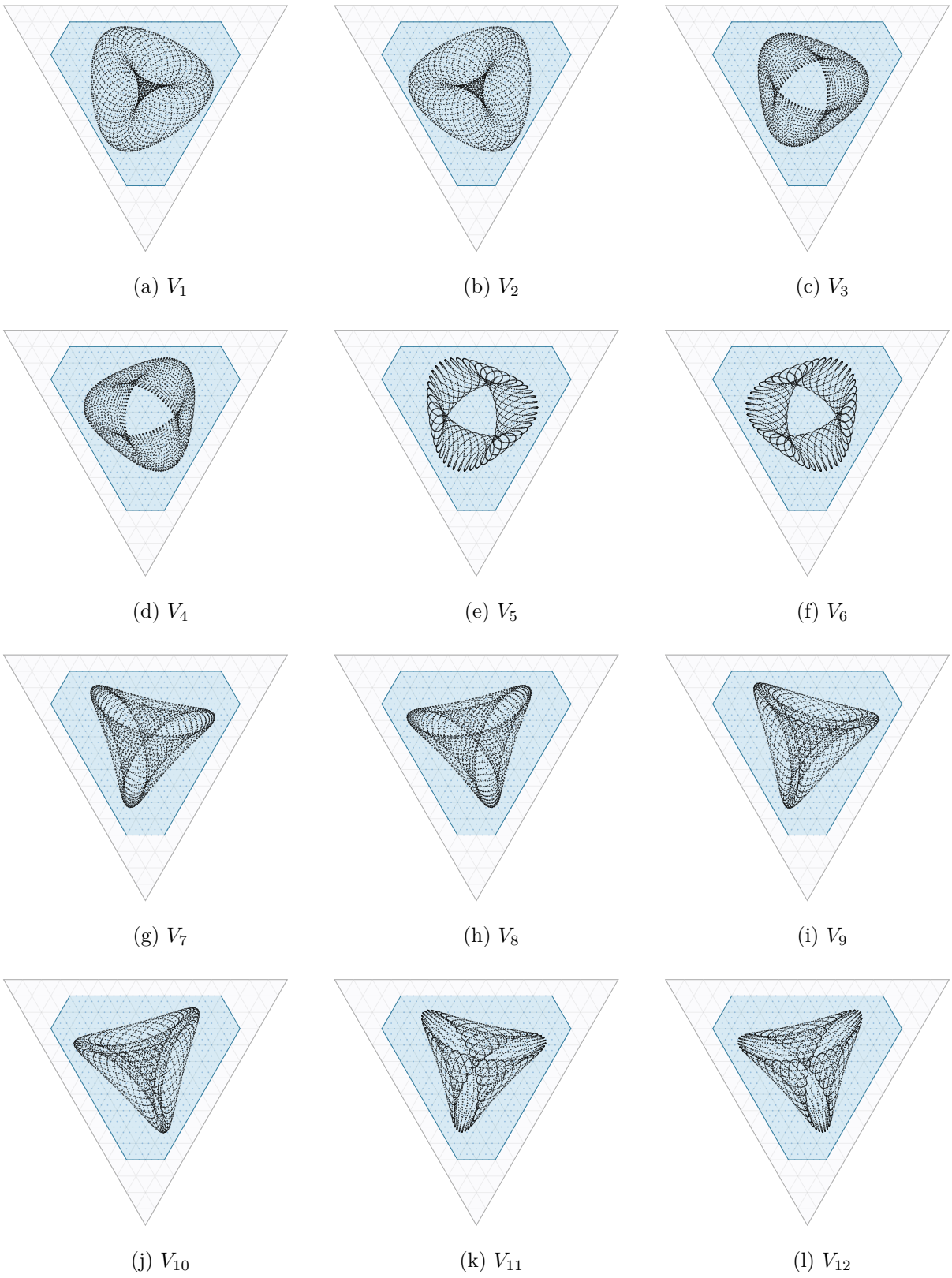
(a) V_1 (b) V_2 (c) V_3

(d) V_4 (e) V_5 (f) V_6

(g) V_7 (h) V_8 (i) V_9

(j) V_{10} (k) V_{11} (l) V_{12}

Figure 19: Classical toroidal map for different frame choices in the balanced configuration $\overline{\boldsymbol{n}} = (1/3, 1/3, 1/3)$, with $\overline{\Lambda} = \boldsymbol{\lambda} = (0.7333, 0.2000, 0.0667)$, $N = 45$, $\Lambda = (33, 9, 3)$, and $U = F_3$. Each subfigure shows the image of the relative-phase torus under $\mathcal{C}_{\overline{\Lambda},V,U}$ for a different frame V_a, with $a = 1, \ldots, 12$. The black points represent the sampled image of the torus, the light-blue region indicates the spectral region associated with $\overline{\Lambda}$, and the outer triangle is the normalized occupation simplex.



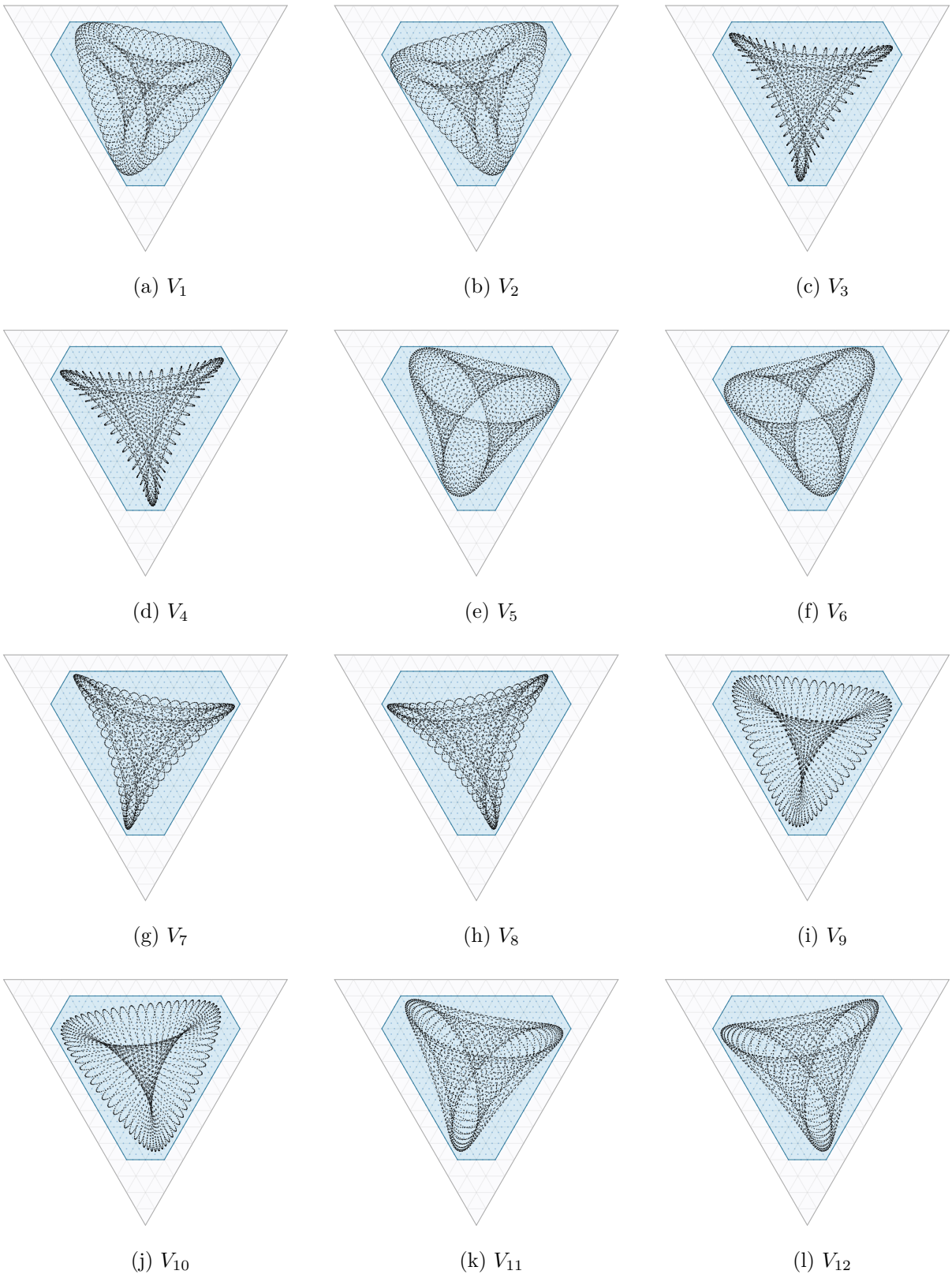
(a) V_1 (b) V_2 (c) V_3

(d) V_4 (e) V_5 (f) V_6

(g) V_7 (h) V_8 (i) V_9

(j) V_{10} (k) V_{11} (l) V_{12}

Figure 20: Classical toroidal map for different frame choices in the partially unbalanced configuration $\overline{\boldsymbol{n}} = (0.6, 0.2, 0.2)$, with $\overline{\Lambda} = \boldsymbol{\lambda} = (0.7333, 0.2000, 0.0667)$, $N = 45$, $\Lambda = (33, 9, 3)$, and $U = F_3$. Each subfigure shows the image of the relative-phase torus under $\mathcal{C}_{\overline{\Lambda},V,U}$ for a different frame V_a, with $a = 1, \ldots, 12$. The black points represent the sampled image of the torus, the light-blue region indicates the spectral region associated with $\overline{\Lambda}$, and the outer triangle is the normalized occupation simplex.



(a) V_1 (b) V_2 (c) V_3

(d) V_4 (e) V_5 (f) V_6

(g) V_7 (h) V_8 (i) V_9

(j) V_{10} (k) V_{11} (l) V_{12}

Figure 21: Classical toroidal map for different frame choices in the asymmetric configuration $\overline{\boldsymbol{n}} = (0.3333, 0.2222, 0.4444)$, with $\overline{\Lambda} = \boldsymbol{\lambda} = (0.7333, 0.2000, 0.0667)$, $N = 45$, $\Lambda = (33, 9, 3)$, and $U = F_3$. Each subfigure shows the image of the relative-phase torus under $\mathcal{C}_{\overline{\Lambda},V,U}$ for a different frame V_a, with $a = 1, \ldots, 12$. The black points represent the sampled image of the torus, the light-blue region indicates the spectral region associated with $\overline{\Lambda}$, and the outer triangle is the normalized occupation simplex.

I.10 Summary

The analysis developed in this appendix identifies the geometry associated with the exponential bound for projected pure-state propagation in a fixed irrep. Starting from the exact probability, the condition for the absence of an exponential penalty leads to the map

$$\mathcal{C}_{\overline{\Lambda},V,U}(\boldsymbol{\phi}) = Q(Ut_{\boldsymbol{\phi}}V)\overline{\Lambda}, \tag{I.112}$$

whose image $\mathcal{R}_{\overline{\Lambda},V,U} = \mathcal{C}_{\overline{\Lambda},V,U}(\mathbb{T}^{M-1})$ defines the classical region associated with the sector under consideration. For a frame compatible with the initial occupation, $Q(V)\overline{\Lambda} = \overline{\boldsymbol{n}}$, the same conditions that remove the exponential penalty also satisfy the stationary condition of the phase.

Together with a subexponential bound on the inverse initial norm, this construction yields exponential suppression of the projected pure-state probability for normalized output occupations separated from $\mathcal{R}_{\overline{\Lambda},V,U}$ by a positive distance. Appendix H provides a Gaussian analysis of the initial norm under its regularity assumptions. The region $\mathcal{R}_{\overline{\Lambda},V,U}$ therefore identifies where the coefficient-level penalty may disappear, but it does not determine the asymptotic probability within that image. The prefactors, the local structure of the stationary points, and interference between distinct contributions remain relevant.

These constructions also relate the geometric description to the physical preparations studied earlier, but the projected orbit state must not be identified with the physical density of a sector. The latter generally contains contributions from several weights, as shown in Appendix G. The Gram matrix Γ and occupation $\overline{\boldsymbol{n}}$ determine the effective single-particle matrix. When its spectrum agrees with $\overline{\Lambda}$, the sectorial map coincides algebraically with the classical map constructed from that matrix. This identifies a precise geometric connection between the two formulations, without transferring the pure-state suppression bound automatically to the physical sectorial or total detection probabilities.

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